MAPPING GEOLOGICAL INFORMATION FLOW IN THE MINING VALUE CHAIN: STRENGTHENING DOWNSTREAM DECISION-MAKING WITH ROBUST OREBODY KNOWLEDGE

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Abstract

A comprehensive understanding of geological information and its application across the Mining Value Chain (MVC), expressed as orebody knowledge (OBK), is essential for improving the reliability, efficiency, and sustainability of mining operations. Geological information comprises data generated through exploration, drilling, and testing, whereas OBK represents the integrated interpretation of that information to support cross-disciplinary decision-making. Despite its central role in risk reduction and value realization, geological data frequently remain fragmented or weakly connected to operational and strategic decisions across the MVC.

This study introduces the Geological Information in the Mining Value Chain Map (GiM Map), a systems-based framework that visualizes how geological information flows, transforms, and integrates into OBK to support decision-making across the mining value chain, from exploration and mineral processing to environmental management and closure. The GiM Map integrates 505 information nodes and 2,786 directed connections, classifying data by type, certainty, and function across the MVC. A proxy-weighted analysis ranks models according to their cumulative input connectivity, indicating that empirically grounded datasets, such as Bond Work Index (BWI), Drop Weight Index (DWI), and flotation kinetics, form the most robust and resilient model foundations. The density of data linkages increases downstream through the MVC, quantifying the progressive transformation of raw geological data into decision-oriented outputs.

The GiM Map provides a reproducible foundation for applying Value of Information (VoI) and Return on Information (RoI) approaches, explicitly linking data quality and connectivity to economic and environmental outcomes. These findings position OBK as a strategic enabler of corporate performance and sustainability. The flow, integration, and quality of geological

information across the MVC emerge as fundamental drivers of value creation, operational transparency, and long-term resilience in the mining industry.

Lay Summary

Mining projects rely on geological information to estimate how much rock must be mined or how much valuable metal can be recovered. These estimates influence nearly every major decision in a mine's life, from early exploration and mine design to processing methods, environmental management, and closure. This research examines how geological information is shared among specialists working on mining projects and how information flows shape decisions across the mining value chain.

The study introduces a framework, the Geological Information in the Mining Value Chain Map, that shows how uncertainty in geological data, such as insufficient rock density measurements, can propagate across multiple stages of a mining project, leading to higher costs, inefficient operations, and increased environmental risk. A key finding is that while modern technology can address many technical challenges, breakdowns in communication, organizational practices, and decision-making processes remain major sources of information loss in the mining industry.

Preface

The work presented in this thesis was prepared and conducted by the author, C. Marin Suárez, under the supervision of Dr. Shaun Barker, co-supervision of Dr. Cassady Harraden, and committee member Dr. Lindsey Heagy. Additional guidance and support were provided by Dr. Haruna Grema.

This thesis fully incorporates a manuscript submitted for publication as a research paper. The authors are Carolina Marin Suarez, Shaun Barker, Cassady Harraden, Lindsey Heagy, Haruna Grema, and Andrea Ruley. This work represents approximately 60.6% of the total content of this thesis. The author of this thesis is the lead and primary contributor to the manuscript, with co-authors providing editorial input and technical guidance.

Chapters 4 and portions of Chapters 3, 5, and 6 are included in the submitted manuscript. The remaining chapters and appendices contain original material developed specifically for this thesis.

AI-based tools—Grammarly, DeepL, OpenAI, and Claude—were used to improve text clarity and, in limited cases, to assist with troubleshooting Python code (Appendix B and E) used in the quantitative analyses presented in Chapter 5. The author retains full responsibility for the content, analyses, interpretations, and conclusions presented in this thesis.

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Electronic Supplementary Material C

file name: ubc_2026_may_marinsuarez_carolina_supp_edges.csv

List of Abbreviations

ABA – Acid–Base Accounting

ARD – Acid Rock Drainage

BWI – Bond Work Index

CCNK – Calcium–Carbonate–Sodium–Potassium geochemical proxy

CSV – Comma-Separated Values

DWI – Drop Weight Index

ESG – Environmental, Social, and Governance

GSI – Geological Strength Index

ICP-MS – Inductively Coupled Plasma–Mass Spectrometry

ICP-OES – Inductively Coupled Plasma–Optical Emission Spectroscopy

LCT – Locked Cycle Test

LIBS – Laser-Induced Breakdown Spectroscopy

LOM – Life of Mine

MVC – Mining Value Chain

NSR – Net Smelter Return

NPV – Net Present Value

OBK – Orebody Knowledge

PLT – Point Load Test

QA/QC – Quality Assurance / Quality Control

RMR – Rock Mass Rating

RQD – Rock Quality Designation

SWIR – Shortwave Infrared

TPH – Throughput

UCS – Uniaxial Compressive Strength

VNIR – Visible Near-Infrared

VoI – Value of Information

XRF – X-ray Fluorescence

XRT – X-ray Transmission

Glossary

ABA	A geochemical method used to predict the acid generation and neutralization potential of waste rock and tailings.
ARD	The outflow of acidic water from sulfide-bearing rocks, typically a key environmental concern in mining.
BWI	A standard laboratory test measuring ore grindability for comminution circuit design.
CCNK	Geochemical proxy representing the ratio of elements used to infer clay alteration (commonly includes Ca, K, Na, and Mg).
CSV	Comma-Separated Values. A file format used for storing and exchanging tabular data between software applications.
DWI	A test quantifying ore resistance to impact breakage, used in comminution modeling.
ESG	Environmental, Social, and Governance criteria—used to evaluate the sustainability and ethical impact of a mining project.
ICP-MS / ICP-OES	Analytical methods for quantifying element concentrations in geological samples.
Gross Revenue	The total income generated from the sale of produced metals or concentrates before deducting operating costs, royalties, taxes, treatment charges, or other expenses.

LCT	A laboratory flotation test designed to simulate continuous circuit performance under steady-state conditions.
LIBS	A sensor-based technology used to analyze the chemical composition of rocks in real time.
LOM	The planned duration over which a mine operates economically, from initial production to closure, based on mineable reserves, production rate, and economic assumptions.
Metallurgical recovery	The percentage of valuable metal or mineral recovered from ore during processing, relative to the amount contained in the feed. A critical parameter affecting revenue and project economics.
MVC	The integrated sequence of processes from exploration to closure, including extraction, processing, logistics, and sales.
NPV	A financial metric that represents the present value of future cash flows from a project, discounted at a specified rate, minus initial capital investment. Used to assess project profitability and value creation
NSR	The revenue from metal sales minus treatment, refining, and transport costs—a key metric for economic evaluation.
OBK	The integrated understanding of orebody characteristics and their impact on mining, processing, and environmental performance.
PLT	A geotechnical test used to estimate rock strength.
QA/QC	Procedures ensuring data integrity, accuracy, and reproducibility.

RMR	Geotechnical indices are used to assess rock mass quality and stability.
SWIR / VNIR	Spectral ranges used in hyperspectral scanning for mineral identification.
TPH	The quantity of material processed by a mining or processing operation per unit time, typically expressed in tonnes per hour (tph) or tonnes per day (tpd). A key driver of production capacity and economics
UCS	A laboratory test measuring the compressive strength of rock samples.
VoI	A quantitative method for assessing how additional data reduces uncertainty and improves decision-making.
XRF	A non-destructive analytical technique for determining elemental composition.
XRT	A sensor-based ore sorting technology that classifies materials based on density and composition.

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Dedication

"Do not judge me by my success, judge me by how many times I fell down and got back up again."

— *Nelson Mandela*

To: Viviana, Ademir, and Paulina

Chapter 1: Introduction

1.1 Introduction

The global transition toward renewable energy and technological advancement has significantly increased demand for critical metals, particularly copper (Cu). While global Cu extraction expanded rapidly in previous decades worldwide (Northey et al., 2014), mine production has recently plateaued, reaching approximately 23.0 million tonnes in 2024 (Natural Resources Canada, 2026). Recent projections indicate that demand will increase to over 50 million tonnes per year by 2050, driven by a compound annual growth rate (CAGR) of roughly 2.0% to 2.6% (BHP, 2024). Meeting this demand presents significant challenges; analysis suggests the industry must add approximately 10 million tonnes of new annual supply within the next decade alone (BHP, 2024). This will require developing several dozen large-scale Cu deposits by mid-century, representing aggregate production capacity comparable to multiple world-class porphyry operations, such as La Escondida in Chile, which currently produces approximately 1.2 Mt of Cu per year (BHP, 2024; Comisión Chilena del Cobre (COCHILCO), 2025). This challenge is compounded by forecasts indicating that more than 200 operating Cu mines may reach depletion before 2035, alongside an insufficient pipeline of new projects to replace lost capacity (Mills, 2024).

More than 60% of global Cu production originates from porphyry Cu deposits (Tabelin et al., 2021). These deposit types are increasingly characterized by declining ore grades and rising water and energy consumption for extraction and processing (Calvo et al., 2016). Calvo et al (2016) reported an average Cu grade decrease of nearly 25% between 2003 and 2013, while energy consumption increased by 46% compared to only a 30% increase in production. More recently,

global Cu head grades declined from 0.56% Cu in 2012 to 0.52% Cu in 2022, with reserve grades declining from 0.54% Cu to 0.50% Cu over the same period (DeLa Cruz, 2024).

Declining grades introduce cascading technical and environmental challenges. As lower-grade porphyry ores are exploited, mineralogical complexities, very fine particle distributions, and concentrate contamination become more prevalent, while operating costs increase due to higher throughput requirements and expanded waste management demands (Flores et al., 2020; Tabelin et al., 2021). The substantial volumes of waste material generated place additional pressure on tailing storage, waste stabilization, and long-term environmental management. Consequently, mining companies have become increasingly concerned with materials that may generate acidification, as well as buffering materials that can mitigate environmental risk during operations and closure (Franks et al., 2011; Frenzel et al., 2023; Tusa et al., 2025).

Empirical studies demonstrate that poor orebody knowledge increases project risk and contributes to industry underperformance. For example, throughput fluctuations during mining operations are often linked to uncharacterized spatial variability in ore properties and insufficient comminution testing (Bearman et al., 1997; Powell, 2013). McCarthy (2003) reported that more than half of feasibility study risks stem from inadequate testing and poor understanding of mining–metallurgy interactions, while Dominy et al (2018) emphasized that limited integration between geology and metallurgy often results in costly design failures. Wood (2018) found that about 70% of mining operations underperform relative to feasibility expectations, largely due to insufficient incorporation of geologic and mineralogical information into mine plans and processing flowsheets.

Gillis et al. (2024) examined 266 asset impairments among Toronto Stock Exchange–listed mining firms (2002–2015) and identified ore grade and access issues in ~20% of cases, with an

additional 5.6% attributed to impurities in the concentrate (e.g., Mercury), and cost inflation. Such technical deficiencies, frequently linked to data uncertainty, limited sampling, or inexperience (Flyvbjerg et al., 2003; Barker et al., 2025), exacerbate financial risks driven by market volatility and commodity prices, the dominant cause of project impairments. Under these combined pressures, improved orebody knowledge across the mining value chain (MVC) has become essential for informed decision-making.

The MVC represents an integrated system in which raw materials sourced from mines are processed and converted into marketable products, encompassing all activities from exploration, extraction, mining, stockpiling, processing, waste management, tailings facilities, marketing, and closure (Dimitrakopoulos & Godoy, 2014; Goodfellow & Dimitrakopoulos, 2017; Hazrathosseini & Moradi Afrapoli, 2023). Key components of the MVC include identifying mineral resources, optimizing operations, managing logistics, and engaging stakeholders to enhance profitability while promoting sustainable mining practices (Hazrathosseini & Moradi Afrapoli, 2023).

Despite its systemic nature, the MVC is inherently multidisciplinary and operationally fragmented. Specialized departments often operate in functional silos, limiting cross-disciplinary communication and integration (Lishchuk et al., 2020; Yildirim et al., 2025). Similar to other complex technical industries, departmental autonomy and differing priorities hinder collaboration (Kunz et al., 2017). Consequently, integration between geology, mining, metallurgy, and environmental management remains limited (Dominy et al., 2018). This fragmentation is reflected in the academic literature, which typically focuses on isolated stages of the value chain rather than on information and knowledge flows across domains.

Concepts such as “mine-to-mill” and “geometallurgy” address some domain gaps but remain restricted in scope. The mine-to-mill concept focuses mainly on the influence of geology

on blasting, mining, and milling performance (Lishchuk et al., 2020; Yildirim et al., 2025). Geometallurgy integrates geology, geotechnical engineering, metallurgy, and geoenvironmental sciences to characterize rock behavior in downstream processes (Dominy et al., 2018a; Navarra et al., 2018; Lakshmanan & Gorain, 2019; Frenzel et al., 2023; Tusa et al., 2025). These approaches establish quantitative links between geological attributes, such as mineralogy, texture, grain size, rock fabric, rock mass behavior, comminution, and liberation. Incorporating such datasets into block models supports predictive modeling of mining and processing performance (Powell, 2013; Dehaine et al., 2020; Olson & Frenzel, 2022).

Building upon these principles, the concept of orebody knowledge (OBK) has emerged over the past 2 decades as a broader, decision-oriented framework. The highly variable nature of ore deposits means that data quality, sampling accuracy, and model precision have a large impact on project outcomes (Carrasco et al., 2004). Data quality and density determine model uncertainty, which ultimately impacts mine development and the financial viability of producing the minerals required to meet the growing demand (Dominy et al., 2018). Unlike manufacturing, mining cannot standardize input materials (McCarthy, 2003), processes must instead adapt to natural geological, geographic, and environmental variability. In accordance with the frameworks established by Platten & Dominy (2009) and Dominy et al., (2018), OBK is defined as the integrated understanding of the orebody's geological, geotechnical, metallurgical, and environmental characteristics that support decision-making across the MVC. Orebody knowledge offers a broader integrative workflow that spans the entire MVC, recognizing that every deposit is geologically and mineralogically unique. By promoting cross-disciplinary communication grounded in quantification of orebody variability, OBK provides a foundation for integrated decision-making throughout the life of a mining project.

Although OBK, geometallurgy, and the MVC are conceptually related, they address distinct but complementary dimensions of mining decision-making (Figure 1.1). Orebody Knowledge focuses on developing an integrated understanding of the orebody itself and provides foundational inputs to MVC activities and processes. Geometallurgy builds upon this understanding by translating geological characteristics into predictions of processing performance. The MVC, in turn, integrates multidisciplinary inputs into operational and economic decision-making across the life of mine. Clarifying these distinctions is essential before examining their integration through the GiM Map.

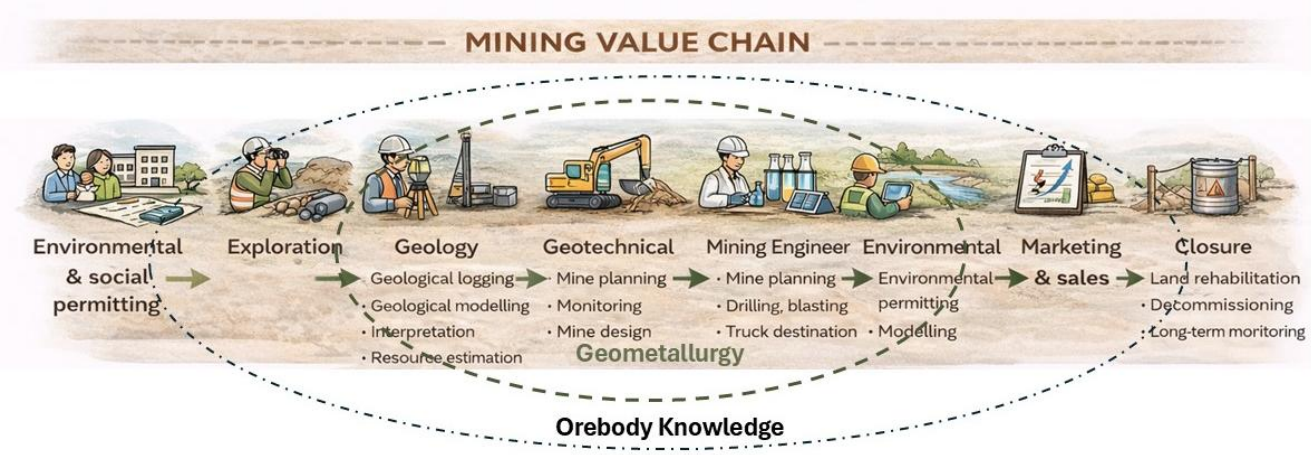


Figure 1.1: Mining Value Chain framework illustrating the sequential stages from Environmental & Social Permitting to Closure. The diagram highlights key functional disciplines: Geology, Geotechnical, Mining Engineering, Environmental, and Marketing & Sales, and emphasizes the role of Geometallurgy and Orebody Knowledge (OBK) as integrative components across the value chain. Green arrows represent the operational flow, while the dashed boundary illustrates cross-disciplinary integration throughout the mine life cycle. Concept developed by the author; base image generated using OpenAI DALL·E (2026) and subsequently modified and expanded by the author (2026).

1.2 Thesis Objectives

The objective of this research is to improve understanding of how geological information contributes to OBK across the MVC, and how this information is transformed, transferred, and utilized by different technical disciplines. The study focuses on tracing how geological data generated during exploration, resource drilling, and production in Cu porphyry mines are integrated into mining, mineral processing, and closure workflows, and how these integrations influence technical outcomes, uncertainty, and project risk.

To support this objective, the Geological Information in Mining Value Chain Map (GiM Map) is developed as a system-based, diagnostic framework that enables visualization and quantification of information flows across the MVC. The GiM Map is not intended as a rigid or prescriptive model, but rather as an interpretive tool to clarify relationships among geological data, technical models, and downstream decisions.

Specifically, this research aims to:

- Map how geological information flows across exploration, mining, mineral processing, and closure, identifying key linkages, bottlenecks, and points where information is transformed or degraded.
- Characterize data quality and quantify connectivity to establish a foundation for probabilistic analysis and future applications of Value of Information (VoI) in mining decision-making.
- Evaluate model reliability and uncertainty by weighting models based on empirical, interpretive, and derived datasets, and assessing how these data types are integrated within geological, geotechnical, and metallurgical models.

- Identify opportunities for improved integration and communication across disciplines, supporting more efficient use of data and risk-informed technical and strategic decisions.

Through these objectives, the GiM Map provides a reproducible and adaptable framework that documents the role and utility of geoscience data across the mining value chain. By highlighting how geological information supports comminution sizing, throughput forecasting, metallurgical recovery assumptions, and long-term mine planning, this research demonstrates how geological data can be transformed from passive records into active decision-support assets, contributing to improved management of geological uncertainty and value creation across mining operations.

1.3 Thesis Organization

This thesis is structured into seven chapters, each focusing on different aspects of the study and the specific areas analyzed:

Chapter 1 introduces the role of geological information in mining decision-making and frames the central problem of information disconnection across the mining value chain. It defines the concept of OBK, outlines the research objectives, and highlights the academic and industry relevance of the study.

Chapter 2 provides context on how mining operations are organized across technical disciplines. It describes how geological information is generated, transferred, and used across different teams, and identifies structural and organizational sources of information loss and misalignment.

Chapter 3 presents the methodology used to develop the GiM Map framework and describes how data classification and model weighting were defined.

Chapter 4 describes the GiM Map framework in detail, with particular emphasis on the main categories that form the map and illustrates how interconnectivity between departments occurs during operations.

Chapter 5 presents the quantitative results for data classification and model weighting. It summarizes the statistical analyses applied to the map data and reports the results of both the data classification and the model weighting processes. Furthermore, the chapter presents a detailed example of the economic impact of the density test.

Chapter 6 interprets the results in terms of OBK, VoI, and decision-making. It discusses the technical, organizational, and policy implications, emphasizing the role of human and institutional factors in the disconnection of geological information.

Chapter 7 summarizes the main findings, contributions, and limitations of the research. It outlines recommendations for improving geological information management and suggests directions for future research.

Chapter 2: Background

2.1 The Mining Value Chain: Definition and Scope

The MVC concept is derived from the broader value chain framework described by Porter (1985), which analyzes the sequence of activities through which companies generate value for customers. In the mining industry, the MVC represents the interconnected set of activities and factors necessary to transform in-situ mineral resources into marketable products delivered to final users (Vorster, 2001; Hazrathosseini & Moradi Afrapoli, 2023; Gholami et al., 2024). All these activities encompass the entire life cycle of mining projects, including prospecting and exploration, mine design and development, extraction (exploitation), mineral processing, extractive metallurgy (smelting and refining), waste management, logistics, and ultimately closure and post-closure management (Goodfellow & Dimitrakopoulos, 2017; Dominy et al., 2018).

The concept of the MVC has evolved from early economic interpretations of the value of mining to a more integrated, system-oriented framework. Lane (1988) laid the foundations by defining cut-off grade as an economic variable rather than a purely geological one, highlighting that value emerges from the interaction between geology, mining, and processing. This perspective was further operationalized through stochastic mine planning and Net Present Value (NPV)-based optimization frameworks, enabling a transition from stage-specific, siloed decision-making to integrated life-of-mine optimization (Dimitrakopoulos & Godoy, 2014).

Although different terms are used in the literature, such as mineral value chain, mine-to-market value chain, or mining supply chain, the underlying concept consistently emphasizes the vertical integration of activities required to deliver a saleable product (Figure 2.1, Vorster, 2001; Gholami et al., 2024). Recent studies (e.g. Hazrathosseini & Moradi Afrapoli, 2023) differentiate the supply chain, which prioritizes efficient physical movement of materials, from the value chain

(Figure 2.1), which focuses on value creation, demand-driven processes, and decision-making across interconnected stages. In this context, the MVC is typically structured by activities, such as drilling and blasting, extraction, processing, and outbound logistics. These activities are supported by enabling functions, including technology development, procurement, infrastructure, and organizational systems, which collectively underpin value creation across the mining project.

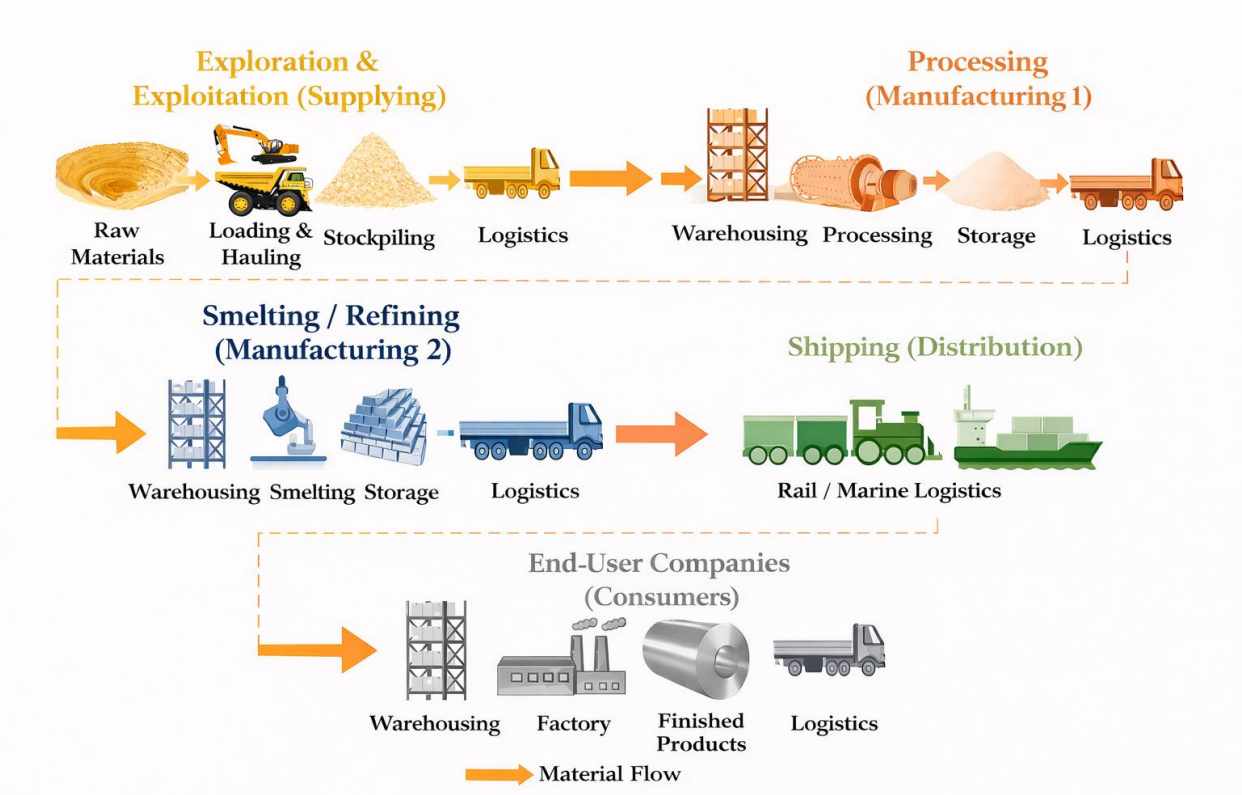


Figure 2.1 Simplified representation of the Mining Value Chain (MVC), including the supply chain, showing the sequence of activities from exploration and exploitation through processing, smelting/refining, distribution, and end-use. The diagram primarily illustrates the physical movement of material and the associated logistics. (Modified from Hazrathosseini & Moradi Afrapoli, 2023).

In this study, the MVC framework is applied to large-scale open-pit mining operations producing sulfide Cu concentrates through flotation circuits. This operational context was selected because open-pit porphyry Cu mines account for the majority of global Cu production

and are characterized by long mine lives, high capital intensity, and strong interdependencies between geology, mine planning, processing, waste management, and closure. These characteristics make them particularly sensitive to the quality, integration, and use of geological information across the MVC.

The focus on sulfide ores reflects their mineralogical complexity and their reliance on comminution and flotation processes, both of which are strongly influenced by geological variability and ore texture. Comminution performance primarily controls throughput capacity, whereas flotation circuits represent a critical value realization stage in Cu operations, where uncertainties in geological inputs propagate into recovery performance, concentrate quality, and ultimately economic outcomes.

While the GiM in the MVC framework is broadly applicable to other deposit types and mining methods, the examples and information flows examined in this thesis are grounded in the porphyry copper model with flotation as the primary ore processing method.

2.2 The Mining Value Chain as an Integrated Information System

In the mining industry, interpreting the MVC only as a linear sequence of material flow oversimplifies the complexity and reality of modern operations. Recent systems-based and optimization-oriented studies conceptualize the MVC as a complex, non-linear system in which strong interdependencies exist between its components, and where value creation depends on how effectively these interactions are managed (Hazrathosseini & Moradi Afrapoli, 2023; Gholami et al., 2024). Furthermore, recent studies emphasize that maximizing the Net Present Value (NPV) of a mining operation requires coordinating components effectively across the entire system, rather than optimizing individual stages in isolation (Goodfellow & Dimitrakopoulos, 2017; Hazrathosseini & Moradi Afrapoli, 2023; Gholami et al., 2024).

In this context, the MVC is defined not only by the physical movement of rock or material flow but also by the interdependencies between extraction stages linked by information flow (Goodfellow & Dimitrakopoulos, 2017; Gholami et al., 2024). Material flow is the movement of raw material from open pits or underground workings through stockpiles, crushing circuits, and processing streams, ultimately producing concentrates or cathode products. Overlaying this physical pathway is an information flow in which upstream decisions, such as extraction sequencing, blasting strategies, or blending plans, directly constrain downstream performance, including mill throughput, metallurgical recovery, energy consumption, and operating stability (Powell, 2013; Dominy et al., 2018; Lishchuk et al., 2020).

While the flow of material through the MVC is largely one-directional (e.g., blasting domain information that feeds blasting operations), the associated information system is inherently iterative. The effectiveness of the MVC depends on how well raw data, such as geochemical assays, drill logs, mineralogical measurements, and sensor outputs, are translated into geological and predictive models that support economic and operational decision-making (Dominy et al., 2018; Lishchuk et al., 2020). This translation is not simply a technical exercise, but a value-adding process. For example, high-quality geological models have been shown to reduce the gap between predicted and actual operational performance (Yildirim et al., 2025). Central to this information system are feedback loops that use operational indicators, including reconciliation results, plant throughput, and metallurgical recovery, to update upstream geological models and planning assumptions. In practice, however, these feedback loops are often incomplete or operate in only one direction. Instead, geological information that feeds drilling and blasting operations should have a feedback loop to improve geological models; however, this information is often not used to improve model performance. This reflects the organizational structures and disciplinary

boundaries that limit the ability of mining systems to learn from operational outcomes and adapt over time (McCarthy, 2003; Wood, 2018).

2.3 Organization of Modern Mining Operations

While the MVC operates as an integrated system, mining organizations are typically structured around discrete technical disciplines. Large-scale operations are commonly divided into Geology, Geotechnical Engineering, Mine Planning, Metallurgy, Hydrogeology, and Environmental Management, each responsible for discipline-specific objectives and performance metrics (Wood, 2018).

In parallel, mining decision-making is structured across temporal planning horizons:

1. **Strategic Planning (Long-term):** This stage centers on the Life-of-Mine (LOM) and the maximization of Net Present Value (NPV) for the asset. It encompasses defining resources and reserves, determining ultimate pit limits, and making capital investment decisions (Dimitrakopoulos & Godoy, 2014).
2. **Tactical Planning (Mid-term):** This planning horizon, typically spanning one to 18 months, bridges strategic objectives and operational execution. It includes annual production scheduling, budget cycles, and the definition of "predictable ore" through refined geological models informed by updated drilling data. (Carrasco et al., 2004; Goodfellow & Dimitrakopoulos, 2017).
3. **Operational Planning (Short-term):** This stage addresses daily to weekly execution, emphasizing the achievement of immediate production targets, grade control, and equipment dispatching. Operational decisions use high-resolution data, such as blast-hole sampling, to minimize variance between planned and actual performance and to enable

prompt processing adjustments based on the characteristics of the incoming feed (Powell, 2013; McCarthy, 2003; Carrasco et al., 2004).

While this organization, along disciplinary lines and across planning time horizons (Figure 1.1), enhances technical specialization and planning clarity, the simultaneous presence of these two organizational layers also creates structural challenges. Functional specialization combined with differing temporal priorities fosters information silos, in which data are generated, interpreted, and optimized to satisfy localized objectives rather than system-wide requirements (Wood, 2018).

2.4 Orebody Knowledge (OBK): Concept and Evolution

Effective management of the MVC depends fundamentally on the quality, scope, and integration of knowledge describing the orebody. As discussed in Chapter 1, OBK provides the conceptual basis for this requirement by recognizing that geological variability exerts a primary control on technical performance, economic outcomes, and risk across all stages of mining operations (Platten & Dominy, 2009; Dominy et al., 2018).

Early interpretations of OBK were largely geology-centric, focusing on spatial distributions of grade, lithology, and structure derived from exploration drilling and resource estimation workflows. Within this traditional paradigm, OBK was frequently implicitly associated with the accuracy of the geological block model and the reliability of resource and reserve estimates (Rossi & Deutsch, 2014). While these elements remain foundational, subsequent research has demonstrated that geological grade alone is insufficient to characterize orebody behavior in downstream mining and processing systems, particularly in deposits exhibiting strong mineralogical, textural, or geomechanical heterogeneity (Carrasco et al., 2004; Emery et al., 2006).

As mining operations have become larger, lower-grade, and more technically complex, the concept of OBK has evolved into a multidisciplinary, systems-oriented framework (Dominy et al.,

2018). Contemporary definitions emphasize that OBK must incorporate not only geological attributes, but also rock mass properties, mineralogical composition, comminution behavior, metallurgical response, contaminant distributions, and environmental characteristics that influence waste management and closure performance (Dominy et al., 2018; Lakshmanan & Gorain, 2019; Barker et al., 2025). In this expanded view, OBK is no longer confined to exploration or resource definition, but represents a continuously evolving body of knowledge that informs strategic, tactical, and operational decisions across the MVC.

A key characteristic of OBK is its direct relationship to uncertainty and risk. Natural geological variability introduces uncertainty, while limitations in sampling density, data quality, and model resolution add epistemic uncertainty. The goal of OBK is not to eliminate uncertainty but to characterize it sufficiently to support informed decision-making and risk management (Rossi & Deutsch, 2014; Dominy et al., 2018; Barker et al., 2025). Better OBK reduces the likelihood of unexpected variability in mining and processing performance, thereby lowering technical risk and enhancing the reliability of economic forecasts. In contrast, gaps in OBK increase the probability that uncertainty will propagate through planning and operational models, affecting production forecasts, recovery assumptions, and long-term value creation.

2.5 Importance of Orebody Knowledge in the Mining Value Chain

Although OBK is widely acknowledged as influential in mining operations, its impact becomes clear when considering how geological information collected throughout the mining life cycle systematically informs each stage of the MVC. Geological data from geophysical surveys and drill core analysis directly inform exploration targets and resource classification, while geotechnical parameters guide mine design and extraction sequencing. During extraction, geological models shape blast design and grade control protocols, and metallurgical test data

combined with mineralogical characterization determine processing parameters and strategies to optimize recovery (Lishchuk et al., 2020).

Beyond active operations, OBK also underpins environmental management, closure planning, and economic decision-making. Environmental management relies on geochemical characterization for waste classification and acid rock drainage prediction. Closure planning requires strategic integration of geological knowledge to ensure long-term geotechnical and geochemical stability of waste rock repositories, tailings storage facilities, open-pit walls, and underground workings, as well as to estimate closure costs (Lotter, 2011; Parbhakar-Fox & Lottermoser, 2015). In addition, OBK incorporates market considerations, as understanding deleterious elements enables identification of penalty sources and the development of selective strategies that minimize economic penalties while maximizing product value (Lotter, 2011). These extended linkages highlight the need for systematic visualization of where geological information influences decisions across the MVC and where it is lost (Dominy et al., 2018; Tusa et al., 2025).

2.6 Geometallurgy: History and Evolution

The evolution of OBK toward a multidisciplinary concept has been closely associated with the development of geometallurgy as a formal framework for integrating geological information into mining and processing. The term geometallurgy was first introduced by McQuiston and Bechaud (1968), who argued that geological and metallurgical information are fundamentally interconnected and should be jointly evaluated to define economically mineable resources.

In the 1980s and 1990s, large mining operations began to implement practical applications of geometallurgy. Holmgren and Marti (1984) introduced metallurgical domains to support production forecasting. Bojcevski (2004) demonstrated how rock texture and mineralogy influence

metallurgical response at the George Fisher mine, Australia. These studies represented the initial operational efforts to map geological attributes to processing behavior.

During the 2000s, geometallurgy evolved from a conceptual link between geology and processing into an integrated framework for value-based decision-making. The AMIRA P843 GeM project provided one of the first structured approaches to linking rock properties with processing performance through spatially constrained models (Keeney, 2011). This was complemented by early efforts to embed metallurgical attributes directly into geological block models (William. R & Richarson, 2004) and to frame geometallurgy as a risk- and value-management tool across the mining value chain (Dunham & Vann, 2007).

More recently, the Primary Resource Efficiency for enhanced Prediction (PREP) (Figure 2.2) framework formalized this integrated perspective by establishing a deposit-agnostic methodology for linking geological, mineralogical, and processing information within a unified predictive system (Lishchuk et al., 2020). Together, these contributions illustrate the progressive shift toward treating geological information as a dynamic, system-wide input rather than a discipline-specific dataset.

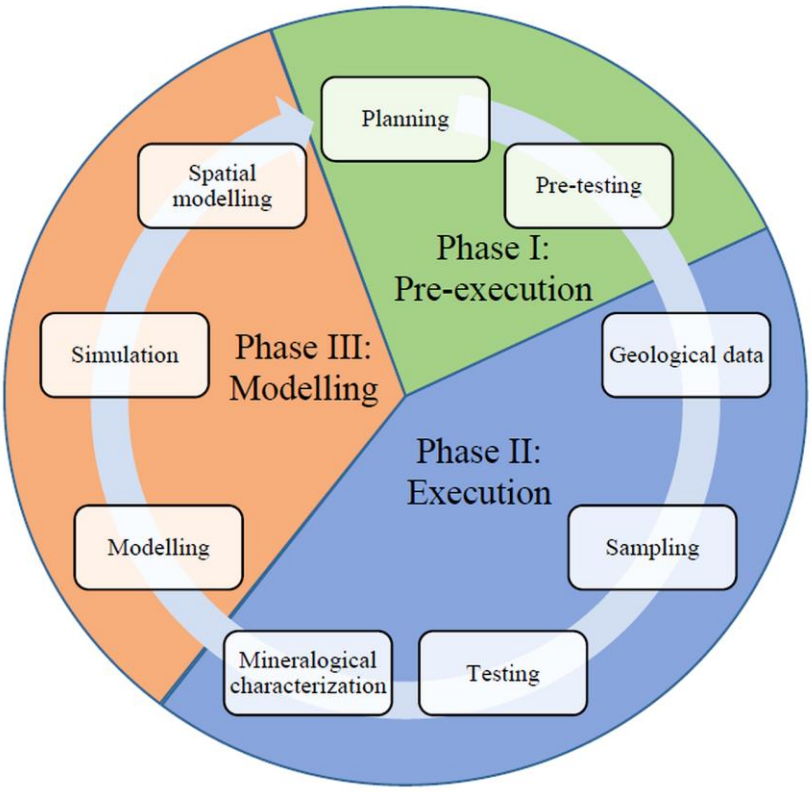


Figure 2.2 Conceptual representation of the main phases involved in the development of a geometallurgical program, integrating the PREP framework. The workflow includes planning, sampling, testing, and spatial modelling. (Lishchuk et al., 2020).

2.7 Geometallurgy and Cross-Disciplinary Integration

According to the evolution of the geometallurgy concept, it is now commonly defined as an integrated approach that combines geological, mineralogical, geochemical, geotechnical, and metallurgical data to predict how different parts of an orebody will behave during mining and processing (Dominy et al., 2018; Navarra et al., 2018; Lakshmanan & Gorain, 2019; Lishchuk et al., 2020; Olson & Frenzel, 2022; Tusa et al., 2025). Central to this approach is the use of geological and mineralogical proxies, such as mineral abundance, texture, grain size, hardness

indices, and chemical signatures, to forecast comminution behavior, liberation characteristics, metallurgical recovery, and process variability (Powell et al., 2013; Lishchuk et al., 2020). By linking these attributes to spatial models, geometallurgy extends OBK beyond resource estimation and enables predictive modeling of operational performance across multiple stages of the MVC.

From an organizational perspective, geometallurgy represents an explicit attempt to bridge traditional disciplinary boundaries by establishing shared datasets, models, and performance metrics between geology, mining engineering, and metallurgy (Figure 2.3). As illustrated in Figure 2.3, rather than treating geological data as static inputs for mine planning, geometallurgical workflows emphasize continuous data acquisition, model updating, and feedback between upstream characterization and downstream performance (Powell et al., 2013; Dominy et al., 2018; Lakshmanan & Gorain, 2019). This integration supports better decisions on domain classification, sample selection, mine sequencing, blending strategies, plant design, throughput optimization, and reconciliation.

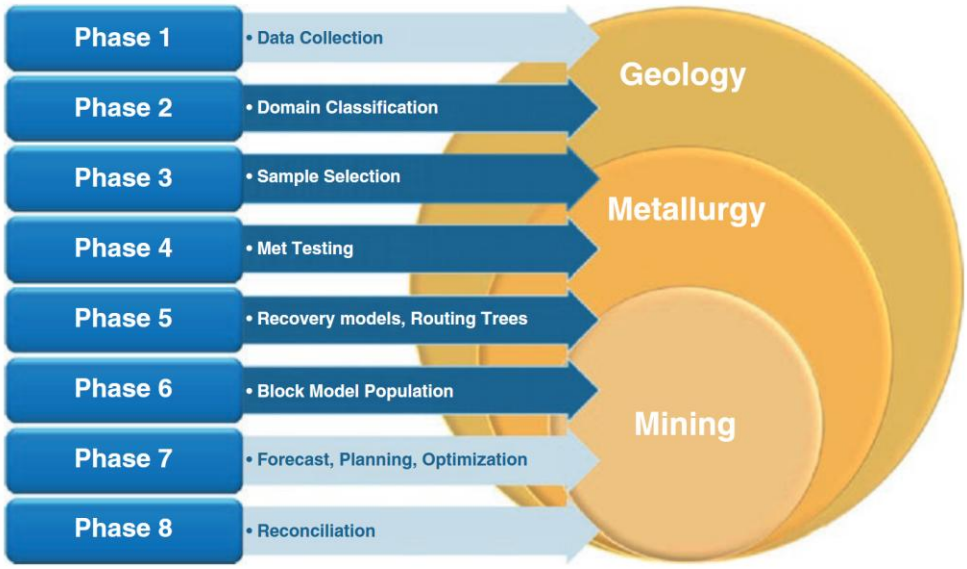


Figure 2.3 Geometallurgical framework showing the interconnections among Geology, Metallurgy, and Mining, along with the processes and activities involved. The image presents the workflow in the mining industry, starting

with data collection by the geology team, followed by processing, and complemented by sample selection and testing by the metallurgy team. All the first stages served as important data inputs for the mining process, such as block modeling, planning, and forecasting. Finally, the flow ends in reconciliation (Lakshmanan & Gorain, 2019).

Despite these advances, the practical implementation of geometallurgy remains uneven across the industry. In many operations, geometallurgical programs are primarily focused on processing performance and plant optimization, with limited integration into broader mine planning, geotechnical design, environmental management, or closure strategies (Navarra et al., 2018; Lakshmanan & Gorain, 2019). As a result, while geometallurgy enhances OBK within specific segments of the MVC, particularly the mine-to-mill interface, it does not fully address the propagation of geological information across all decision-making horizons or organizational departments.

This limitation underscores the need for approaches that move beyond isolated workflows and instead evaluate geological information as a system-wide input. Orebody Knowledge and geometallurgical frameworks offer technical means to characterize, integrate, and propagate geological variability across the MVC. However, without explicit mechanisms to assess the economic relevance of this information, improvements in data quality may be regarded as technical refinement rather than value-generating investments.

2.8 Value of Information (VoI) in Mining Decisions

This gap motivates applying the VoI framework to quantify why, where, and to what extent improved geological information influences mining decisions. The VoI provides a formal framework for quantifying how improvements in information reduce uncertainty and support better decision-making under conditions of risk. The concept was first formally introduced by Ronald A. Howard in 1966, and later expanded within decision theory to support rational

decision-making under uncertainty (Howard, 1966; Eidsvik et al., 2015). In mining systems, VoI enables translating improvements in OBK and geometallurgical characterization into measurable economic and operational benefits across the MVC.

Mining decisions are inherently uncertain due to geological variability, limited sampling, and model simplifications. Improvements in the characterization of orebody attributes can reduce uncertainty, improve forecast reliability, and mitigate technical risk, ultimately enhancing expected economic outcomes (Dimitrakopoulos & Godoy, 2014; Dominy et al., 2018).

Although the VoI is highly relevant, it is rarely applied to geological information within operational mining contexts. Geological data are typically regarded as mandatory technical inputs rather than as economic assets whose value can be assessed against the acquisition, processing, and interpretation costs (McCarthy, 2003; Carrasco et al., 2004). Furthermore, the significance of geological information beyond its initial generation is often underappreciated, and its transfer throughout the MVC is frequently neglected. Consequently, valuable information is lost as data transitions between disciplines and models, which restricts its impact on decision-making and increases uncertainty in subsequent technical and economic outcomes (Eidsvik et al., 2015; Goodfellow & Dimitrakopoulos, 2017; Dominy et al., 2018).

2.9 Knowledge Gaps in Geological Information Flow

Although the importance of OBK and VoI for integrated decision-making is widely recognized, significant gaps remain in understanding how geological information moves across the MVC. A main gap is the lack of structured frameworks that show where geological information is generated, how it is transformed, and where it is applied across disciplines and planning horizons. In current mining practice, systematic mapping of geological information flows is uncommon. Initial efforts to address this have emerged in the geometallurgical domain as

described in Section 2.6 (Figure 2.3), but their scope is limited. These approaches usually focus on specific stages of the value chain and do not capture the continuity of geological information flow from early exploration to mine closure.

There is also limited guidance for assessing the influence, reliability, and interdependencies of models that link data and decisions. This lack of system-level visibility constrains the practical use of VoI concepts, because the value of information cannot be effectively evaluated without understanding how it moves, evolves, and influences decisions across the MVC.

Despite substantial progress in understanding OBK and its role in mining decisions, a system-level, reproducible framework that explicitly maps geological information, its transformations, and its decision relevance across the entire mining value chain is still lacking. This limitation motivates the present study. By mapping geological information flows across the MVC, this research provides a structured basis for evaluating OBK, model reliability, and the Value of Information (VoI).

The Geological Information in the Mining Value Chain Map (GiM Map) novel framework developed to visualize and systematically map how geological information is generated, transformed, and used across the entire mining value chain. The map shows where information enters the system, how it moves between departments, how it changes through models and reports, and where it ultimately supports operational and economic decisions.

Until now, no framework has explicitly mapped the full flow of geological information from exploration to processing, environmental management, and final product. Geological data has always been essential in mining, but its movement across disciplines has remained largely invisible. The GiM Map makes this flow visible, measurable, and analyzable.

This work addresses key gaps identified in the previous chapter: the lack of traceability of geological inputs, the fragmentation between disciplines, and the absence of a system-level perspective on how uncertainty propagates through models and decisions. By mapping these connections, the GiM Map provides a foundation to better manage uncertainty, improve communication between departments, and evaluate the value of information in economic terms.

Chapter 3: Methodology and Data Compilation

To link geological data generated during exploration and resource definition to downstream models and decision-making, this study traces how information is generated, transformed, and transferred across the mining value chain. A mixed-method approach was used, integrating (i) the author's operational experience, (ii) a structured literature review and gap analysis, (iii) consultation and peer review with industry domain experts, and (iv) iterative development and digitization of the Geological Information in the GiM Map. The workflow comprised four phases: (1) an initial map hypothesis informed by industry practice, (2) literature-based refinement, (3) expert validation and revision, and (4) graph-based digitization and interactive visualization (Figure 3.1). While development was anchored in a generalized open-pit porphyry Cu operation producing sulfide concentrate via flotation, the approach is modular and can be adapted to other deposit types, mining methods, and processing routes.

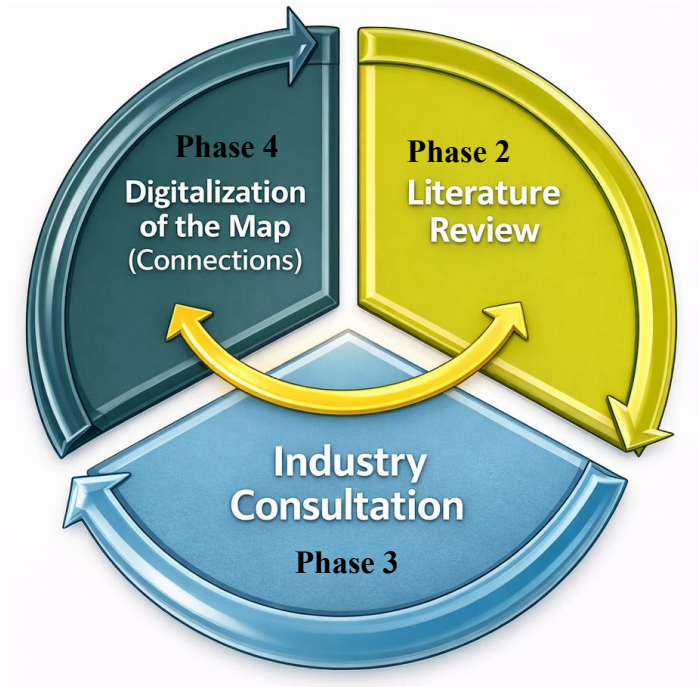


Figure 3.1 The flowchart illustrates the iterative process for constructing GiM Map.

3.1 Initial Framework Development

To establish a practical foundation for understanding geological information flows, an initial framework was developed outlining how geological information is generated and distributed across different operational areas. This framework was developed in Miro® Board and drew upon the Author's comprehensive professional experience across multiple mining disciplines, providing the integrated perspective necessary to map complex interdepartmental relationships. This corresponds to a flow chart with edges and nodes.

This foundational understanding was acquired through the author's direct experience in exploration (both brownfield and greenfield), mine geology, and a comprehensive professional training program. The training included rotations through mine planning, production operations, operational metallurgy, metallurgical planning, environmental management, marketing departments, and human resources. This year-long exposure to diverse mining functions provided an integrated perspective of the complete mining business cycle, which proved essential for identifying potential information pathways that might not be apparent from literature alone. To reduce author bias, phase 1 outputs were treated as hypotheses and were revised only when supported by literature evidence or expert confirmation in Phases 2–3 (Figure 3.1).

3.2 Literature Sources

After developing the initial framework, extensive research was conducted using academic literature and textbooks to examine how geological data is generated and utilized across various departments within mining operations. The comprehensive literature review revealed well-documented workflows in several key areas of mining operations, as summarized in Table 3.1. This information was integrated into the GiM Map structure.

Table 3.1 Summary references used to construct the GiM Map, separated by Operational Units

Operational Units	Key Focus Areas	Primary References
Exploration and Geology	Stages of exploration, sampling methods, geochemical analysis, and grade control.	(Marjoribanks, 2010; Haldar, 2018c)
Geometallurgy	Test programs, main activities in geology, metallurgy, and OBK approaches	(Lund & Lamberg, 2014; Dominy et al., 2018a; Lakshmanan & Gorain, 2019; Dehaine et al., 2020; Lishchuk et al., 2020; Tusa et al., 2025)
Resource Estimation & Planning	Data collection, geostatistical analysis, reserve classification	(Abzalov, 2013; Rossi & Deutsch, 2014; Haldar, 2018e; Lakshmanan & Gorain, 2019)
Mine Operations	Blast design, drilling optimization, geological-operational integration	(Ma et al., 2016; Haldar, 2018a; Pomasoncco-Najarro et al., 2022; Wang et al., 2023)
Geotechnical Engineering	Slope design, structural data integration, safety-critical applications	(Hoek & Bray, 1981; Read & Stacey, 2009; Carpinteri, 2021)
Metallurgical Processing	Process workflows, crushing to thickening, information flow patterns	(Haldar, 2018d; Bascur & Soudek, 2019; Lakshmanan & Gorain, 2019)
Environmental Management	Geological characterization, compliance requirements	(Haldar, 2018b; Tost et al., 2018; Makhathini et al., 2023)
Production Control	Operational optimization, geological data integration	(Dessureault et al., 2001; Marjoribanks, 2010)
Hydrogeology	Sampling process, hydrological modeling	(Harter, 2003; Houben et al., 2018)
Economic Evaluation	Net Smelter Return calculations, financial assessments	(Wellmer, 1989)
Core Scanning Technologies	Use of hyperspectral scanning, XRF, and LIBS in exploration and geometallurgy	(Harmon et al., 2019; Harraden et al., 2019; Lishchuk et al., 2020; Olson & Frenzel, 2022; Harmon, 2024)

In addition to peer-reviewed journal articles, technical reports on porphyry Cu deposits were used to provide industry insights not captured in the literature review (Table 3.2). These reports were particularly insightful, providing operational information that is rarely reported in academic literature.

Table 3.2 Technical Reports Reviewed for Map Enhancement

Technical Reports	Location	Focus Areas	Reference
Porphyry Cu Operations	Canada	Mineral processing connections, environmental management	(Stewart et al., 2021)
Mining Consulting Reports	Various	Geological data workflows, operational integration	(McCarthy et al., 2020; AGP Mining Consultants Inc., 2023)
Technical Assessments	South America	Environmental-geological connections, compliance workflows	(Evans & Di-Prisco, 2023)
Project Evaluations	South America	Metallurgical applications, process design,	(Marinho et al., 2023; Arseneau et al., 2024)

Although the literature and technical reports offered valuable context, gaps in the connectivity of geoscience data persisted. These were most apparent in midterm mining models and short-term forecasting.

3.3 Industry Expert Consultation and Peer Review

To address gaps identified in the literature review (e.g., transfer of information between departments, reporting, and use of models) and validate the theoretical structure with practical knowledge, additional insights were gathered through discussions with industry professionals. These consultations involved conversations with operational metallurgists, senior metallurgists, senior mining engineers, principal hydrologists, mining geologists, geophysicists, and exploration

geologists across different producing mining companies. These discussions focused on collaboratively constructing workflow diagrams that mapped the specific roles and activities of each expert and, based on their knowledge and experience, the ways geological data were used within their respective organizational units. The objective was to identify how information was generated, transferred, and applied across departments.

Conversations were conducted through a combination of in-person meetings (including whiteboard sessions), virtual meetings (Teams), and follow-up email exchanges. The interviews began with a set of standardized guiding questions to ensure consistency across participants, and were subsequently adapted with more targeted questions tailored to each expert's level of experience and functional responsibilities (Appendix A). Emphasis was placed on understanding data connectivity, specifically how geological information was shared, transformed, and integrated into downstream workflows.

Peer reviews of the GiM map were undertaken by industry professionals, including principal geologists, a manager in Technical Services, a manager in geometallurgy and mineralogy, and a director of an orebody knowledge group. The feedback from these experts clarified interdepartmental processes and identified opportunities to strengthen multidisciplinary communication and interaction across the MVC. The professionals involved in this stage of the research are summarized in Table 3.3.

Through these conversations with specialists, it became apparent that although the established GiM Map captured many of the key connections between geoscience and the MVC, full traceability was not achievable for all production-level pathways due to proprietary workflows and undocumented informal practices. For example, highly specific production-level inputs derived from geological models were often not documented in the existing map. This challenge

highlights the complexity of information pathways in modern mining operations and the need for further research in this field.

Table 3.3 Summary of industry experts involved in the review and validation of the GiM Map.

Expert Role / Discipline	Organizational Context	Number of People	Focus of Review	Contribution to GiM Map Development
Operational Metallurgists	Producing mining operations	3	Use of geological inputs in plant operations; short-term decision-making	Validated downstream use of geological and geometallurgical data; identified gaps between geological models and operational reporting
Senior Metallurgists	Corporate and site-level operations	2	Metallurgical modeling, recovery drivers, and performance reconciliation	Clarified model dependencies on geological data; supported mapping of geometallurgical feedback loops
Senior Mining Engineers	Open-pit mining operations	3	Mine planning, production scheduling, and model usage	Validated use of geological and block models in planning horizons; highlighted information loss between planning stages
Senior Geotechnical Specialists	Mine operations and technical services	2	Geotechnical data generation, rock mass characterization, and model usage	Validated geotechnical data flows and clarified integration of geotechnical models into planning and operational workflows
Principal Geotechnical Engineers	Corporate technical services	1	Geotechnical modeling frameworks and interdepartmental data transfer	Provided high-level validation of geotechnical information pathways and identified opportunities to improve cross-disciplinary communication
Principal Hydrologists	Technical Services / Hydrogeology teams	1	Hydrogeological data generation and model integration	Confirmed limited integration of hydrogeological data into broader MVC workflows; supported inclusion of hydrological models as parallel data streams
Mining Geologists	Mine geology and grade control teams	4	Geological logging, domaining, and reporting workflows	Refined representation of geological data generation and interpretation stages
Geophysicists	Exploration and resource development teams	1	Upstream geophysical data use and interpretation	Supported early-stage data flows; clarified transition from exploration to operational geology
Environmental Specialists	Environmental and permitting teams	1	Environmental data collection, monitoring, and reporting	Supported inclusion of environmental datasets and clarified feedback loops between environmental

				monitoring and operational decision-making
LIBS and Hyperspectral Data Specialists	Geochemistry and sensing workflows	2	Review of methodologies used for geochemical and spectral data acquisition	Validated information pathways and workflows defined in the GiM Map
Exploration Geologists	Brownfield and greenfield exploration teams	3	Data transfer from exploration to development	Identified discontinuities in data continuity across project stages
Resource Estimation Specialists	Resource geology and technical services	2	Resource estimation workflows, domaining, and classification	Validated geological inputs to resource and block models; identified gaps in documentation of data transfer to downstream models
Principal Geologists	Corporate or site-level technical leadership	1	Overall structure and completeness of the GiM Map	Provided high-level validation of geological data flows and model interactions
Manager, Technical Services	Corporate technical services and Peer Review	1	Cross-departmental information transfer and reporting	Identified opportunities to improve documentation of interdepartmental workflows
Manager, Geometallurgy & Mineralogy	Corporate or site-based geometallurgy teams and Peer review	1	Integration of mineralogical and geometallurgical models	Strengthened representation of mineralogical data pathways and feedback mechanisms
Director, Orebody Knowledge Group	Corporate OBK leadership and Peer review.	1	Conceptual framework and OBK alignment	Validated overall structure and confirmed relevance to OBK strategy and decision-making
Total		**29**		Industry professionals formally involved

In addition to formal meetings and structured consultations, the industry engagement also included informal discussions with mining professionals not listed in Table 3.3.

3.4 Map Development and Digitization

Integrating insights from all previous phases, the GiM Map was developed through a systematic digitization process. Multiple versions of the map were produced at this stage, and the outputs were exported as PDF files. Due to the complexity of the framework, several attempts at digitization were required. The final version was formally digitized using Gephi® software, which enabled the creation of a structured database composed of nodes (Electronic Supplementary

Material B), directed edges (Electronic Supplementary Material C), and associated attributes (Figure 3.2).

In this context, nodes represent discrete information entities (e.g., datasets, models, and documents), and edges represent directional dependencies through which information is transferred or transformed. Node colors indicate functional type (e.g., input data, derived outputs, block models, operational plans, analytical products); color has no additional semantic meaning. The edges represent a strictly unidirectional flow of information; however, they are classified as IN and OUT edges, representing the data input and data output flows, respectively. These concepts will be explained in more detail in the following section.

Finally, the resulting database was exported as CSV files, enabling targeted visualization and analysis of specific subsections within the broader GiM Map. This approach enabled rapid generation of focused diagrams while maintaining consistency with the overall system architecture. Additionally, the database structure enabled quantitative assessment of data interconnectivity and the relative impact of information across the GiM Map.

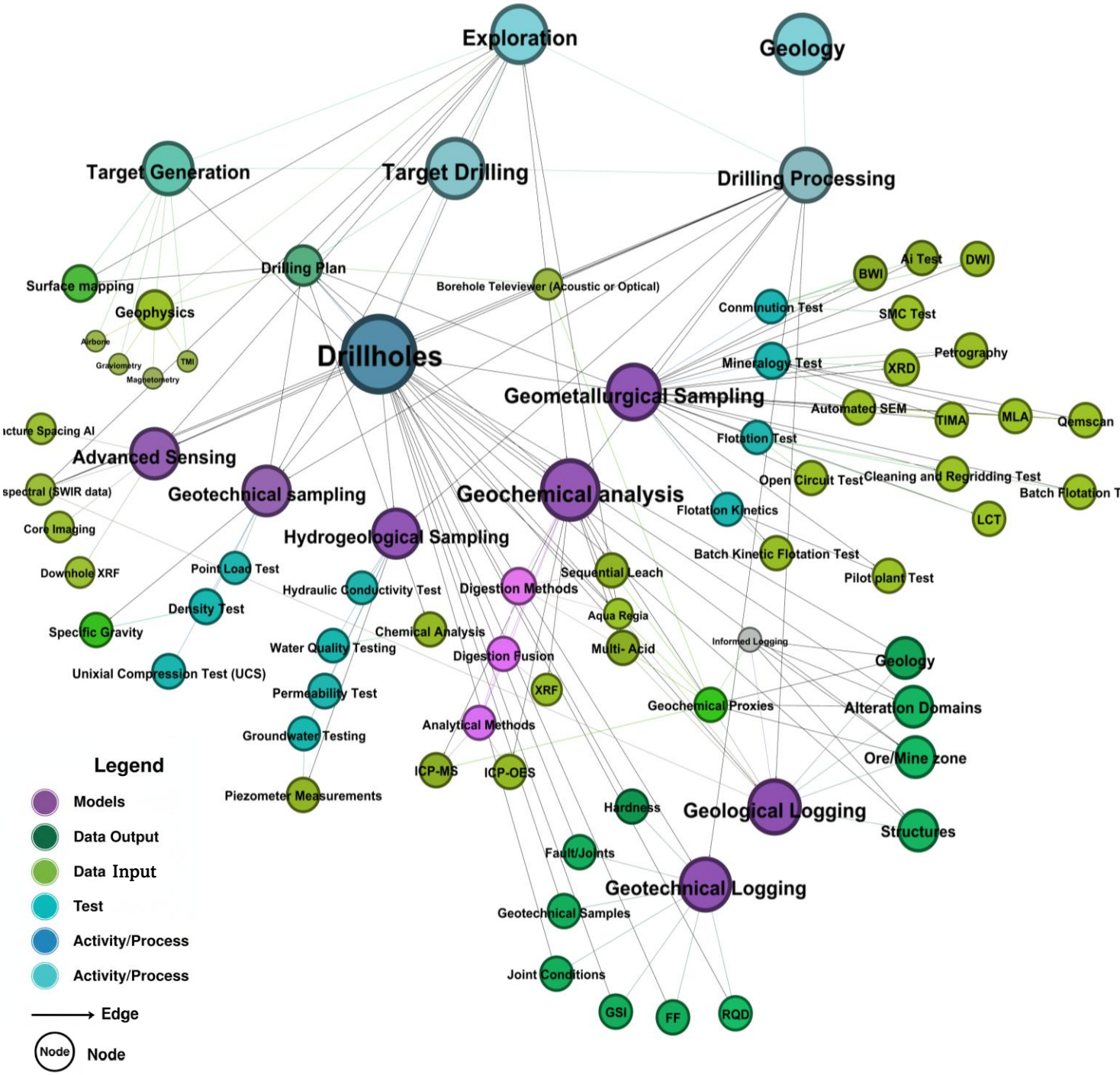

Exploration
Geology
Target Generation
Target Drilling
Drilling Processing
Drilling Plan
Surface mapping
Geophysics
Airborne
Gravimetry
TMI
Magnetometry
Borehole Televiewer (Acoustic or Optical)
Drillholes
Geometallurgical Sampling
Advanced Sensing
Geotechnical sampling
Geochemical analysis
Hydrogeological Sampling
Core Imaging
Downhole XRF
Point Load Test
Density Test
Specific Gravity
Unixial Compression Test (UCS)
Hydraulic Conductivity Test
Water Quality Testing
Permeability Test
Groundwater Testing
Piezometer Measurements
Chemical Analysis
Digestion Methods
Digestion Fusion
XRF
Analytical Methods
ICP-MS
ICP-OES
Sequential Leach
Aqua Regia
Multi- Acid
Comminution Test
BWI
Ai Test
DWI
SMC Test
Mineralogy Test
XRD
Petrography
Automated SEM
TIMA
MLA
Qemscan
Flotation Test
Open Circuit Test
Cleaning and Regridding Test
Batch Flotation T
LCT
Flotation Kinetics
Batch Kinetic Flotation Test
Pilot plant Test
Informed Logging
Geology
Alteration Domains
Ore/Mine zone
Geochemical Proxies
Geological Logging
Structures
Hardness
Fault/Joints
Geotechnical Logging
Geotechnical Samples
Joint Conditions
GSI
FF
RQD
Legend
Models
Data Output
Data Input
Test
Activity/Process
Activity/Process
Edge
Node
Node

Figure 3.2 Initial version of the GiM Map digitized using Gephi software, representing the main category of Exploration and Geological Optimization. Nodes correspond to individual data sources, processes, or geological attributes, while edges represent the relationships and information flow between them.

3.5 Data Classification Approach

Once the map was digitized, all data were classified using a multi-layered classification scheme to enhance data traceability, highlight uncertainty, and clarify how information evolves across the MVC. For example, some geological information is qualitative (e.g., alteration logging or rock type), and its quality is often difficult to assess since these datasets are collected by multiple geologists over (often significant) periods of time (Lishchuk et al., 2015). Other datasets (e.g., Cu and Au assays) are quantitative, with precision and accuracy metrics that can be easily calculated and reported. At the highest level, information is classified as either input or output based on its origin:

- **Input data** refers to information collected through physical sampling, testing, logging, geochemical assays, direct geological observations, and advanced sensors or instruments prior to any processing or modeling (Dominy et al., 2018; Tusa et al., 2025). Examples include core sample assay results (e.g., Cu %, Au ppm via ICP-MS or XRF) and geological logging entries (e.g., lithology, alteration, mineralization zone).
- **Output data** is generated after analysis, typically derived from a combination of qualitative and quantitative modelling, calculations, interpretations, or simulations rather than direct measurements (Dominy et al., 2018). For example, predicted values of montmorillonite and kaolinite obtained with machine learning algorithms, or structural interpretations and alteration domains derived from interpolations in the block model.

It is important to note that the role of data throughout the MVC is not static; data transformations occur as information originally classified as output (e.g., estimated clay percentages from a block model) may later serve as input for subsequent processes (e.g., mine planning, blasting, or drilling index calculations). This highlights the dynamic and multi-functional nature of data flow in mining systems.

3.5.1 Certainty-Based Classification of Data

Input and output data were classified based on the degree of human interpretation and the certainty of measurement. This study adopts three categories: interpretive, empirical, and calculated/derived, each associated with a corresponding level of certainty. These definitions are summarized in Table 3.4. The distinction between interpretive and calculated/derived categories lies in subjectivity: interpretive data depend primarily on human judgment (e.g., geological logging), whereas calculated/derived data rely on algorithms that, while systematic, still embed expert-defined parameters or assumptions.

Table 3.4 Categorization of Information

Category	Definition	Example Source	Certainty Level
Interpretive	Based primarily on geological interpretation, experience, or expert judgment	Lithology, alteration, and mine zone logging	Low
Calculated/Derived	Quantitative data produced through algorithms using empirical inputs or calculations. Interpolation or estimation is included in the process.	Block model, Net Smelter Return (NSR), %Cu derived from the block model.	Moderate
Empirical	Quantitative data from measurement, with uncertainty due to method or sampling	Assays, hyperspectral data, sequential Cu tests	High (method-dependent)

In this study, measurements generated from geotechnical, geochemical, geometallurgical, mineralogical, and hydrogeological tests are collectively referred to as test-based data. This category encompasses parameters obtained from comminution tests (e.g., BWI, DWI, SMC), flotation and leaching experiments, geochemical assays, mineralogical analyses, rock mass classification systems (e.g., RQD, GSI), hydrogeological assessments (e.g., packer testing, permeability, water chemistry), and processing surveys. These test-derived parameters quantify the physical, chemical, and metallurgical behavior of the orebody and provide essential inputs for operational and strategic decision-making. However, the quality and reliability of test-based data can vary significantly. While most tests are empirical, their results are often affected by external conditions and execution-related variability. This variability may arise from factors such as borehole conditions, instrumentation limitations, geological heterogeneity, sample scale, and field execution constraints, particularly in hydrogeological tests like packer testing, permeability assessments, and water quality evaluations (Lishchuk et al., 2015; Dominy et al., 2018a; Pells, 2024).

For this reason, test-based data is classified based on robustness and method complexity, providing a certainty level, as shown in Table 3.5:

Table 3.5 Classification of Test Types

Test-based Data Classification	Description	Examples	Certainty Level	Reference
Highly Robust Empirical	Lab-based, standardized tests with low interpretive input	UCS, BWI, DWI, Abrasion Index (Ai), Specific Gravity, Water Chemistry	Very High	(Bieniawski, 1989; Mwanga et al., 2015; Yildirim et al., 2025)

Empirical	Empirical tests with scale or sampling variability	Geochemical analysis Batch Flotation, Flotation Kinetics,	High	(Dominy et al., 2018a; Lishchuk et al., 2015)
Semiquantitative / multi-step	Composite or multistep empirical methods	Batch Kinetic Flotation, LCT, Mineral Liberation Size	Moderate	(Lishchuk et al., 2015)
Field-controlled Tests	Less controlled, influenced by the environment or execution	Packer Testing, Permeability, Water Quality	Low	(Dominy et al., 2018; Pells, 2024)
Interpretive / Mixed	Dependent on expert judgment, qualitative scoring	GSI	Very Low	(Bieniawski, 1989; Adeli et al., 2017; Yildirim et al., 2025)

UCS = Uniaxial/Unconfined Compressive Stress; BWI = Bond Mill Working Index; DWI = Drop-Weight Index; LCI = Lock-Cycle Test; GSI = Geologic Strength Index.

3.5.2 Data Classification Criteria

To classify test-based data into the five categories outlined in the previous section, primary evaluation criteria were selected through a literature review and by reference to established industry practices. These criteria were designed to systematically assess the robustness, reliability, and methodological complexity of each test. A summary of the criteria and their corresponding descriptions is provided in Table 3.6.

Table 3.6 The main criteria used to classify the information coming from Tests.

Criteria	Description	References
1. Degree of Standardization	Tests that follow well-established, repeatable procedures under lab conditions are highly robust.	(Bieniawski, 1989; Mwanga et al., 2015)
2. Sampling and Scale Sensitivity	Tests sensitive to sample quality, representativity, or scale effects are considered less robust.	(Lund & Lamberg, 2014; Dominy et al., 2018a)

3. Interpretive vs. Measured Data	Tests relying on expert judgment or indirect proxies are less reliable.	(Bieniawski, 1989; Dominy et al., 2018a)
4. Multi-step / Composite Methods	Tests involving multiple variables, proxies, or derived indices are considered semi-quantitative.	(Lund & Lamberg, 2014)
5. Environmental or Operational Conditions	Field tests influenced by in-situ conditions or processing plant environment tend to be less predictable.	(Harter, 2003; Dominy et al., 2018a)

After defining the classification criteria, test-based data were classified into the categories outlined in Table 3.6 using the decision tree shown in Figure 3.3. This structure provides a transparent, reproducible framework for consistently applying these criteria across all test-based data nodes in the GiM Map.

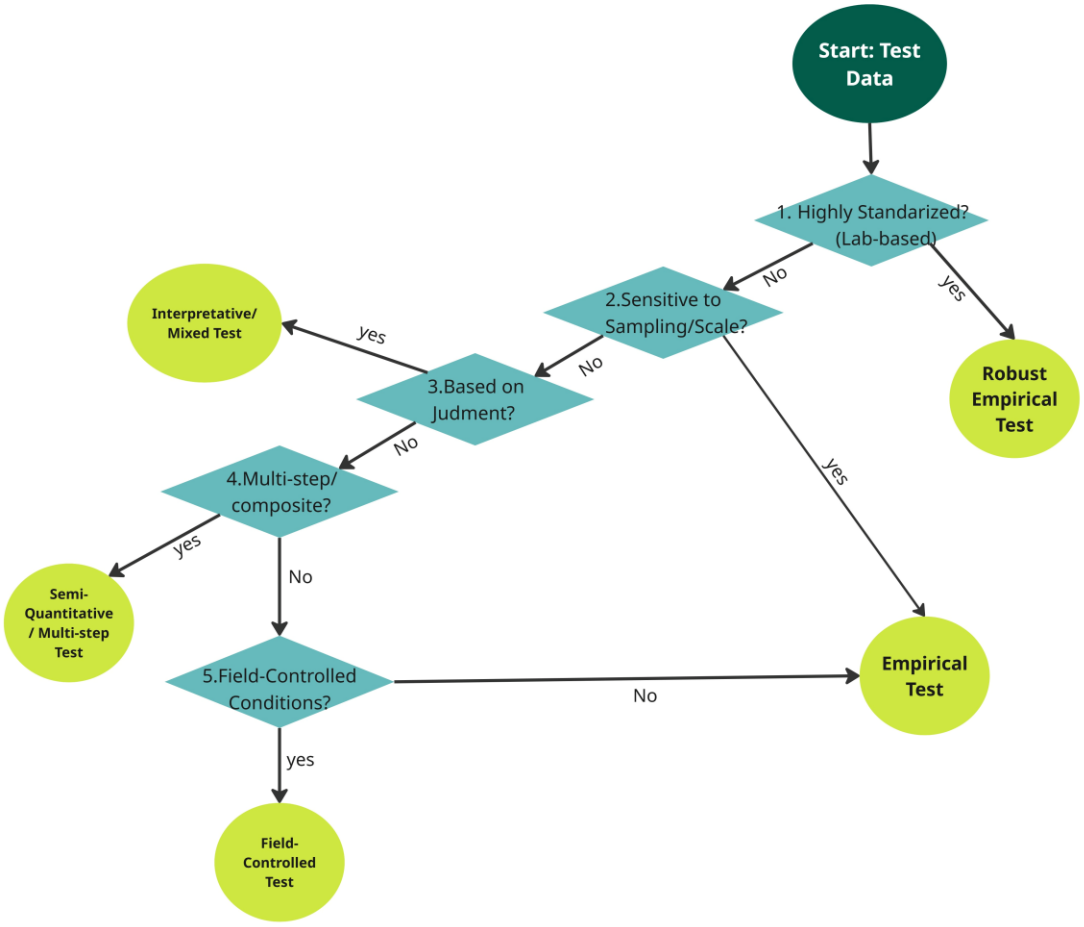


Figure 3.3 Decision tree for classifying test-based data robustness in the GiM Map. Illustration of the decision scheme used to categorize all the information derived from testing and analytical programs.

3.5.3 Data Weighting Approach

To enable quantitative assessment of data within the GiM Map, each certainty level (Tables 3.4–3.5) was assigned a numerical weight ranging from 1 (Very Low certainty) to 5 (Very High certainty) (Figure 3.4). This approach was used as a weighting proxy to convert qualitative confidence assessments into quantitative indicators of data reliability, following the logic of VoI and risk-weighting approaches widely applied in geometallurgical and resource modeling (Goodfellow & Dimitrakopoulos, 2017; Lishchuk et al., 2015; Dominy et al., 2018a).

A total of 26 models listed in the GiM map were selected for evaluation. These models represent the more common models in a porphyry Cu mine, and these include geological, metallurgical, geotechnical, and environmental models. For each model, a cumulative model input score was calculated and used as a proxy for robustness. Python-based processing was used to calculate the total number of connections associated with each model (Appendix B). This score corresponds to the sum of all connections ("edges") linking a model node, and their corresponding weighting. These calculations were carried out on different stages:

(i) All edges connected to the different models were identified and counted.

(ii) The connections were separated into OUT and IN edges. As mentioned in section 3.4, IN edges indicate that the information feeds the model, while OUT edges indicate that the model is the source of the information that feeds another model or process.

(iii) Only IN edges were used to determine the cumulative model input score. These connections were divided into data or test-based data.

(iv) The data-based inputs were classified by data category: empirical, semi-quantitative, and interpretive (Table 3.4), while the test-based data were separated according to the categories described in Table 3.5.

(v) After the classification and according to the level of certainty described in Figure 3.4, the weighting proxy was applied to all the connections.

(vi) The final score of each model corresponds to the total sum of its input connections and the weighting assigned to each data type.

Consequently, models supported by multiple high-certainty, empirically derived inputs receive higher cumulative scores, whereas those reliant on fewer or more interpretive inputs

receive lower scores. In practical terms, the cumulative nature of the score means that each input contributes additively to the total level of informational support for a given model.

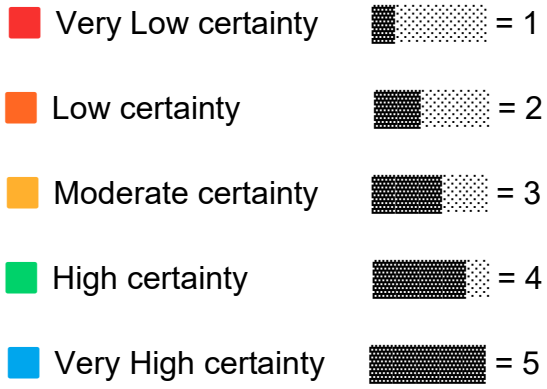


Figure 3.4 Levels of Data Certainty and Assigned Weighting Values. Representation of the five data-certainty levels used in the GiM Map weighting classification.

After the weighting proxy was evaluated and applied, the models were ranked by cumulative input scores; the final scores range between 15 and 231. The score thresholds were defined empirically based on the quartile distribution of the weighted scores, using the 25th and 75th percentiles as classification cutoffs.: High Input Support (≥ 75th percentile; ≥ 136 score), Moderate Input Support (25th–75th percentile; 36–136 score), and Low Input Support (< 25th percentile; ≤36 score). This percentile-based calibration reduces subjectivity and allows site-specific adjustment by comparison with validated operational performance (Goodfellow & Dimitrakopoulos, 2017).

Chapter 4: Mapping Geological Information across the Mining Value Chain

This chapter presents the results of mapping the geological information in MVC and the development of the GiM Map. Map-derived observations are combined with published literature and industry practice to describe how geological information is generated, transformed, and transferred across the MVC.

4.1 GiM Map Structure and Composition

The completed GiM Map (Electronic Supplementary Material A) is a flowchart comprising 505 nodes and 2,786 directed edges that illustrates the extent of interdependencies between geological datasets and downstream decision points. The network of geological information flows across mining operations, from exploration on the left through mining operations to mineral recovery on the right. The map employs a four-category organizational structure (Electronic Supplementary Material D) reflecting key MVC categories: (1) Exploration and Development, (2) Geology Process Optimization, (3) Mine Plan Optimization, and (4) Metallurgical Process Optimization. Each category is further subdivided into specific activities or processes that are interconnected and interact with activities in other categories.

Because the map is large, detailed, and complex, a summary is provided in Figure 4.1, which presents a hierarchical breakdown of the four main categories into their respective activities and sub-activities. The figure also shows the number of connections per category (Pie charts in Figure 4.1), along with estimated timeframes indicating how long each data category usually takes to be completed or processed. The figure illustrates the cascading nature of information flow, where output data from one activity often serves as input to subsequent processes, as in geological models.

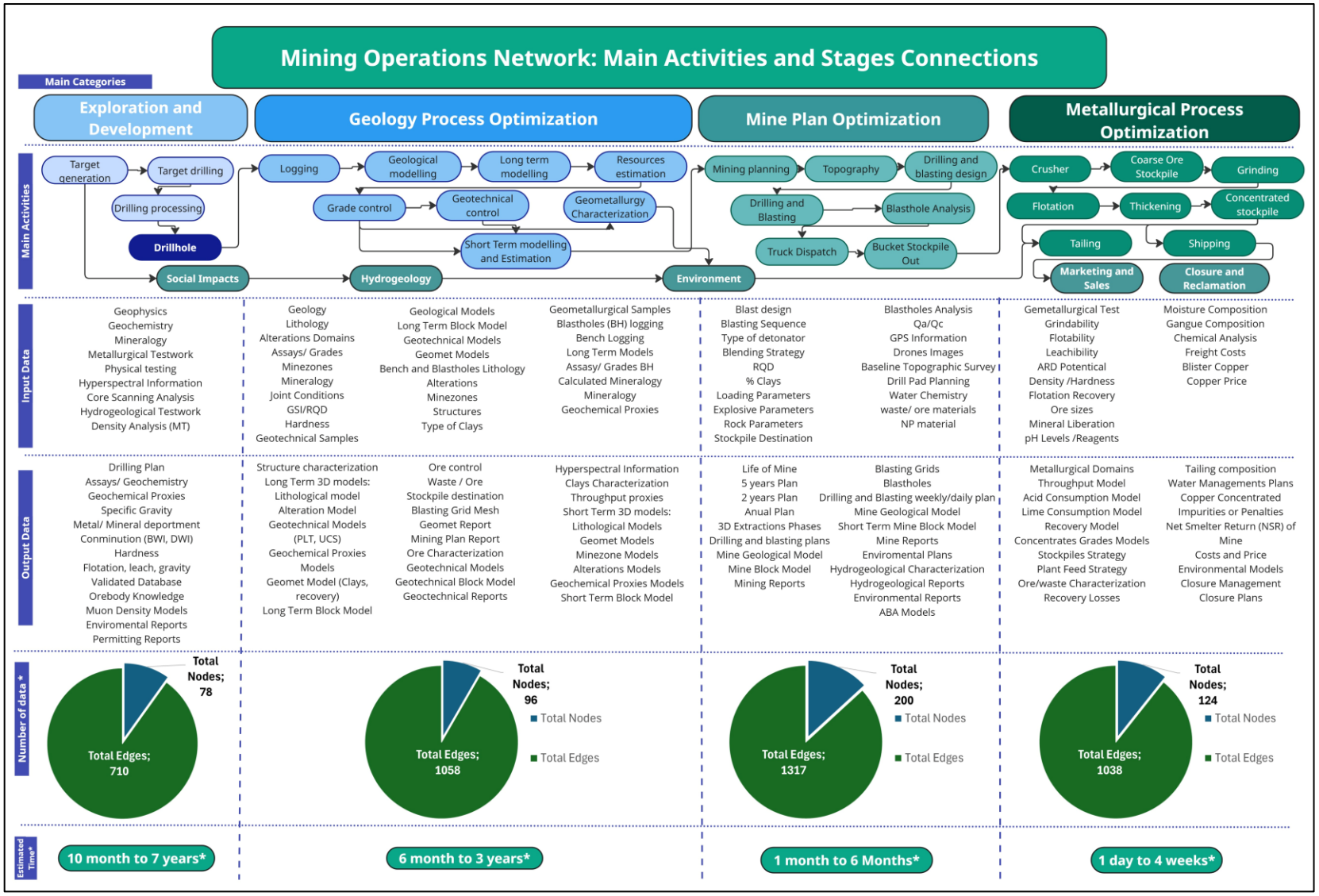

Mining Operations Network: Main Activities and Stages Connections
Main Categories
Exploration and Development
Geology Process Optimization
Mine Plan Optimization
Metallurgical Process Optimization
Main Activities
Target generation
Target drilling
Drilling processing
Drillhole
Logging
Geological modelling
Long term modelling
Resources estimation
Grade control
Geotechnical control
Geometallurgy Characterization
Short Term modelling and Estimation
Mining planning
Topography
Drilling and blasting design
Drilling and Blasting
Blasthole Analysis
Truck Dispatch
Bucket Stockpile Out
Crusher
Coarse Ore Stockpile
Grinding
Flotation
Thickening
Concentrated stockpile
Tailing
Shipping
Marketing and Sales
Closure and Reclamation
Social Impacts
Hydrogeology
Environment
Input Data
Geophysics
Geochemistry
Mineralogy
Metallurgical Testwork
Physical testing
Hyperspectral Information
Core Scanning Analysis
Hydrogeological Testwork
Density Analysis (MT)
Geology
Lithology
Alterations Domains
Assays/ Grades
Minezones
Mineralogy
Joint Conditions
GSI/RQD
Hardness
Geotechnical Samples
Geological Models
Long Term Block Model
Geotechnical Models
Geomet Models
Bench and Blastholes Lithology
Alterations
Minezones
Structures
Type of Clays
Geometallurgical Samples
Blastholes (BH) logging
Bench Logging
Long Term Models
Assasy/ Grades BH
Calculated Mineralogy
Mineralogy
Geochemical Proxies
Blast design
Blasting Sequence
Type of detonator
Blending Strategy
RQD
% Clays
Loading Parameters
Explosive Parameters
Rock Parameters
Stockpile Destination
Blastholes Analysis
Qa/Qc
GPS Information
Drones Images
Baseline Topographic Survey
Drill Pad Planning
Water Chemistry
waste/ ore materials
NP material
Gemetallurgical Test
Grindability
Flotability
Leachibility
ARD Potentical
Density /Hardness
Flotation Recovery
Ore sizes
Mineral Liberation
pH Levels /Reagents
Moisture Composition
Gangue Composition
Chemical Analysis
Freight Costs
Blister Copper
Copper Price
Output Data
Drilling Plan
Assays/ Geochemistry
Geochemical Proxies
Specific Gravity
Metal/ Mineral deportment
Conminution (BWI, DWI)
Hardness
Flotation, leach, gravity
Validated Database
Orebody Knowledge
Muon Density Models
Enviromental Reports
Permitting Reports
Structure characterization
Long Term 3D models:
Lithological model
Alteration Model
Geotechnical Models (PLT, UCS)
Geochemical Proxies Models
Geomet Model (Clays, recovery)
Long Term Block Model
Ore control
Waste / Ore
Stockpile destination
Blasting Grid Mesh
Geomet Report
Mining Plan Report
Ore Characterization
Geotechnical Models
Geotechnical Block Model
Geoctechnical Reports
Hyperspectral Information
Clays Characterization
Throughput proxies
Short Term 3D models:
Lithological Models
Geomet Models
Minezone Models
Alterations Models
Geochemical Proxies Models
Short Term Block Model
Life of Mine
5 years Plan
2 years Plan
Anual Plan
3D Extractions Phases
Drilling and blasting plans
Mine Geological Model
Mine Block Model
Mining Reports
Blasting Grids
Blastholes
Drilling and Blasting weekly/daily plan
Mine Geological Model
Short Term Mine Block Model
Mine Reports
Enviromental Plans
Hydrogeological Characterization
Hydrogeological Reports
Environmental Reports
ABA Models
Metallurgical Domains
Throughput Model
Acid Consumption Model
Lime Consumption Model
Recovery Model
Concentrates Grades Models
Stockpiles Strategy
Plant Feed Strategy
Ore/waste Characterization
Recovery Losses
Tailing composition
Water Managements Plans
Copper Concentrated
Impurities or Penalties
Net Smelter Return (NSR) of Mine
Costs and Price
Environmental Models
Closure Management
Closure Plans
Number of data *
Total Nodes; 78
Total Edges; 710
Total Nodes; 96
Total Edges; 1058
Total Nodes; 200
Total Edges; 1317
Total Nodes; 124
Total Edges; 1038
Total Nodes
Total Edges
Estimated Time*
10 month to 7 years*
6 month to 3 years*
1 month to 6 Months*
1 day to 4 weeks*

Figure 4.1 Mining Operations Network and GiM Map Integration. Overview of the main activities, data types, and information connections, from exploration to metallurgical optimization.

Using the data in Figure 4.1, it is possible to observe varying degrees of complexity across operational domains, with Mine Plan Optimization showing the highest level of connectivity (200 nodes, 1317 connections), followed by Metallurgical Process Optimization (124 nodes, 1038 connections). The geology optimization process (96 nodes and 1058 edges) and Exploration and development (78 nodes and 710 edges) have significantly fewer nodes. The temporal dimensions observed in Figure 4.1 indicate a clear transition from long-term strategic activities (10 months to 7 years in exploration) to shorter-term operational timeframes (1 day to 4 weeks in metallurgical processes). This visualization effectively demonstrates the progressive increase in information density and decision-making frequency as operations progress from conceptual exploration to daily production.

4.2 Exploration and Development: The Foundation of Geological Information

The Exploration and Development stage is the first major operational department of the GiM Map, where information collected early in a project's life is combined to establish the empirical foundation for all subsequent mining phases. As illustrated in Figure 4.2, the flow of information at this stage is inherently iterative rather than linear: data acquired for one purpose (such as target generation) often feed into downstream models (such as grade models) while simultaneously providing feedback that informs and refines exploration strategies and conceptual understanding of the deposit.

During target generation, surface geological mapping, geophysical surveys, and geochemical sampling are integrated to delineate prospective zones for mineralization (Figure 4.2A). A broad range of airborne and ground-based geophysical techniques is applied at this stage,

including potential-field methods (magnetometry and gravimetry), electrical and electromagnetic surveys, and, where appropriate, seismic and radiometric methods. These approaches are increasingly complemented by advanced technologies such as muon tomography and drone-based acquisition systems (Lechmann et al., 2021). In addition, improvements in inversion algorithms, machine-assisted processing, and integrated modelling workflows have significantly enhanced the capacity of these methods to resolve deep, subtle, or structurally complex orebodies (Lelièvre & Farquharson, 2016; Lechmann et al., 2021).

The resulting geophysical anomalies, surface geochemical patterns, and geological interpretations are synthesized into a drilling plan to test potential targets. This plan defines drill density, drilling type (e.g., reverse circulation vs diamond core), hole orientation and depth, and analytical priorities, directly influencing data quality, spatial representativity, and project economics. The use of downhole survey and logging tools further enhances positional accuracy and geological characterization, making drilling design both a scientific and economic decision-making process that determines how effectively subsurface information informs subsequent resource modeling and the quality of the underlying data. Beyond target generation, geophysical methods also support downstream stages of the MVC. Downhole geophysical logging, including gamma density, resistivity, magnetic susceptibility, acoustic televiewer, and full-waveform sonic, provides key subsurface constraints that improve lithological boundaries, structural interpretations, and geotechnical domains (Li et al., 2013).

An essential parallel activity during target generation is environmental and social assessment (Figure 4.2B). Baseline environmental studies, permitting assessments, and community-impact analyses ensure that exploration feasibility aligns with regulatory requirements and social expectations (Franks et al., 2011; Kemp & Owen, 2013; Owen & Kemp, 2013). Early

coordination among geological, environmental, and social-performance teams safeguards project continuity and facilitates proactive identification of social impacts and geoenvironmental risks associated with exploration activities and potential future development.

Once drilling commences, drillholes become one of the most important sources of geological knowledge and a source of physical subsurface material (until mining activities begin), generating the most valuable long-term datasets within the MVC. The utilization of physical drill core samples extends beyond exploration to underpin resource estimation, geotechnical studies, mine planning, metallurgical modeling, and closure studies (Dominy et al., 2018). The reliability of data derived from drill core samples (and their use in subsequent models) depends on drill hole survey accuracy, sampling accuracy, analytical precision, and robust QA/QC protocols.

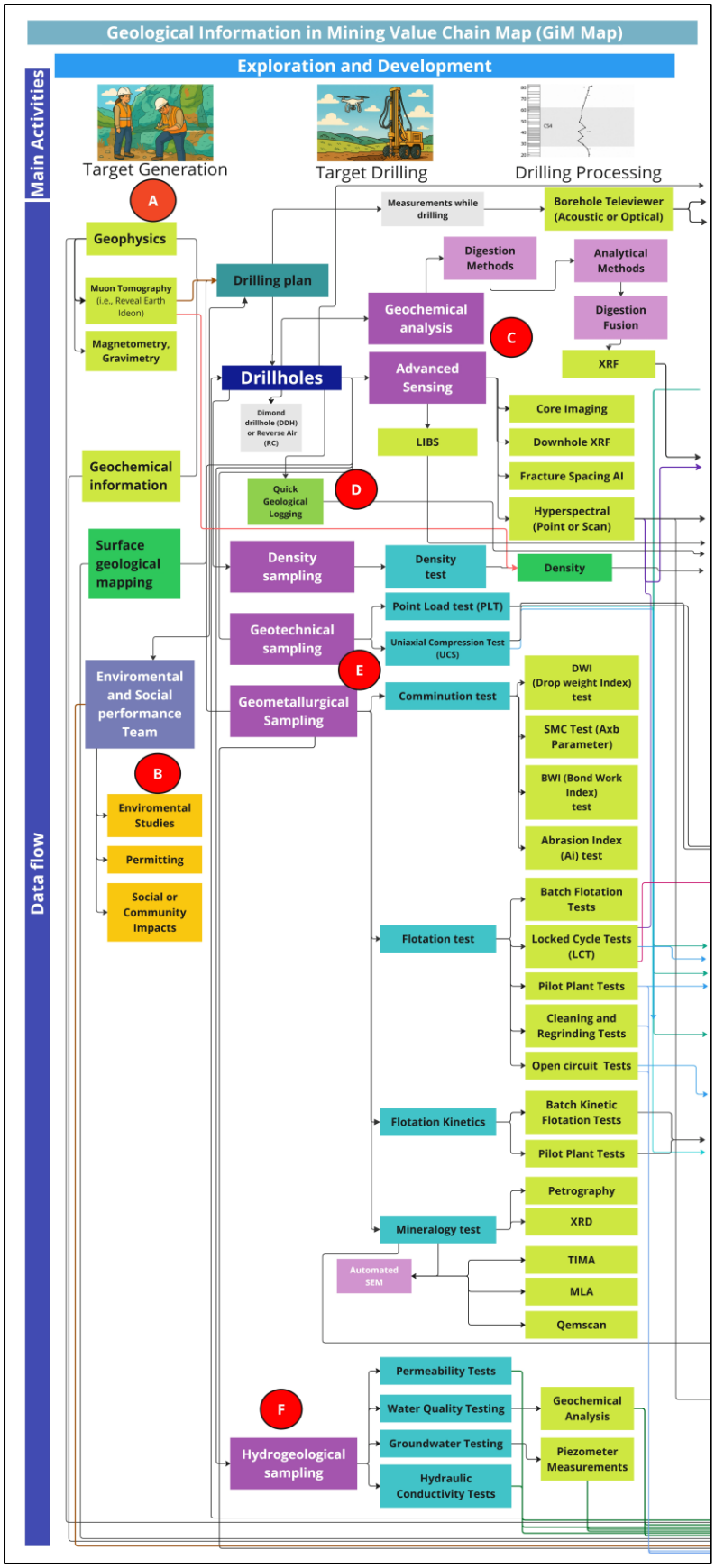

Geological Information in Mining Value Chain Map (GiM Map)
Exploration and Development
Main Activities
Data flow
Target Generation
Target Drilling
Drilling Processing
A
Geophysics
Muon Tomography (i.e., Reveal Earth Ideon)
Magnetometry, Gravimetry
Geochemical information
Surface geological mapping
Enviromental and Social performance Team
B
Enviromental Studies
Permitting
Social or Community Impacts
Drilling plan
Drillholes
Dimond drillhole (DDH) or Reverse Air (RC)
Quick Geological Logging
D
Density sampling
Geotechnical sampling
Geometallurgical Sampling
E
Measurements while drilling
Borehole Televiewer (Acoustic or Optical)
Digestion Methods
Analytical Methods
Geochemical analysis
C
Digestion Fusion
XRF
Advanced Sensing
LIBS
Core Imaging
Downhole XRF
Fracture Spacing AI
Hyperspectral (Point or Scan)
Density test
Density
Point Load test (PLT)
Uniaxial Compression Test (UCS)
Comminution test
DWI (Drop weight Index) test
SMC Test (Axb Parameter)
BWI (Bond Work Index) test
Abrasion Index (Ai) test
Flotation test
Batch Flotation Tests
Locked Cycle Tests (LCT)
Pilot Plant Tests
Cleaning and Regrinding Tests
Open circuit Tests
Flotation Kinetics
Batch Kinetic Flotation Tests
Pilot Plant Tests
Mineralogy test
Petrography
XRD
Automated SEM
TIMA
MLA
Qemscan
F
Hydrogeological sampling
Permeability Tests
Water Quality Testing
Groundwater Testing
Hydraulic Conductivity Tests
Geochemical Analysis
Piezometer Measurements

Figure 4.2 Geological information flows in the exploration and development stage of the MVC. Summary of the GiM Map for early mining stages of exploration.

During core processing, a broad spectrum of analytical and sensing techniques is employed to extract maximum information (Figure 4.2C). Multi-element geochemical analyses (also called assays), including aqua regia and multi-acid digestions, sequential leach tests, and instrumental methods such as ICP-MS, ICP-OES, and portable XRF, yield compositional data for major and trace elements and can be used to delineate mineralogical domains. Advanced sensing technologies such as hyperspectral imaging, laser-induced breakdown spectroscopy (LIBS), and downhole sensors capture mineralogical and structural variability at high spatial resolution, complementing geological logging and geochemical assay analysis. These tools can provide near-real-time feedback that enhances decision-making and prioritization of drilling targets (Marjoribanks, 2010; S. C. Dominy et al., 2018a; Olson & Frenzel, 2022) and can be used to improve geological logging. Rapid-logging programs (Figure 4.2D) deliver real-time data for immediate decision support, whereas detailed logging enhances geological interpretation. The extent of these activities varies with project objectives, phase, and resources. Some operations undertake detailed logging only after geochemical results are available, as explained in section 4.3. Similarly, early-stage projects may defer geotechnical logging due to limited budgets or scale, reflecting corporate strategy and project maturity.

As projects advance toward resource definition, geotechnical and geometallurgical sampling is introduced according to deposit type and development stage (Figure 4.2E). In porphyry Cu systems, these may proceed concurrently with exploration, emphasizing early integration of mechanical and metallurgical parameters in economic assessments. Tests such as UCS, PLT, and DWI quantify rock strength and comminution behavior, while flotation and pilot-plant trials

evaluate metallurgical performance. Automated mineralogical systems, TIMA (TESCAN Integrated Mineral Analyzer), QEMSCAN (Quantitative Evaluation of Minerals by Scanning Electron Microscopy), and MLA (Mineral Liberation Analyzer), provide quantitative insights into mineral associations and textures at micro-scales, bridging geological characterization with metallurgical behavior (Carrasco et al., 2004; Lamberg, 2011).

Finally, hydrogeological sampling forms a further critical dimension, encompassing permeability, groundwater, and hydraulic-conductivity testing (Figure 4.2F). Early acquisition of these data enables the construction of hydrogeological models that inform pit-slope design, dewatering strategies, and environmental management. Integrating hydrological characterization and permitting at this stage ensures that water resources are accurately defined and sustainably managed before mining operations begin.

4.3 Geology Process Optimization: Geological data processing

Systematic collection of geological information occurs during core logging, where lithology, alteration, mineralization, and structural features are described and classified (Figure 4.3A).

Once geological, geotechnical, and geochemical datasets are collected (as described in section 4.2) and validated, they are transferred to the modeling teams responsible for constructing three-dimensional geological representations. These geological and long-term models (including structural, alteration, and ore models; Figure 4.3B) integrate empirical data and transform it into calculated data, using domain expert knowledge to define lithology, alteration, and mineralization domains. The definition of domains is a pivotal interpretive step, as these 3D volumes are based on detailed, measured data, whereas expert interpretation captures the conceptual understanding of the orebody's geometry and internal variability (Emery et al., 2006). Well-defined domains,

grounded in lithological and structural observations and analytical evidence, produce robust estimates and modelling of grade and tonnage (Abzalov, 2016). However, it is important to recognize the scale contrast involved in resource modeling. A density test is performed on only a few centimeters of core, yet after interpolation, that value is used to represent a much larger rock volume, typically a block of 15 × 15 × 10 m. This upscaling process assumes that the small sample is representative of the larger rock mass. If sampling is limited or biased, small measurement errors can propagate across large volumes and significantly influence tonnage, metal estimates, and economic outcomes. Conversely, oversimplified or inconsistent domaining propagates uncertainty or errors through subsequent stages of resource estimation and mine planning. The modelling process is inherently collaborative, with close communication between field geologists and modelers to ensure that the resulting models reflect plausible geological conditions rather than just computational interpolation.

Following the construction of geological domains, the validated datasets are transformed into derived information through resource estimation, producing numerical outputs that represent the spatial distribution of grades and other variables of interest. The Resource Block Model is one of the most critical outcome of this workflow (Figure 4.3C). A block model is a 3D rock grid of cells or blocks used to represent an orebody, where each block contains specific information, including geological domains, metal grade, and density (Dunham & Vann, 2007). The importance of this model lies in its integration of empirical measurements, analytical data, and interpretive domains into a unified spatial model that underpins mine planning, geotechnical design (e.g., slope design), and metallurgical prediction. However, not all relevant geological or geotechnical information can be directly incorporated into a conventional block model structure. Some data types are inherently incompatible with the scalar, block-based representation due to their geometric

complexity or their dependence on structural orientation rather than volumetric properties. A key example is discontinuity orientation data collected by geotechnical teams through televiewer surveys, oriented core logging, or outcrop structural mapping. Discontinuity sets (characterized by dip, dip direction, spacing, persistence, and roughness) exert a fundamental control on bench stability, and overall pit slope configuration (Hoek & Bray, 1981; Read & Stacey, 2009: Verma, 2005).

In a porphyry Cu deposit, variables such as Cu grade, alteration intensity, density, and clay content define the economic, geotechnical, and environmental characteristics of each block (Journel & Kyriakidis, 2004; Emery et al., 2006). The reliability of this model depends not only on the quality of incoming data and the density of data across the deposit, but also on the internal consistency and accuracy of the domaining.

The transition from geological long-term models to operational activities occurs through grade control (Figure 4.3D), which represents the interface between geology and daily mine production. At this stage, information from blast-hole drilling (see Section 4.4 for details) provides chemical and mineralogical data that affect material handling. Geological logging of blast holes plays a critical role in incorporating geological information into mid- and short-term workflows (Figure 4.3E). Bench-scale geological mapping, supported by structural measurements and lithological verification, further validates geological short-term models and ensures that production boundaries align with actual mineralization patterns. Breakdowns in information flow at this stage, such as the absence of fault mapping or misidentification of clay-rich zones. This can lead to operational inefficiencies, including suboptimal fragmentation and material misclassification, with direct repercussions for plant performance.

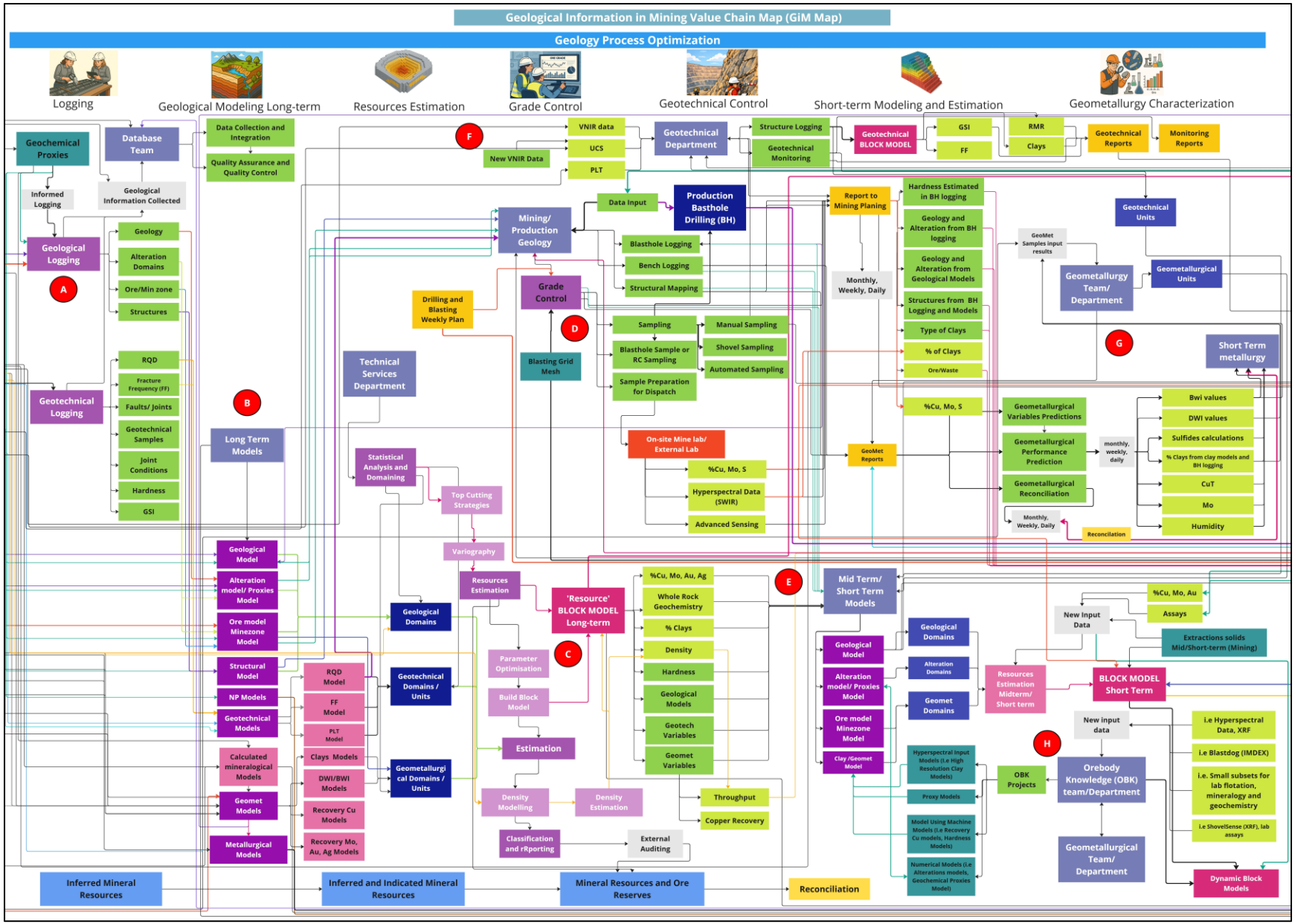

Geological Information in Mining Value Chain Map (GiM Map)
Geology Process Optimization
Logging
Geological Modeling Long-term
Resources Estimation
Grade Control
Geotechnical Control
Short-term Modeling and Estimation
Geometallurgy Characterization
Geochemical Proxies
Database Team
Informed Logging
Geological Information Collected
Geological Logging
Geology
Alteration Domains
Ore/Min zone
Structures
Geotechnical Logging
RQD
Fracture Frequency (FF)
Faults/ Joints
Geotechnical Samples
Joint Conditions
Hardness
GSI
Data Collection and Integration
Quality Assurance and Quality Control
Long Term Models
Geological Model
Alteration model/ Proxies Model
Ore model Minezone Model
Structural Model
NP Models
Geotechnical Models
Calculated mineralogical Models
Geomet Models
Metallurgical Models
RQD Model
FF Model
PLT Model
Clays Models
DWI/BWI Models
Recovery Cu Models
Recovery Mo, Au, Ag Models
Technical Services Department
Statistical Analysis and Domaining
Top Cutting Strategies
Variography
Geological Domains
Geotechnical Domains / Units
Geometallurgical Domains / Units
Resources Estimation
'Resource' BLOCK MODEL Long-term
Parameter Optimisation
Build Block Model
Estimation
Density Modelling
Density Estimation
Classification and rRporting
External Auditing
Inferred Mineral Resources
Inferred and Indicated Mineral Resources
Mineral Resources and Ore Reserves
Reconciliation
VNIR data
UCS
PLT
New VNIR Data
Geotechnical Department
Structure Logging
Geotechnical Monitoring
Geotechnical BLOCK MODEL
GSI
FF
RMR
Clays
Geotechnical Reports
Monitoring Reports
Data Input
Production Basthole Drilling (BH)
Mining/ Production Geology
Blasthole Logging
Bench Logging
Structural Mapping
Drilling and Blasting Weekly Plan
Grade Control
Blasting Grid Mesh
Sampling
Manual Sampling
Shovel Sampling
Automated Sampling
Blasthole Sample or RC Sampling
Sample Preparation for Dispatch
On-site Mine lab/ External Lab
%Cu, Mo, S
Hyperspectral Data (SWIR)
Advanced Sensing
%Cu, Mo, Au, Ag
Whole Rock Geochemistry
% Clays
Density
Hardness
Geological Models
Geotech Variables
Geomet Variables
Throughput
Copper Recovery
Report to Mining Planing
Monthly, Weekly, Daily
GeoMet Reports
Hardness Estimated in BH logging
Geology and Alteration from BH logging
Geology and Alteration from Geological Models
Structures from BH Logging and Models
Type of Clays
% of Clays
Ore/Waste
%Cu, Mo, S
Geometallurgical Variables Predictions
Geometallurgical Performance Prediction
Geometallurgical Reconciliation
monthly, weekly, daily
Monthly, Weekly, Daily
Reconciliation
Bwi values
DWI values
Sulfides calculations
% Clays from clay models and BH logging
CuT
Mo
Humidity
GeoMet Samples input results
Geometallurgy Team/ Department
Geotechnical Units
Geometallurgical Units
Short Term metallurgy
Mid Term/ Short Term Models
Geological Model
Alteration model/ Proxies Model
Ore model Minezone Model
Clay /Geomet Model
Geological Domains
Alteration Domains
Geomet Domains
Hyperspectral Input Models (i.e High Resolution Clay Models)
Proxy Models
Model Using Machine Models (i.e Recovery Cu models, Hardness Models)
Numerical Models (i.e Alterations models, Geochemical Proxies Model)
Resources Estimation Midterm/ Short term
OBK Projects
%Cu, Mo, Au
Assays
New Input Data
Extractions solids Mid/Short-term (Mining)
BLOCK MODEL Short Term
New input data
Orebody Knowledge (OBK) team/Department
Geometallurgical Team/ Department
i.e Hyperspectral Data, XRF
i.e Blastdog (IMDEX)
i.e. Small subsets for lab flotation, mineralogy and geochemistry
i.e ShovelSense (XRF), lab assays
Dynamic Block Models
A
B
C
D
E
F
G
H

Figure 4.3 Geological Information Flow in Geology Process Optimization Stage of the MVC. Red circles with numbers show the relative locations of the information described in the text.

Parallel to these processes, geotechnical information captured during the sampling stage (Figure 4.2E) is processed and transformed for operational use. Generally, the technical services team converts geotechnical information (e.g., PLT, FF, or GSI) from empirical or interpretive data into calculated or derived values. This workflow begins with characterizing rock-mass quality using some formulas and calculations, followed by modeling and geotechnical domaining using lithological, alteration, and structural boundaries to define homogenous zones. Subsequently, geotechnical parameters are estimated within these domains using spatial interpolation methodologies to populate the Block Model (Figure 4.3F). This information is usually transferred to the geotechnical operational teams in 3D models. In some operations, the geotechnical teams maintain their own geotechnical block model, which is fed daily with operational information.

In addition to rock mass strength parameters, structural information plays a critical role in slope design. Fault and joint orientations control potential failure kinematics and are therefore key considerations in structurally controlled failures. Structurally controlled mechanisms govern bench and interramp slope angle design, while block models of rock mass strength parameters (e.g., PLT, RMR, or GSI) generally govern ultimate pit slope stability (Keaton, 2007; Hoek & Brown, 2019). Pit design is therefore an iterative process that must integrate both structurally controlled stability at the bench and interramp scale and rock mass–controlled stability at the ultimate slope scale. This distinction becomes particularly critical in deep open pits, where both structural geometry and rock mass strength interact to define overall slope performance (Read & Stacey, 2009; Keaton, 2007; Hoek & Brown, 2019)

Long-term geotechnical information is complemented by operational data, including radar monitoring, geotechnical monitoring, and operational sampling. This information feeds slope stability models relevant to the mining operation and significantly impacts mining planning, slope design, bench, and pit stability. Furthermore, the geotechnical team received data from the hydrological teams, including water levels, pore pressures, and dewatering pressures, which significantly impact rock quality.

On the other hand, downhole geophysics provides continuous physical-property profiles that complement geological and geotechnical characterization. Downhole geophysical data provide continuous physical property profiles, such as televiewer data (which provides structural orientations and rock-mass characteristics), while density, resistivity, and sonic logs help constrain domain boundaries, reducing uncertainty where core recovery is poor, or contacts are unclear (Williams & Johnson, 2004; Li et al., 2013). Collaboration among geological, geotechnical, and hydrogeological teams ensures that structural and hydromechanical conditions are represented consistently, improving the reliability of slope-stability analyses and mine-safety assessments (Hoek & Bray, 1981). Information then flows toward geometallurgical characterization, where the geometallurgical or production-geology teams synthesize lithological, textural, and mineralogical information to predict throughput, recovery, and concentrate quality (Figure 4.3G) (Lamberg, 2011).

Finally, many mining operations have adopted OBK (Figure 4.3H) or Integrated Geoscience Teams, emphasizing a more holistic and cross-disciplinary approach to orebody understanding. The use of advanced computational tools, including machine learning algorithms, sensor-based scanning, and automated mineralogical characterization, has greatly enhanced these teams' ability to generate high-resolution, data-driven models. These technologies enable more

statistically robust prediction of geological variability, improved domaining, and dynamic updates to block models that reflect real-time conditions in the pit or underground environment.

4.4 Mine Plan Optimization: From Planning to Real-Time Execution

This unit marks the transition from geological (long and short-term), geotechnical, and geometallurgical models into execution focused decision making (Figure 4.4). As illustrated in Figure 4.4, mine planning functions as the operational bridge between resource models and active production. Long- and mid-term block models inform short-term extraction plans, which define bench heights, drilling layouts, daily sequences, and production targets (Figure 4.4A). As mining and processing advance, reconciliation against production data iteratively calibrates the short-term models and quantifies uncertainty in the underlying resource estimates (Figure 4.4B). Thus, effective planning at this scale relies on continuous feedback among modeling and production data, enabling the system to incorporate new (often higher density) information and reduce uncertainty in short-term execution (Richard & Sulemana, 2015; Wang et al., 2023).

Geological domains model feed into mine geological models or mine block models (Figure 4.4C), and, depending on the site or project, this transfer of information varies in detail, resolution, and update frequency. Drilling and blasting designs integrate geological inputs with geotechnical parameters, including rock mass strength, structural fabric, joint orientation, and fracture spacing. In more complex deposits, geometallurgical attributes such as alteration intensity, clay content, or moisture sensitivity may also be incorporated to adjust designs and anticipate blasting or handling challenges. Together, these datasets support the determination of burden and spacing, powder factors, and explosive loads, all of which influence fragmentation quality, slope stability, and downstream comminution performance (Thornton et al., 2002). During drilling and blasting, topographic datasets converge in real time. High-resolution terrain models derived from drone photogrammetry or LiDAR complement blast design by validating drilling layouts, updating pit geometry, and quantifying material movement after each blast (Figure 4.4D). Because surface

topography changes continually in active pits, these updates ensure that planning and reconciliation models remain spatially accurate. Frequent terrain surveys are essential for safety monitoring, supporting pit wall inspections, and identifying slopes or bench instabilities.

Once the designs are ready, the operation begins drilling according to specific sequences. After drilling is completed, the blast holes become the main source of operational information for this stage. Each blast is evaluated through blast hole analysis workflows that integrate pre- and post-blast datasets. Blast-hole samples are typically sent to an on-site laboratory for geochemical analysis, and these assays provide essential grade-control information that refines ore-waste boundaries and updates short-term models in near-real time. Post-blast datasets, including fragmentation distributions, vibration monitoring, and the heterogeneity of the blasted material, are also incorporated into these workflows. Information generated during drilling, blasting, and blast-hole analysis supports ore sorting, linking pit operations with plant performance. As shown in Figure 4.4E, some operations include additional technologies within their ore sorting systems, such as X-ray transmission (XRT), XRF (e.g., ShovelSense), hyperspectral sensors, or LIBS, to classify material streams in real time (Lessard et al., 2014; Robben & Wotruba, 2019). At some sites, technological and geochemical data are integrated in near-real-time, enabling immediate validation of ore boundaries and rapid adjustment of drilling patterns when unexpected structures or lithological changes are encountered.

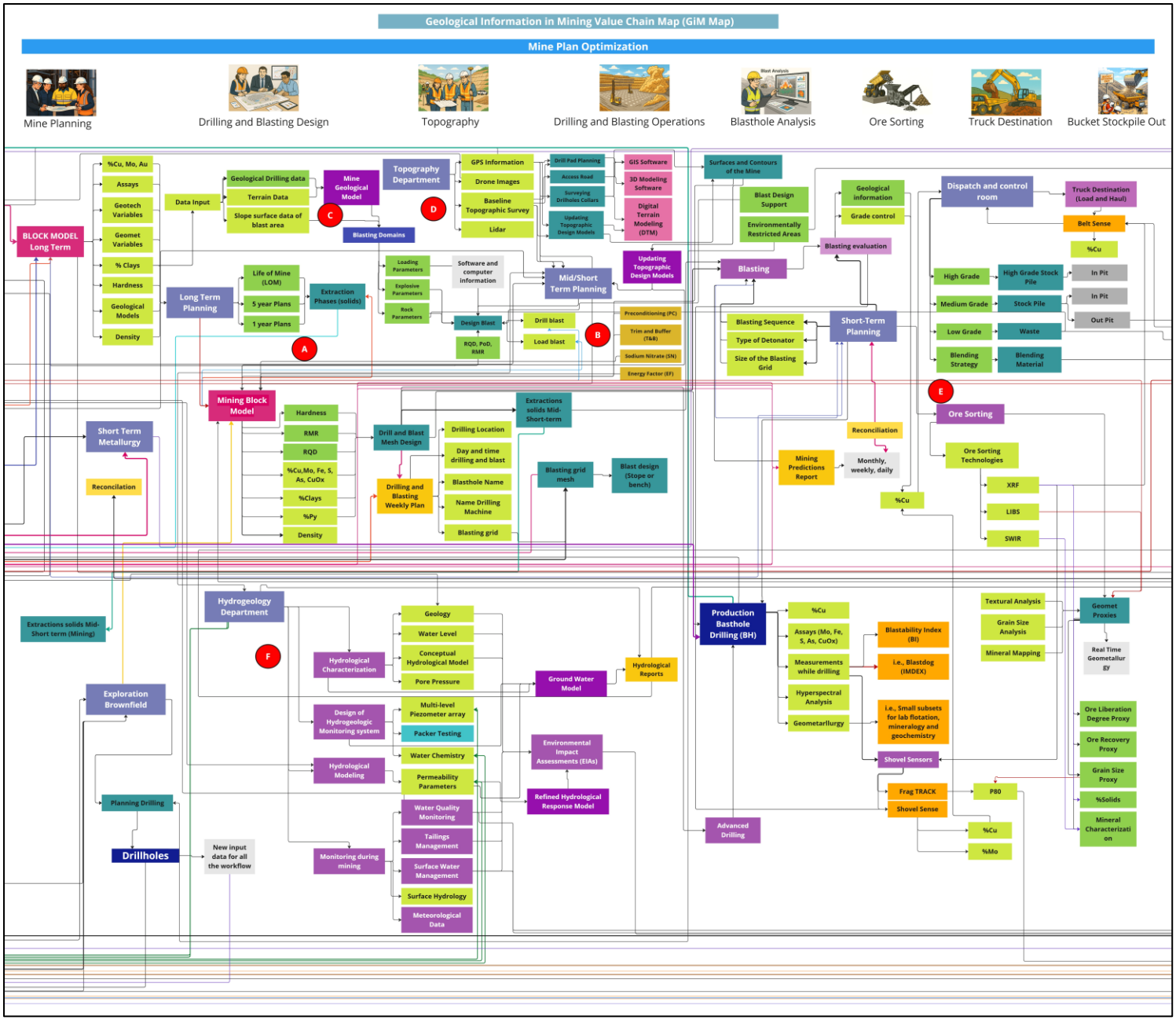

Geological Information in Mining Value Chain Map (GiM Map)
Mine Plan Optimization
Mine Planning
Drilling and Blasting Design
Topography
Drilling and Blasting Operations
Blasthole Analysis
Ore Sorting
Truck Destination
Bucket Stockpile Out
BLOCK MODEL Long Term
%Cu, Mo, Au
Assays
Geotech Variables
Geomet Variables
% Clays
Hardness
Geological Models
Density
Data Input
Geological Drilling data
Terrain Data
Slope surface data of blast area
Mine Geological Model
Blasting Domains
Topography Department
GPS Information
Drone Images
Baseline Topographic Survey
Lidar
Drill Pad Planning
Access Road
Surveying Drillholes Collars
Updating Topographic Design Models
GIS Software
3D Modeling Software
Digital Terrain Modeling (DTM)
Updating Topographic Design Models
Surfaces and Contours of the Mine
Blast Design Support
Environmentally Restricted Areas
Geological information
Grade control
Blasting evaluation
Blasting
Dispatch and control room
Truck Destination (Load and Haul)
Belt Sense
%Cu
High Grade
High Grade Stock Pile
In Pit
Medium Grade
Stock Pile
In Pit
Low Grade
Waste
Out Pit
Blending Strategy
Blending Material
Long Term Planning
Life of Mine (LOM)
5 year Plans
1 year Plans
Extraction Phases (solids)
Loading Parameters
Explosive Parameters
Rock Parameters
Software and computer information
Mid/Short Term Planning
Design Blast
Drill blast
Load blast
RQD, PoD, RMR
Preconditioning (PC)
Trim and Buffer (T&B)
Sodium Nitrate (SN)
Energy Factor (EF)
Blasting Sequence
Type of Detonator
Size of the Blasting Grid
Short-Term Planning
Mining Block Model
Hardness
RMR
RQD
%Cu,Mo, Fe, S, As, CuOx
%Clays
%Py
Density
Extractions solids Mid-Short-term
Drill and Blast Mesh Design
Drilling Location
Day and time drilling and blast
Blasthole Name
Name Drilling Machine
Blasting grid
Drilling and Blasting Weekly Plan
Blasting grid mesh
Blast design (Stope or bench)
Short Term Metallurgy
Reconciliation
Mining Predictions Report
Monthly, weekly, daily
Reconciliation
%Cu
Ore Sorting
Ore Sorting Technologies
XRF
LIBS
SWIR
Extractions solids Mid-Short term (Mining)
Hydrogeology Department
Geology
Water Level
Conceptual Hydrological Model
Pore Pressure
Hydrological Characterization
Ground Water Model
Hydrological Reports
Design of Hydrogeologic Monitoring system
Multi-level Piezometer array
Packer Testing
Water Chemistry
Permeability Parameters
Hydrological Modeling
Environmental Impact Assessments (EIAs)
Refined Hydrological Response Model
Water Quality Monitoring
Tailings Management
Surface Water Management
Surface Hydrology
Meteorological Data
Monitoring during mining
Exploration Brownfield
Planning Drilling
Drillholes
New input data for all the workflow
Production Basthole Drilling (BH)
%Cu
Assays (Mo, Fe, S, As, CuOx)
Measurements while drilling
Hyperspectral Analysis
Geometarllurgy
Blastability Index (BI)
i.e., Blastdog (IMDEX)
i.e., Small subsets for lab flotation, mineralogy and geochemistry
Shovel Sensors
Frag TRACK
Shovel Sense
P80
%Cu
%Mo
Advanced Drilling
Textural Analysis
Grain Size Analysis
Mineral Mapping
Geomet Proxies
Real Time Geometallurgy
Ore Liberation Degree Proxy
Ore Recovery Proxy
Grain Size Proxy
%Solids
Mineral Characterization
A
B
C
D
E
F

Figure 4.4 Geological Information Flow in the Mine Plan Optimization Stage of the MVC. The illustration shows the main information pathways and interconnections between planning, drilling, blasting, and operational stages. Red circles with numbers show the relative locations of the information described in the text.

Grade control and ore sorting information feeds directly to dispatch systems and truck allocation. Modern fleet management platforms integrate geological and geometallurgical information with sensor-derived data (e.g., %Cu, clay content, deleterious elements) to direct haul trucks to their destinations. These destinations correspond to high, medium, or low-grade stockpiles, blending areas, waste dumps, or direct plant feed (Wang et al., 2023). Together, all the previous stages and information form a loop among design, execution, and performance, enabling continuous improvement in blasting practices and material movement.

In some mines, operational geophysics also supports short-term mine planning. Seismic and micro seismic monitoring provide real-time insight into rock-mass response during blasting, improving fragmentation and hazard management (Mendecki, 1997). Ground-penetrating radar (GPR) and seismic refraction help detect voids, weak zones, and water-bearing structures, reducing geotechnical risks and improving digability forecasting (Tichauer et al., 2020).

Hydrogeology contributes essential information during this stage. The workflows begin with the hydrogeological characterization of the deposit, which is achieved by integrating hydrogeological sampling (Figure 4.2F) with multidisciplinary inputs: geological (e.g., Lithology, clays, alterations), geotechnical (e.g., RQD, FF, GSI), and environmental parameters (e.g., ARD or AP). This characterization feeds the conceptual hydrogeological model, which provides a comprehensive understanding of rock-mass permeability and groundwater flow within the deposit. Subsequently, this information is used to generate calculated data using hydrological software that converts conceptual models and operational testing (e.g., Packer Lugeon Test, sensor data) into

numerical groundwater models. The primary output is a pore pressure grid, which is transferred as a critical input to the geotechnical, mining planning, and mining geology teams.

Finally, hydrogeological models and water quality monitoring (Figure 4.4F) support regulatory compliance, guide pit dewatering strategies, and ensure responsible management of water reuse throughout the operation (Beale & Read, 2013). Regular updates of groundwater conditions and slope stability models enable early detection of potential instabilities, ensuring that production plans remain aligned with safety requirements and environmental standards (Franks et al., 2011). At this stage, collaboration between hydrogeological and geotechnical teams is essential, as their combined datasets provide the operational constraints required to safely manage mining activities while minimizing environmental impacts.

4.5 Metallurgical Process Optimization

The final main operational unit of the GiM Map corresponds to Metallurgical Process Optimization (Figure 4.5), where geological and geometallurgical information converge into operational information, ultimately influencing environmental management and financial outcomes.

As depicted in Figure 4.5A, the flow of information and material begins with crushing, where ore hardness, clay content, and mineralogical composition, derived from geological and geometallurgical models, determine crushing efficiency and energy consumption. Parameters such as the Drop Weight Index (DWI), Bond Work Index (BWI), and percentage of clays are used to predict comminution performance and to design crushing circuits that minimize energy use while preserving particle size distributions favorable to subsequent grinding. In different parts of the circuit, mechanical crushing begins to separate ore mineral phases from gangue (or waste) mineral phases. The crushed material then advances to the coarse ore stockpile, where blending strategies

are informed by short-term geological and geometallurgical data. Here, integrating long-term models (representing average ore characteristics from lower-density data) with short-term metallurgical models is critical to achieving consistent feed to the grinding circuit.

In the grinding stage, the influence of ore mineralogy, texture, and alteration becomes more pronounced. Variables such as clay content, sulfide mineralogy, and grain-size distribution affect grinding kinetics, pulp viscosity, and liberation degree (Figure 4.5B; Lamberg, 2011). The iterative feedback between plant data and geological input ensures that throughput and particle size targets are maintained (which ensures liberation of ore-containing minerals), reducing energy losses, and stabilizing downstream recovery processes.

Following grinding, the material proceeds to flotation (Figure 4.5C), where mineralogical and chemical variability directly influences recovery and concentrate quality. At this stage, mineralogy models derived from assay data, hyperspectral proxies, LIBS/XRF data, and mineral liberation analysis (MLA, TIMA, QEMSCAN) become essential for predicting flotation performance. Variables such as %Cu, %S, clay mineral proportions, and deleterious elements (e.g., As, Sb, Pb) control reagent consumption and froth stability, which directly affect recovery. Short-term metallurgical domains are continuously updated using laboratory test results (such as batch flotation or locked-cycle tests) and plant performance data, refining recovery and selectivity predictions. Feedback between short- and long-term models enables proactive adjustments to reagent dosing and blending strategies, reducing variability in concentrate grade.

The thickening stage (Figure 4.5D) further integrates geometallurgical and environmental information. Parameters such as particle size distribution, slurry density, and water chemistry, derived from earlier mineralogical and geochemical datasets, determine thickening efficiency and water-recovery potential. Thickening data also serves as a bridge between processing optimization

and water management planning, contributing to hydrogeological models and site-wide water balance assessments (Parbhakar-Fox & Lottermoser, 2015).

Subsequently, the concentrate is transferred to the concentrate stockpile, representing the upgraded material, typically around 20–30 wt.% Cu for primary-sulfide (chalcopyrite-rich) concentrates, and up to 40 wt.% Cu for secondary-sulfide concentrates, that is ready for transport to the smelter (International Cu Association, 2014). At this stage, metallurgical reconciliation between predicted and actual concentrate grades is performed, integrating both laboratory and online sensor data. The stockpile acts as a buffer to stabilize plant throughput and feed consistency.

Environmental teams provide essential information throughout the mine's life, and although they appear near the end of the GiM Map, their role remains critical through closure. Waste material generated during flotation and thickening is directed to tailings storage facilities, where tailings characterization is used to evaluate ARD potential, neutralization capacity, and deleterious element mobility—parameters derived from geochemical assays and mineralogical tests (Figure 4.5E). These assessments, together with waste-rock classification and monitoring programs, inform geoenvironmental management plans aimed at preventing acid generation, metal leaching, and other long-term risks (Parbhakar-Fox & Lottermoser, 2015). Information from these facilities is continuously fed back into the Environmental Department and into long-term metallurgical and environmental models, ensuring that short-term operational decisions remain aligned with sustainability and closure objectives.

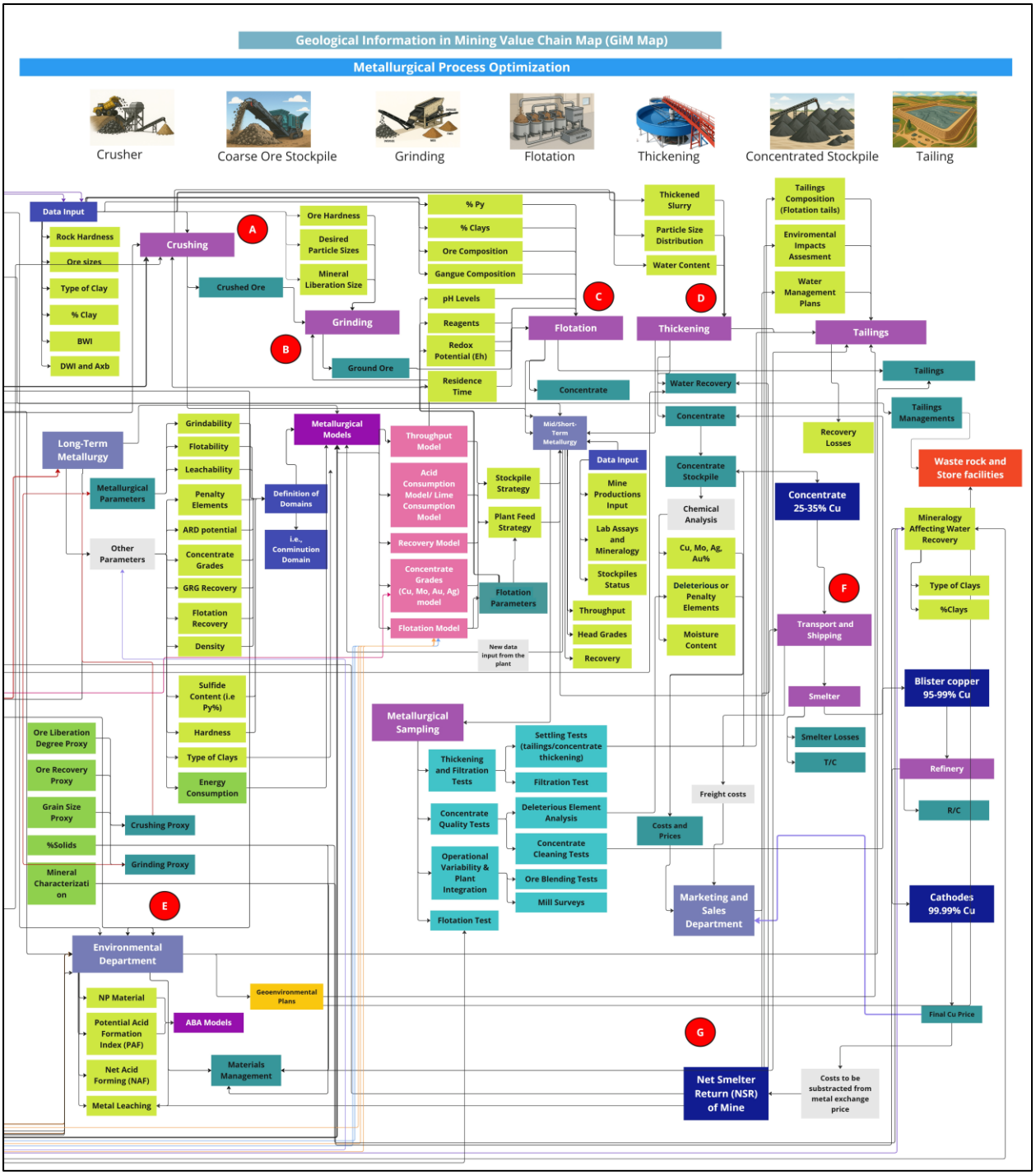


Figure 4.5 Geological information flows in the metallurgical process optimization stage of the MVC. The illustration summarized the data connections between crushing, grinding, flotation, thickening, and tailings management. The information flow highlights how geological and mineralogical inputs influence metallurgical performance and product value.

Beyond the processing plant, the flow of data continues toward transport and shipping (Figure 4.5F), where measurements of concentrate moisture, deleterious-element content, and tonnage are critical for logistics planning and commercial transactions. Analytical data are transferred to the Marketing and Sales Department, which integrates metallurgical and economic variables (such as concentrate grade, impurities, smelter deductions, and freight costs) into financial models. These models are used to determine the Net Smelter Return (NSR), the derived economic expression of geological and metallurgical information (Figure 4.5 G). Essentially, NSR quantifies the monetary return from metal sales after deducting transportation, treatment, and refining costs, directly linking the quality of geological and metallurgical data to the mine's financial performance (Emery et al., 2006).

Because this operational unit incorporates all the data and models represented in the GiM Map, the Metallurgical Process Optimization stage demonstrates the power and necessity of incorporating all the orebody knowledge across the MVC (Figure 4.5). This operational unit integrates long-term geological and geometallurgical understanding with short-term operational feedback, connecting orebody characterization, processing efficiency, and economic performance within a single data-driven structure. The inclusion of environmental and waste-management parameters ensures that metallurgical optimization aligns not only with production goals but also with sustainability and closure requirements.

4.6 Hyperspectral data: One data type with multiple uses and users

The previous sections describe the connections between various datasets within the MVC in relation to operational units and stages of the mining process. To provide further insight into the interconnections among specific datasets, this subsection traces hyperspectral data across the GiM. Among the multiple datasets analyzed in the GiM Map, hyperspectral data (specifically the visible near infrared and shortwave infrared system, VNIR–SWIR) stands out for its broad applicability. This technology stands out as one of the few that directly measure mineralogy and is especially well-suited to identifying clay minerals, which are notoriously difficult to detect visually and chemically. When appropriately calibrated and validated, VNIR–SWIR hyperspectral datasets provide mineralogical information, particularly for clays and carbonates, that are difficult to characterize consistently using visual logging alone. Figure 4.6 traces hyperspectral information from exploration through geological and mine planning workflows to metallurgical process optimization.

During exploration, point- or scan-based hyperspectral analyses of drill cores or chips during logging generate VNIR–SWIR spectra that can, in turn, be interpreted for mineralogy and then used for alteration vectoring and geological or alteration-domain definition (Hunt, 1977; Kruse, 2012; Navarra et al., 2018). When collected systematically and supported by robust quality control, these datasets become foundational for subsequent geological modeling and feed directly into Geology Process Optimization, where they contribute to the construction of long-term geological, alteration, and clay models used in resource estimation, geometallurgical domain definition, and block model development (Figure 4.6A; Journel & Kyriakidis, 2004). A key application is the generation of calculated mineralogy models derived from hyperspectral indices calibrated against mineral standards or laboratory mineralogy. These models are particularly

valuable for quantifying clay mineral assemblages, which strongly influence processing performance, pit stability, and rock strength. Because thresholds for "high clay content" (that is, concentrations of clay minerals that interfere with processing and recovery) vary across deposits, hyperspectral data enable site-specific calibration and improve the consistency of domain boundaries. These same datasets support predictive geometallurgical models, helping to distinguish soft versus hard ore zones and refine geometallurgical domains. Once integrated into block modeling workflows, clay abundance can be interpolated using geostatistical techniques such as ordinary kriging or nearest neighbor (Emery et al., 2006). The resulting outputs (expressed as the percentage of clays per block) are essential for both long-term planning and short-term operational models (Figure 4.6B).

In Mine Plan Optimization, hyperspectral data informs blast design, mine scheduling, and hydrological modeling. Clay distribution maps and alteration intensity indices enable engineers to delineate blasting domains, optimize explosive loading, and minimize fines generation, oversize material, and unintended dilution (Kanchibotla, 2003). It is important to recognize that the classification of material as "soft" or "hard" differs between geological and engineering contexts. From a geological perspective, these terms often relate to alteration intensity, mineralogical composition, or weathering grade, whereas in engineering practice, they are expressed in terms of fragmentation behavior, compressive strength, or diggability indices. Hyperspectral-derived proxies can therefore be translated into quantitative parameters (e.g., percentage of "soft" material within a block or domain) to align geological interpretation with operational requirements. This translation facilitates communication between geologists and mining engineers, improving blast design accuracy and ensuring that domain boundaries defined in geological models are operationally meaningful during extraction and material handling.

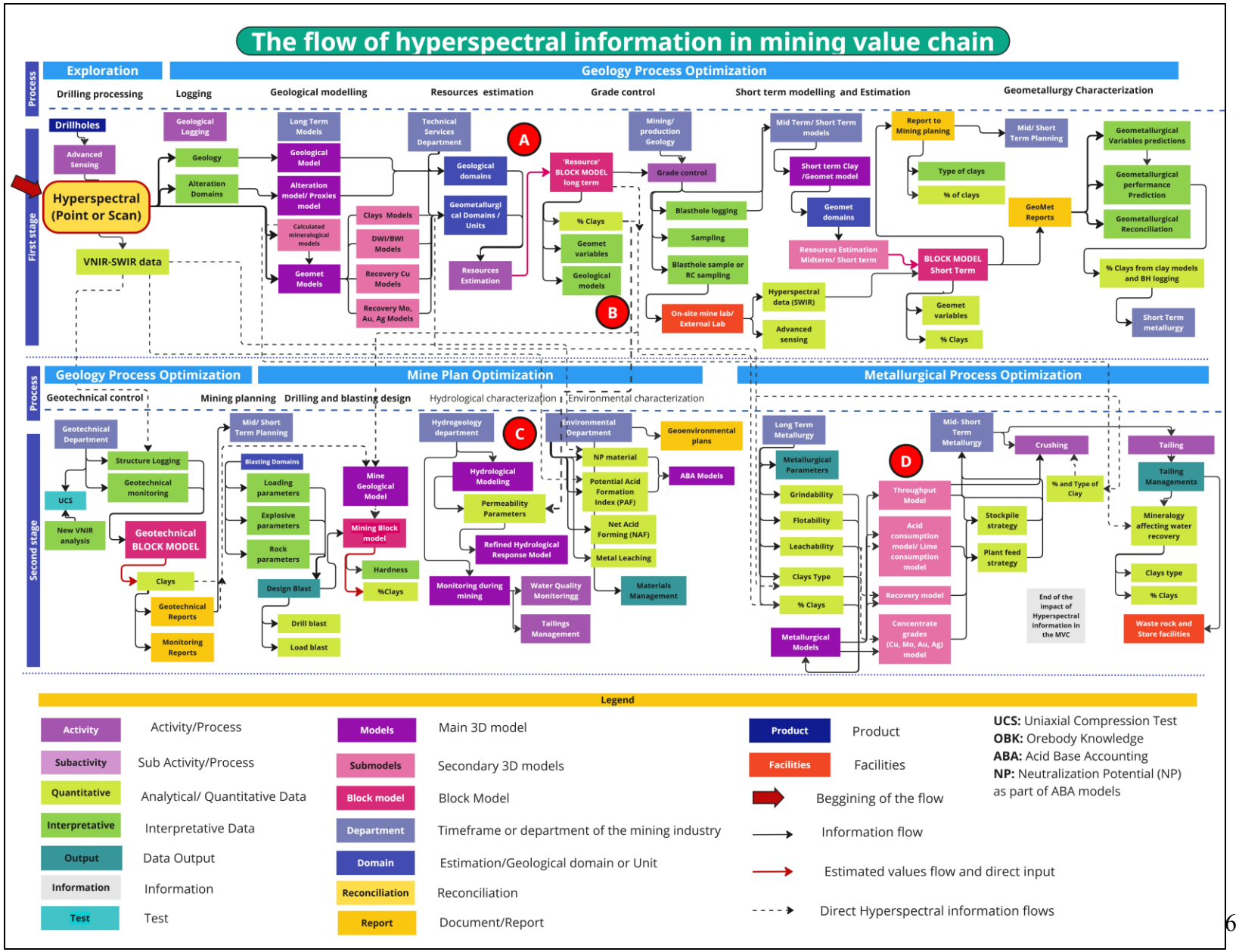

The flow of hyperspectral information in mining value chain
Process
First stage
Second stage
Exploration
Geology Process Optimization
Drilling processing
Logging
Geological modelling
Resources estimation
Grade control
Short term modelling and Estimation
Geometallurgy Characterization
Drillholes
Advanced Sensing
Hyperspectral (Point or Scan)
VNIR-SWIR data
Geological Logging
Geology
Alteration Domains
Long Term Models
Geological Model
Alteration model/ Proxies model
Calculated mineralogical models
Geomet Models
Clays Models
DWI/BWI Models
Recovery Cu Models
Recovery Mo, Au, Ag Models
Technical Services Department
Geological domains
Geometallurgical Domains / Units
Resources Estimation
A
'Resource' BLOCK MODEL long term
% Clays
Geomet variables
Geological models
B
Mining/ production Geology
Grade control
Blasthole logging
Sampling
Blasthole sample or RC sampling
On-site mine lab/ External Lab
Mid Term/ Short Term models
Short term Clay /Geomet model
Geomet domains
Resources Estimation Midterm/ Short term
Hyperspectral data (SWIR)
Advanced sensing
Report to Mining planing
Type of clays
% of clays
GeoMet Reports
BLOCK MODEL Short Term
Geomet variables
% Clays
Mid/ Short Term Planning
Geometallurgical Variables predictions
Geometallurgical performance Prediction
Geometallurgical Reconciliation
% Clays from clay models and BH logging
Short Term metallurgy
Geology Process Optimization
Mine Plan Optimization
Metallurgical Process Optimization
Geotechnical control
Mining planning
Drilling and blasting design
Hydrological characterization
Environmental characterization
Geotechnical Department
Structure Logging
Geotechnical monitoring
UCS
New VNIR analysis
Geotechnical BLOCK MODEL
Clays
Geotechnical Reports
Monitoring Reports
Mid/ Short Term Planning
Blasting Domains
Loading parameters
Explosive parameters
Rock parameters
Design Blast
Drill blast
Load blast
Mine Geological Model
Mining Block model
Hardness
%Clays
Hydrogeology department
C
Hydrological Modeling
Permeability Parameters
Refined Hydrological Response Model
Monitoring during mining
Water Quality Monitoringg
Tailings Management
Environmental Department
NP material
Potential Acid Formation Index (PAF)
Net Acid Forming (NAF)
Metal Leaching
Materials Management
Geoenvironmental plans
ABA Models
Long Term Metallurgy
Metallurgical Parameters
Grindability
Flotability
Leachability
Clays Type
% Clays
Metallurgical Models
D
Throughput Model
Acid consumption model/ Lime consumption model
Recovery model
Concentrate grades (Cu, Mo, Au, Ag) model
Mid- Short Term Metallurgy
Stockpile strategy
Plant feed strategy
Crushing
% and Type of Clay
End of the impact of Hyperspectral information in the MVC
Tailing
Tailing Managements
Mineralogy affecting water recovery
Clays type
% Clays
Waste rock and Store facilities
Legend
Activity
Activity/Process
Subactivity
Sub Activity/Process
Quantitative
Analytical/ Quantitative Data
Interpretative
Interpretative Data
Output
Data Output
Information
Information
Test
Test
Models
Main 3D model
Submodels
Secondary 3D models
Block model
Block Model
Department
Timeframe or department of the mining industry
Domain
Estimation/Geological domain or Unit
Reconciliation
Reconciliation
Report
Document/Report
Product
Product
Facilities
Facilities
Beggining of the flow
Information flow
Estimated values flow and direct input
Direct Hyperspectral information flows
UCS: Uniaxial Compression Test
OBK: Orebody Knowledge
ABA: Acid Base Accounting
NP: Neutralization Potential (NP) as part of ABA models

Figure 4.6 The flow of hyperspectral information across the MVC. Representation of hyperspectral data integration within the GiM Map, illustrating how VNIR–SWIR information collected during exploration and drilling propagates through geological, geometallurgical, and metallurgical models to support process optimization and value prediction across the MVC.

Hyperspectral-derived mineralogy is also crucial in informing hydrogeological and environmental characterization (Figure 4.6C). Clay and alteration mineral distributions, identified by hyperspectral scanning, can indicate relative permeability contrasts and fluid pathways, particularly in fractured and altered rock masses, supporting hydrogeological conceptual models (Beale & Read, 2013; Kamath et al., 2024; Lehmann et al., 2021; Thiele et al., 2025). For environmental teams, hyperspectral data are used to identify minerals that influence acid-neutralization processes, such as carbonates (e.g., calcite, dolomite), and to develop neutralization potential (NP) models. Hyperspectral data can be integrated with sulfide-mineralogy proxies (e.g., assays or downhole XRF or LIBS) to strengthen acid-based accounting (ABA) models and enhance predictive accuracy (Jackson et al., 2018). These integrated datasets can also guide the design of waste rock dumps, stockpiles, and tailings facilities, reducing long-term environmental liabilities (Parbhakar-Fox & Lottermoser, 2015).

In the final stage, Metallurgical Process Optimization, clay minerals influence flotation performance, grindability, and water recovery efficiency (Lishchuk et al., 2015; Lotter, 2011). By integrating hyperspectral datasets and derived mineralogical proxies into plant feed strategies, stockpile management, and mid- to short-term metallurgical models, operators can anticipate mineralogical variability and proactively adjust processing parameters (Figure 4.6D). Adjustment strategies can include anticipating feed blending and ore routing, enabling metallurgists to match

ore characteristics to plant conditions (thereby improving overall Net Smelter Return (NSR)) and reducing reagent consumption.

As observed in Figure 4.6, following the path of hyperspectral information reveals how a single dataset interconnects multiple stakeholders, geologists, geotechnical engineers, metallurgists, and environmental specialists across the MVC. When collected systematically and calibrated against laboratory standards, hyperspectral data provide moderate to high certainty, offering a high-resolution, accurate, and continuous information layer that bridges early exploration with downstream processing. This layered visualization highlights how an early-stage dataset can generate cascading value throughout the mine lifecycle, linking geological understanding with plant performance.

Chapter 5: Assessing data interaction and model quality across the MVC

5.1 Data Classification Results and Distribution Across the MVC

Up to this point, the research has been primarily qualitative, using the GiM Map to describe information flows and identify key data connections. To quantify how information is transformed across the MVC, all nodes were classified (Section 3.5) and assigned certainty weights using the proxy described in Section 3.5.3. Figure 5.1 summarizes the distribution of all the empirical, interpretive, and calculated/derived information by operational unit. This illustrates how the balance of measurement, interpretation, and model-derived outputs changes across the value chain. Figure 5.1 summarizes the data classification statistics by operational units and provides domain-specific breakdowns. The figure shows the relative proportions of empirical, interpretive, and calculated data highlighting distinct stages of data generation across the MVC. Each stage exhibits a characteristic balance reflecting its operational objectives and degree of model integration.

5.1.1 Exploration and Development (63% empirical, 35% interpretive, 2% calculated)

This stage is dominated by direct measurements (including geochemical assays, geophysical surveys, and field logging), making it the primary source of empirical information. Interpretive data (such as geological and lithological logging) complement these observations by delineating lithological and alteration domains. The limited proportion of calculated data indicates minimal transformation of observations and measurements into predictive models. This suggests an opportunity to enhance early-stage forecasting by using more geochemical proxies for targeting or combining them with spectral parameters to obtain calculated mineralogy, to support geological logging.

5.1.2 Geology Process Optimization (20% empirical, 38% interpretive, 42% calculated)

In this stage, a transformation of the data occurs. Around 40% of information is processed and transformed into calculated data. Empirical inputs from drilling and laboratory analyses are transformed into models and proxies (e.g., grade shells, alteration domains) through geostatistical and analytical processes. The substantial interpretive component (such as lithological domain definition and structural classification), driven by subject matter experts, introduces inherent subjectivity that propagates uncertainty into subsequent modeling and decision-making steps.

5.1.3 Mine Plan Optimization (45% empirical, 39% interpretive, 16% calculated)

Data diversity is highest in this domain, where empirical inputs from grade control drilling, blast hole analyses, and sensor monitoring are continually reconciled with interpretive and derived information used in blast design, short-term modeling, and ore routing. The prevalence of interpretive data reflects ongoing geological re-evaluation at operational scales, while fewer calculated data suggest greater potential for integrating predictive and optimization models.

5.1.4 Metallurgical Process Optimization (59% empirical, 16% interpretive, 25% calculated):

Empirical and calculated data predominate in this phase, underscoring the highly quantitative nature of metallurgical operations. Most inputs derive from plant monitoring, testwork, and sensor systems, which are subsequently incorporated into models for throughput, recovery, and acid consumption. The limited interpretive content reflects reduced geological influence on processing decisions, which may be a limitation when mineralogical variability significantly affects metallurgical performance.

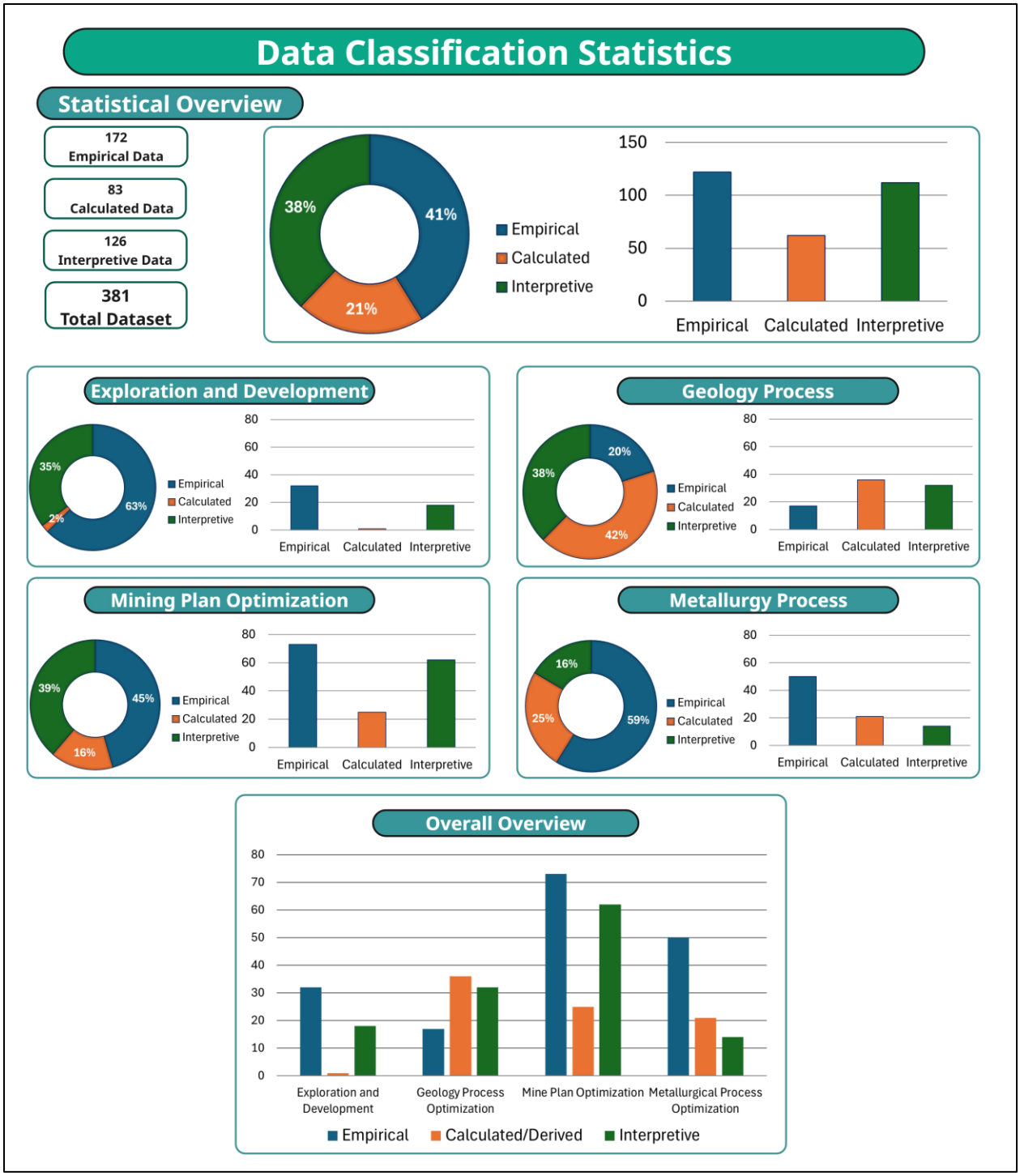


Figure 5.1 Data classification statistics across the MVC. Visualization of the distribution of empirical, calculated/derived, and interpretive data within the GiM Map.

Collectively, these patterns demonstrate a progressive evolution from observation-based practices in early exploration toward model-based integration in later operational and metallurgical stages. The shifting balance between empirical, interpretive, and calculated datasets reflects both the increasing maturity of information management systems and the growing interdisciplinarity of decision-making processes. Understanding these proportions is therefore essential for evaluating data robustness, informing model-weighting strategies, and guiding efforts to improve data standardization and interoperability across the MVC.

5.2 Test-based Data Classification and Distribution Across the MVC

The classification of test-related data inputs across the MVC was conducted following the structure described in Section 3.5.3. A complete list of tests and their assigned categories is summarized in Table 5.1 with the overall statistical distribution presented in Figure 5.2.

Table 5.1 Classification of test-based data

Type of Test	Classification
Density Test	Empirical
Point Load Test (PLT)	Empirical
Flotation Test	Empirical
Geochemical Analysis	Empirical
Uniaxial Compression Test (UCS)	Empirical
Comminution Test	Empirical
Flotation Kinetics	Empirical
Mineralogy Test	Empirical
Permeability Test	Field-Controlled Test
Water Quality Test	Field-Controlled Test
Groundwater Test	Empirical
Hydraulic Conductivity Test	Empirical
X-Ray Fluorescence (XRF)	Empirical
Drop Weight Index (DWI)	Highly Robust Empirical
SAG Mill Comminution (SMC) Test	Empirical
Bond Ball Mill Work Index (BWI)	Highly Robust Empirical

Abrasion Index (Ai) Test	Highly Robust Empirical
Batch Flotation Test	Empirical
Leach Column Test (LCT)	Semi-Quantitative / Multi-Step
Pilot Plant Test	Empirical
Cleaning and Regrinding Test	Empirical
Open Circuit Test	Empirical
Batch Kinetic Flotation Test	Semi-Quantitative / Multi-Step
Rock Quality Designation (RQD)	Field-Controlled Test
Geological Strength Index (GSI)	Interpretive / Mixed
Packer (Lugeon) Test	Field-Controlled Test
Water Chemistry	Highly Robust Empirical
Blast-holes Analysis	Empirical
Impact Brokenness (Axb)	Empirical
Mineral Liberation Size	Semi-Quantitative / Multi-Step
Thickening Tests	Empirical
Settling Tests (tailings/concentrate thickening)	Empirical
Filtration Tests	Empirical
Concentrate Quality Tests	Empirical
Deleterious Element Analysis	Highly Robust Empirical
Concentrate Cleaning Tests	Empirical
Operational Variability & Plant Integration	Field-Controlled Test
Ore Blending Tests	Field-Controlled Test
Mill Surveys (plant sampling, metallurgical balance)	Field-Controlled Test
Flotation Survey	Field-Controlled Test
Dewatering/Pumping Test	Field-Controlled Test
Airlift Testing	Field-Controlled Test
Vibrating Wire Piezometers (VWPs)	Field-Controlled Test
ABA (Acid-Base Accounting) Test	Empirical
Net Acid Generator (NAG) test	Empirical
Metal Leaching Test (dynamic)	Semi-Quantitative / Multi-Step
Kinetic Leaching Test (HCT/Column)	Semi-Quantitative / Multi-Step
Vibrating Wire Piezometers (VWPs)	Field-Controlled Test
Metal Leaching Test (Static)	Highly Robust Empirical

A total of 49 distinct test types were analyzed, encompassing geological, geotechnical, metallurgical, and environmental measurements. These samples can be collected more than once in different parts of the MVC. Instead, the Packer Tests are collected during exploration and also are routinely collected during Mining planning optimization stage as part of hydrogeological monitoring. The test types listed in Table 5.1 are the most common in the porphyry Cu mine.

As illustrated in Figure 5.2, the dataset is dominated by Empirical Tests (47%), followed by Highly Robust Empirical Tests (22%), Interpretive or Mixed Tests (11%), Semi-Quantitative or Multi-Step Tests (9%), and Field-Controlled Tests (11%). This distribution reflects a methodological structure across the MVC (as captured in the GiM Map) that is largely grounded in standardized, directly measurable procedures, complemented by a smaller proportion of interpretive and field-dependent methods.

Across the mining value chain, data generation is dominated by empirical and highly robust empirical tests, which account for approximately 69% of the datasets used. These include geochemical assays, mineralogical analyses, flotation and density measurements, as well as mechanical indices such as UCS, PLT, DWI, BWI, and the Abrasion Index (Ai). Their high repeatability and strong procedural standardization make them foundational for model calibration, comminution modeling, and integration across technical domains.

A smaller proportion of the data, around 9%, is derived from semi-quantitative and multi-step tests, including liberation size determinations and Locked Cycle Tests. These methods combine empirical measurements with iterative interpretation and provide valuable proxies for mineral liberation and circuit optimization. However, their multi-stage structure introduces additional variability that must be explicitly considered when incorporating results into predictive models.

Field-controlled tests, representing approximately 11% of the datasets, include permeability measurements, packer tests, and groundwater assessments. Because these measurements are strongly influenced by site-specific and temporal conditions, they are associated with higher uncertainty. Despite this, they remain essential for hydrogeological characterization and for informing mine design and environmental management decisions.

The remaining 11% of the data consists of interpretive or mixed tests, such as Rock Quality Designation (RQD) and the Geological Strength Index (GSI). These measures rely heavily on expert judgment during field logging and classification. While they are critical inputs for geotechnical design, their subjective component limits comparability across deposits and highlights the need for calibration and contextual interpretation.

Taken together, this distribution underscores the mining sector's strong reliance on standardized, quantitative datasets, while also emphasizing the continued importance of expert knowledge and uncertainty management where field-controlled and interpretive methods remain unavoidable.

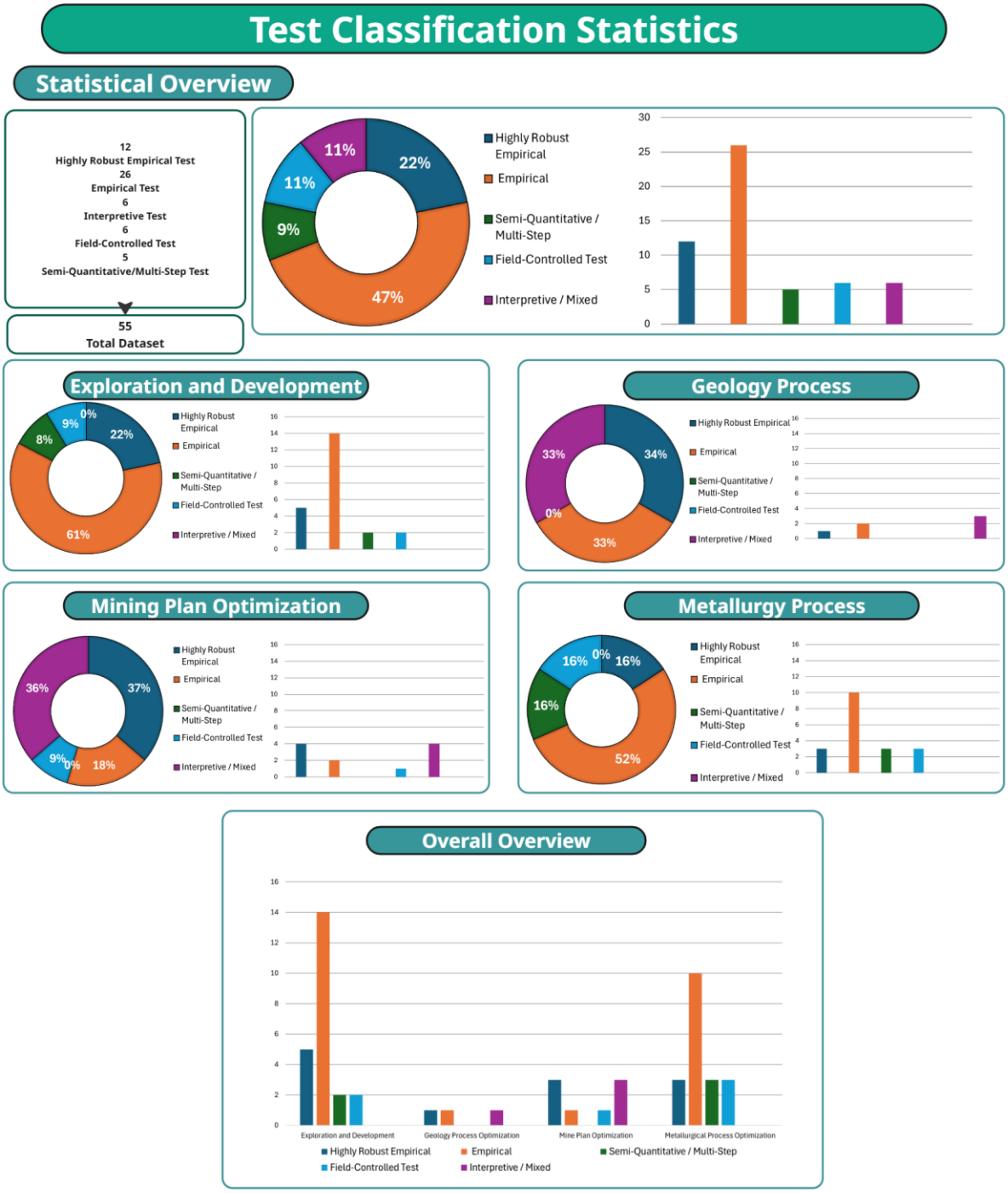


Figure 5.2 Test-based data classification statistics across the MVC. Visualization of data derived from tests and their distribution within the GiM Map, showing the relative proportions of highly robust, empirical, semi-quantitative, field-controlled, and interpretive/mixed tests.

5.3 Model Weighting and Ranking across the Mining Value Chain

Following the data- and test-based classification approach described in Section 3.5.3, the models were ranked and classified based on their final scores. A summary of the results is presented in Table 5.2, and more details are in Appendix C. The ranking includes:

Models with high input support scores (≥136) included Recovery (231 points), Throughput (227 points), and Resource Block (long-term, 198 points) (Table 5.2). These three models demonstrate the strongest integration of empirical and test-based data and combine laboratory assays, mechanical indices (e.g., BWI, DWI, Ai), and real-time plant feedback, creating closed calibration loops between prediction and measurement. Their high connectivity, as measured by proxy and data density, makes them reliable models for forecasting operational and economic performance.

Moderate input support models (36–136) include the Geotechnical Block Model (126 points), and Geomet Model (105 points) (Table 5.2). These models exhibit strong empirical foundations but have fewer direct testing datasets to support them. They partially rely on indirect predictors, such as alteration indices or hyperspectral scalars, which introduce interpretive bias. Although they serve as key intermediaries between the geological and metallurgical domains, their reliability depends on how often they are validated and calibrated.

Models with low input support (≤36) include Lithological (32 points), Alteration (32 points), and RQD (24 points), which are predominantly interpretive and field-based, with limited laboratory verification. Their qualitative nature reduces confidence in quantitative measures, although they remain fundamental for defining spatial and structural context. The Density Model ranked with the lowest confidence of all models evaluated, with only 15 points.

Overall, the weighting results indicate that, according to the metric, proxy scores tend to increase toward the downstream stages of the MVC, where more diverse empirical and test-based inputs are present. In contrast, upstream geological models, although rich in interpretive content, receive lower scores in the framework because they rely more heavily on subjective interpretation and indirect validation, and incorporate fewer discrete data inputs.

This pattern illustrates the progressive increase in data standardization and integration, as shown in the GiM Map (Figure 4.1), and how the metric captures input density and data types.

Table 5.2 Model weighting results based on cumulative input scores. Summary of robustness-proxy scores derived from the GiM Map weighting process, ranking models according to their total data connections, test-based inputs, and resulting ranking categories.

Rank	Model_Label	Total Connect	OUT	IN	Data class	Test class	Test Score	Data Total Score	Final Score	Proxy Category
1	Recovery Models	83	19	64	57	7	28	203	231	High input support
2	Throughput Model	93	23	70	60	8	29	198	227	High input support
3	Resources Block Model Long Term	97	34	63	53	9	33	165	198	High input support
4	Dynamic Block Model	80	19	61	52	8	29	158	187	High input support
5	Mining Block Model	81	23	58	55	2	2	177	179	High input support
6	Flotation Model	61	18	43	36	7	26	150	176	High input support
7	ABA models	51	13	38	33	5	19	117	136	High input support
8	Block Model Short Term	69	21	48	47	1	4	131	135	Moderate input Support
9	Geotechnical Block Model	78	26	52	46	6	12	114	126	Moderate input Support
10	Geomet Models	46	14	32	23	9	36	69	105	Moderate input Support
11	Hydrogeological Modeling	37	9	28	25	3	9	85	94	Moderate input Support
12	Mine Geological Model	44	15	29	28	1	1	90	91	Moderate input Support
13	Clays Model	67	49	18	16	2	8	68	76	Moderate input Support
14	NP Model	32	11	21	18	3	12	64	76	Moderate input Support
15	Geotechnical Models	51	24	27	23	4	10	57	67	Moderate input Support
16	Calculated Mineralogical Models	68	55	13	12	1	4	50	54	Moderate input Support
17	Ore/Minezone Model	45	26	19	18	1	4	50	54	Moderate input Support
18	Proxy Models	38	26	12	11	1	4	47	51	Moderate input Support
19	Structural Model	54	42	12	12	0	0	36	36	Moderate input Support
20	DWI/BWI Models	29	18	11	6	5	20	16	36	Moderate input Support
21	Lithological Model	47	36	11	10	1	4	28	32	Low input support
22	Alteration/Proxies Model	56	45	11	10	1	4	28	32	Low input support
23	RQD Model	41	31	10	8	2	2	22	24	Low input support
24	PLT Model	27	19	8	7	1	4	19	23	Low input support
25	FF Model	25	19	6	6	0	0	16	16	Low input support
26	Density Model	18	12	6	5	1	4	11	15	Low input support

5.4 Case Study: Economic Impact Calculations for Density Uncertainty

Despite its fundamental importance to tonnage estimation (Downing & Bakker, 2025; Parrish, 1993), the density model ranks lowest in cumulative input score (15 points) presented in the previous section. In practice, density data are often collected opportunistically rather than systematically, typically from competent, least altered, and easily accessible core segments. This practice introduces a systematic bias toward higher-density samples (Abzalov, 2016; Lipton, 2001). While companies collect tens of thousands of assay samples, density measurements, typically obtained through laboratory tests such as Archimedes' water-displacement or wax-coated immersion methods, may number only dozens for the same volume of material. This disparity persists despite density influencing tonnes per modeled volume, while grade controls metal per tonne. Both enter multiplicatively into contained metal estimates (tonnage × grade = contained metal, see example calculations in Downing & Bakker, 2025; Lipton, 2000). Even within feasibility studies conducted under regulatory frameworks such as NI 43-101 (Canada) and the JORC Code (Australia), grade data are typically prioritized over density measurements, and uncertainty analyses frequently overlook the impact of density variance on resource estimation and project valuation (Lipton, 2001).

To have an overview of this context, this section presents calculations using static and dynamic sensitivity analyses that demonstrate how density measurement errors propagate from resource estimation to economic impacts. All calculations use the base-case scenario, with transparent assumptions derived from published industry data and technical literature. The objective is to demonstrate that small systematic density errors can produce material first-order impacts on tonnage and revenue projections at the stated scale, motivating more rigorous density characterization as an information investment.

5.4.1 Error Mechanism

Systematic sampling bias commonly occurs in density measurement programs because samples are preferentially collected from competent, intact core intervals that represent the highest-density material. This creates overestimation because:

1. Altered zones are undersampled: Phyllic and argillic alteration reduces rock density through clay mineral formation and void space creation, yet these intervals often exhibit poor core recovery and are systematically avoided in density sampling.

2. Fractured intervals are undersampled: Structural zones with reduced density due to fracturing and brecciation are difficult to sample using standard geotechnical methods.

3. Weathered horizons are undersampled: Oxidized and weathered near-surface materials exhibit significantly reduced density but are often excluded from sampling protocols.

4. Research demonstrates that this preferential sampling creates systematic bias of 2-5% toward higher densities in porphyry systems (Abzalov, 2016; Dominy et al., 2018).

5.4.2 Base Case Scenario Parameters: Orebody Characteristics

For this illustrative analysis, a representative porphyry Cu deposit was defined based on a set of physical, metallurgical, and economic parameters. The parameters selected for this study represent a standard, mid-sized porphyry Cu operation, calibrated against global and regional benchmarks. These parameters are described below.

5.4.2.1 Parameter Justification and Sources

Orebody Volume and Tonnage: The model utilizes a volume of 37.3 Mm^3. When combined with a bulk density of 2.68 t/m^3, this results in a 100 Mt tonnage, a scale representative of numerous porphyry deposits globally (Singer et al., 2008; Sillitoe, 2010).

Cu Grade: A grade of 0.50% Cu was selected as the median value for porphyry systems. This falls within the global range of 0.30% to 1.20% and aligns with recent discoveries (0.40%–0.60%) and the average for Chilean operations (0.48% Cu) (Abzalov, 2013; Comisión Chilena del Cobre (COCHILCO), 2025).

Bulk Density: The value of 2.68 t/m³ is typical for competent, fresh intermediate-to-felsic intrusive rocks. This reflects standard measurements for unaltered core samples within the published industry range of 2.50–2.85 t/m³ (Abzalov, 2016).

Metallurgical Recovery: A recovery rate of 85% was applied, representing a conservative mid-range value for primary sulfide flotation circuits. This aligns with industry practices for similar deposits, which generally demonstrate rates between 80% and 92% (Zanin et al., 2009; Comisión Chilena del Cobre (COCHILCO), 2025).

Cu Price: The long-term planning price of US$8,000/t reflects a conservative mid-cycle assumption. This value is consistent with the 10-year London Metal Exchange (LME) average (US$7,200–$8,500/t) and typical feasibility study trailing averages (Arseneau, 2014; McCarthy et al., 2020).

5.4.3 Static Sensitivity Analysis

The following section presents a static sensitivity analysis to quantify the impact of discrete fluctuations in bulk density on the project's overall economic valuation. This evaluation employs fixed error margins of 3% and 5% for density values, assuming a uniform, constant deviation across the characterized rock mass. By applying these deterministic scenarios, the analysis isolates the project's financial sensitivity to specific sampling inaccuracies, establishing a necessary baseline before implementing more complex stochastic simulations.

5.4.3.1 Scenario 1: 3% Density Error

As a first case scenario, a systematic density overestimation of 3% is evaluated to illustrate how relatively small errors in bulk density propagate through mineral resource estimation. The parameters used: Assumed density: 2.68 t/m³; Actual density: 2.560 t/m³, resulting in a 2.99% (~3%) systematic overestimation. Because volume, grade, recovery, and price are treated as constant, only density is considered a variable. This ~3% bias propagates linearly through tonnage, contained metal, recoverable metal, and gross revenue. This resulted in a US$102M overestimation of gross revenue, representing the difference between anticipated and actual revenue generation capacity. This difference in gross revenue can affect project valuation, financing arrangements, and investment decisions. Calculations are presented in Appendix D.

The key insight from this calculation sequence is that density error propagates linearly through all downstream estimates. These calculations are summarized in Table 5.3.

Table 5.3 3% Density Error Scenario and propagation through Mining Value Chain

Parameter	Assumed	Actual	Error	% Error
Bulk density	2.68 t/m³	2.60 t/m³	+0.08 t/m³	+3.0%
Tonnage	100.0 Mt	97.0 Mt	+3.0 Mt	+3.0%
Contained Cu	500,000 t	485,000 t	+15,000 t	+3.0%
Recoverable Cu	425,000 t	412,250 t	+12,750 t	+3.0%
Gross revenue	US$3,400M	US$3,298M	**+US$102M**	+3.0%

5.4.3.2 Scenario 2: 5% Density Error

The second scenario evaluates the impact of a moderate increase in density bias, consistent with the upper end of the systematic error range commonly reported in the literature for porphyry systems. The parameters used: Assumed density: 2.68 t/m³, Actual density: 2.546 t/m³ (5% lower) and density error: 0.134 t/m³. This scenario demonstrates that a 5% systematic overestimation of

density results in an approximate US$170 million overstatement of gross revenue for the evaluated deposit scale. Calculations are presented in Appendix D.

Table 5.4 Density Error scenario 5% and propagation through the Mining Value Chain

	Estimated	Actual	Error	% Error
Bulk density	2.68 t/m³	2.546 t/m³	+0.134 t/m³	+5.0%
Tonnage	100.0 Mt	95.0 Mt	+5.0 Mt	+5.0%
Contained Cu	500,000 t	475,000 t	+25,000 t	+5.0%
Recoverable Cu	425,000 t	403,750 t	+21,250 t	+5.0%
Gross revenue	US$3,400M	US$3,230M	**+US$170M**	+5.0%

These scenarios are not intended to represent a final financial assessment, but rather to provide a high-level view of how sampling errors can potentially impact project economics. These calculations provide a static sensitivity analysis that allows the impact of data precision to be evaluated as a baseline before introducing more complex, stochastic simulations.

5.4.4 Dynamic Uncertainty Analysis and Monte Carlo Simulation

While the previous error-propagation analysis provided a useful baseline for static deviations, these fixed values do not fully capture the inherent variability of density in a porphyry Cu system. To bridge this gap between these deterministic error margins and the inherent variability of an orebody, the following section introduces a dynamic uncertainty analysis. By employing Monte Carlo simulations, the same parameters utilized in section 5.4.1 for the static evaluation are expanded to explore a range of probabilistic outcomes. This approach allows for a more comprehensive understanding of how uncertainty propagates when values are either underestimated or overestimated, simulating a realistic range of operational fluctuations.

5.4.4.1.1 Monte Carlo Simulation Results

To transition from static benchmarks to a probabilistic risk assessment, a Monte Carlo simulation was executed using 10,000 iterations (Appendix E). In this simulation, a 1% measurement-error boundary was used to account for variability in the density sampling.

Figure 5.3 shows a Monte Carlo simulation where bulk density uncertainty (±1%) is propagated into projected revenue. The normal distribution of blue areas represents the probability of revenue outcomes around the baseline estimate. Red-shaded areas indicate downside risk from density overestimation, while green-shaded areas show potential upside from underestimation. The vertical dashed lines mark fixed density scenarios.

Consistent with the static analysis, the baseline revenue is US$3,399M at a reference bulk density of 2.68 t/m³ (black line in Figure 5.3). A 3% density overestimation (Scenario 1, orange line in Figure 5.3) results in an expected gross revenue of approximately US$3,300M, representing a reduction of about US$100M relative to the baseline. A larger overestimation of 5% (Scenario 2, red line) further reduces projected gross revenue to US$3,230M, corresponding to a decrease of approximately US$170M.

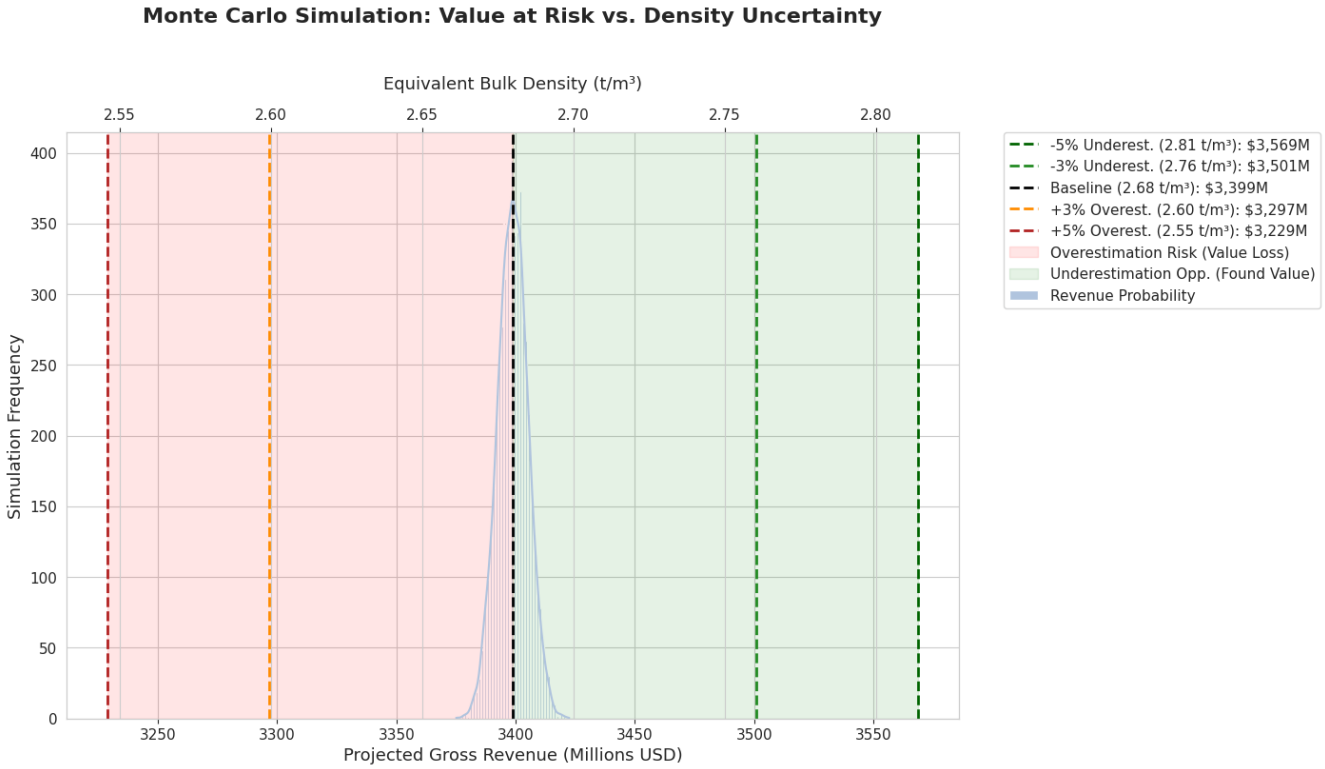


Figure 5.3 Probabilistic impact of bulk density uncertainty on projected revenue derived from a Monte Carlo simulation.

However, this type of analysis also allows the impact of density underestimation to be evaluated, rather than limiting the assessment to a single fixed error. For example, a 3% underestimation of bulk density (green fine line) resulted in a projected gross revenue of approximately US$3,500M, representing an increase of about US$100M relative to the baseline.

While this dynamic analysis specifically isolates density as a single stochastic variable, the framework is designed to incorporate additional complexities in future iterations. Because bulk density is a fundamental driver of gross revenue, expanding this Monte Carlo approach to include variables such as Cu price volatility and metallurgical recovery would provide a more holistic view of project risk. By analyzing these multiple layers simultaneously, it can better quantify the combined impact of geological sampling errors, market fluctuations, and operational recoveries.

In this context, the GiM Map plays a critical role prior to formal risk evaluation by identifying the key nodes where uncertainty and potential errors are most likely to originate and propagate across the MVC. Once these primary nodes and their connecting edges are identified, probabilistic modelling can be applied to assign uncertainty distributions to critical inputs and simulate how variability propagates through interconnected models. This would allow quantification of uncertainty amplification, identification of the most sensitive decision points, and estimation of potential economic impacts, thereby improving our understanding of where additional information would generate the highest VoI.

Chapter 6: Discussions

6.1 Analysis of Data Classification and Model Weighting

When viewed together, the results presented in Sections 5.1, 5.2, and 5.3 point to a consistent pattern in how geological information is generated, transformed, and ultimately becomes operationally relevant across the MVC. Rather than serving as independent findings, the observed distributions of data types, test classifications, total connections, and model input weightings collectively describe a connected progression from initial data generation through validation to operational use.

A critical finding of this integrated analysis is that a high total connection volume (Total Connect in Table 5.2) does not necessarily correspond to a high cumulative input score. This discrepancy is driven by the flow of calculated/derived data rather than by a simple count of linkages or weighting. Some calculated or derived outputs, once transferred to another department or model, undergo a second stage of transformation as additional derived data are generated. Following this second calculation step, the newly derived outputs can be reintegrated into the original source model. This process creates iterative information loops, in which data are repeatedly transformed and reintroduced rather than remaining linear. For instance, the Resources Block Model (score 198, Total Connect 97), as the central repository for geological, geotechnical, geometallurgical, and geoenvironmental variables, amplifies connectivity through iterative 'information loops' rather than solely through high-quality empirical inputs. These loops occur when a primary model, for example, the Throughput Model (score 227, Total Connect 83), receives geological or geotechnical domains from the block model, and includes High Empirical data (e.g., BWI test) to perform complex calculations, only to feed the resulting information back into the Resources Block Model to populate individual blocks.

Consequently, the block model functions simultaneously as both a feeder and a recipient of information, creating a recursive data flow that increases connectivity without necessarily increasing the model's primary empirical support.

In this context, it is important to note that similar information loops frequently occur in operational models (e.g., Block Models Short Term) that often rely on lower-quality or field-controlled data. To fully understand this connectivity, further analysis is required to categorize the data by specific developmental stage. This is because data undergoes multiple transformations at different technical levels throughout the mining cycle.

6.2 Information Value and Decision Leverage

Chapters 4–5 present a scheme for mapping the flow of information and evaluating the data that supports many of the models used throughout the MVC. Assessing each model's ranking relative to its underlying data identifies several key models that, despite being routinely used to inform economic assessments, mine planning, processing, and tailings and waste management, are based on data of lower confidence. In addition to the assessment for each model and the detailed calculations of density-sampling error (section 5.4), the discussion below focuses on the economic consequences of inaccuracies arising from inefficient sampling or from connectivity between departments. It demonstrates how mischaracterization of this fundamental parameter can propagate substantial errors throughout the value chain.

6.2.1 Cascading Impacts of Density Beyond Resource Estimation and improvements

The economic influence of density uncertainty is illustrated through calculations presented in Chapter 5 (Section 5.4). These calculations adopt conservative assumptions and demonstrate that relatively small percentage errors in density can translate into large absolute economic

impacts. Even density uncertainties commonly considered acceptable (3–5%) correspond to economic uncertainty exceeding US$100 million for medium-sized deposits.

However, density mischaracterization propagates through multiple stages of mine planning, producing compounding effects on waste management and tailings infrastructure. A modest 3.7% density error (2.70 vs. 2.60 t/m³) can increase handled volume by approximately 1.5 million m³ (Appendix D), exceeding design capacity and imposing additional pressure on tailings storage facilities (TSFs). Historical analyses of major TSF failures between 1990 and 2010 revealed average losses of US$543 million per incident, with many failures occurring at facilities "pushed to unplanned heights to accommodate additional production not anticipated when tailings dams were originally designed" (Bowker & Chambers, 2015).

The 2014 Mount Polley tailings dam failure in British Columbia, Canada, exemplified how unexpected amount of material can trigger catastrophic outcomes and underscored the growing global environmental risk posed by increasing waste-to-ore ratios and vulnerability to extreme hydrometeorological events (Hudson-Edwards et al., 2011; Independent Expert Engineering Investigation and Review Panel, 2015; Byrne et al., 2018). The collapse was ultimately attributed to foundation failure through an unrecognized glaciolacustrine (GLU) clay/silt layer underlying the embankment, a weak geological horizon that had not been adequately characterized during design (Independent Expert Engineering Investigation and Review Panel, 2015). Thus, additional geological and geophysical data at the geotechnical characterization stage, including detailed subsurface logging, continuous sampling, and geophysical profiling, could have helped identify this hazard earlier and potentially prevented the disaster.

Unplanned TSF expansions typically require new permitting approvals, redesign, and significant capital investment, all of which increase schedule and cost risks (Bowker & Chambers,

2015). Mitigating these possible cascading risks requires treating density as a first-order geological variable, subject to the same sampling rigor and QA/QC as assay data. Recent advancements in hyperspectral scanning, gamma-ray attenuation, and X-ray computed tomography provide opportunities to collect continuous bulk-density profiles (Tonai et al., 2019; Bauer et al., 2021; Kamath et al., 2024; Thiele et al., 2025), while 3D scanning combined with mass measurements can yield volumetric estimates (Betlem et al., 2020). XRF and LIBS enable composition-based density proxies through normative mineralogical modeling (Abers & Hacker, 2016; Escolme et al., 2019). Emerging technologies such as muon tomography can image subsurface density variations over volumes far exceeding drill-hole coverage and provide meter-scale opportunities to assess density for orebody and geotechnical characterization (Zhang et al., 2020). This technology has been deployed at Rio Tinto's Bingham Canyon Mine to "achieve measurable improvement in density estimation at meter-scale precision over tens of millions of tonnes" (Ideon, 2024). More broadly, emerging geophysical technologies, such as portable high-quality electromagnetic (EM) systems, are expanding near-surface and deep orebody characterization capabilities, enabling rapid acquisition of resistivity and conductivity data that are valuable for geotechnical and geological modelling. In parallel, advances in inversion algorithms and large-scale geophysical processing are significantly improving the resolution and interpretability of subsurface imaging (Lelièvre & Farquharson, 2016). Together, these innovations illustrate how emerging sensing and computational technologies can enhance orebody knowledge by providing higher-resolution, spatially continuous subsurface information.

The density model example provides a concrete illustration of how the GiM Map can be applied to identify high-leverage geological inputs and trace their impact on downstream calculations. Although the quantitative outcomes are specific to the case considered, the example

validates the practical utility of the framework as a decision-support tool for prioritizing data acquisition and refinement. The broader GiM Map structure and workflow are therefore intended to be transferable, even though individual results remain site-specific.

6.2.2 Economic Implications and the Value of Information Approach

The gaps identified in this study translate directly into the economic inefficiencies observed through the MVC by numerous authors (e.g., Dominy, 2016; McCarthy, 2003; Wood, 2018). A quantitative assessment of how well integrated a dataset is (measured as the number of effective connections per data node) increases from 2.9 during exploration to 6.6 during mine plan optimization, reflecting greater data integration as operations progress from conceptual to production stages.

This increasing connectivity is not merely descriptive; it represents increasing potential for uncertainty propagation. When geological and operational data are effectively shared across multiple models, their decision-support capacity increases, but so does their systemic influence. Conversely, weakly supported inputs in highly connected nodes amplify uncertainty across the value chain.

The cumulative model input score analysis (Table 5.2) pinpoints where strategic investments in data quality can yield the highest returns. Low-confidence domains, such as density models (15 points), have disproportionate economic leverage, as demonstrated in the tonnage example outlined above. Therefore, improving the amount and quality of data supporting density models can reduce reconciliation errors equating to millions of tonnes (and dollars). Similarly, improving clay models (76 points) through enhanced spectral or XRF analysis can reduce variability in mineral processing and unplanned downtime.

These relationships establish a quantitative basis for assessing the Return on Information (RoI) of geological data collection. Within the GiM framework, RoI is understood as the economic benefit derived from strengthening the input support of models that exhibit high downstream connectivity. Higher input support models (e.g., recovery and throughput) directly impact costs and revenue. However, lower-ranked but highly influential models (e.g., density, RQD, clay proxies) may exert systemic control over tonnage estimation, slope stability, environmental risk, and infrastructure design. Enhancing support for these models mitigates uncertainty amplification, improves tonnage accuracy, reduces environmental liabilities, and increases financial confidence. Integrating model weighting and connectivity into economic evaluations bridges technical uncertainty with financial performance, positioning geological information as a strategic asset rather than a purely technical input (Goodfellow & Dimitrakopoulos, 2017).

While outside the scope of this thesis, future work could operationalize the Value of Information (VoI) approach developed here. Unlike traditional VoI applications that evaluate isolated parameters, the GiM-based VoI framework treats geological inputs as interconnected uncertainty-bearing nodes within the MVC network. The approach would involve four key steps:

1. Mapping model inputs to uncertainty priors, classifying models according to the strength and quantity of supporting data, and calibrating them using reconciliation information.
2. Linking models to economic outcomes, tracing uncertainties in parameters such as density, recovery, and geotechnical parameters propagate through tonnage, revenue, cost models, and risk.
3. Quantifying uncertainty propagation via sensitivity or Monte Carlo analysis to determine its effect on Net Present Value (NPV) confidence intervals.

4. Calculating information value by comparing uncertainty distributions before and after data-quality improvements, e.g., enhanced density sampling, Bayesian approaches, or expanded metallurgical testing.

This system enables evidence-based prioritization of data investments, focusing financial and human resources on reducing uncertainties that exert the greatest financial, environmental, and community impacts. The GiM Map's quantified connections reveal where uncertainty amplification occurs, particularly in high-connectivity nodes such as grade, density, and recovery, thereby identifying leverage points for maximum information value.

VoI methodologies have been applied in isolated domains such as exploration (Eidsvik et al., 2015), grade control (Dimitrakopoulos & Godoy, 2014), and metallurgy (Lishchuk et al., 2020). The GiM Map extends these approaches by offering a holistic, value-driven perspective for the entire MVC, providing insights into value and risk beyond isolated domains. When calibrated with site-specific reconciliation and performance data, the VoI approach, combined with the learnings from GiM, enables information integration to become an instrument of operational optimization, aligning technical certainty with financial, environmental, and social performance and sustainable capital allocation.

6.3 Integration Challenges across the Mining Value Chain

While the GiM Map enabled the identification of formal connections between data, models, and processes, interviews with industry experts were essential for understanding how these connections function, or fail to function, in practice. Through these discussions, several integration challenges emerged consistently across departments (geology, mine planning, metallurgy, geotechnics, and environmental management) and across different levels of expertise. The

repeated appearance of these challenges suggests that they are not isolated issues, but rather structural and organizational barriers within mining operations.

A key insight emerging from this research is that successful integration of orebody knowledge depends less on technological capability and more on overcoming organizational and cultural constraints. Mining companies have invested heavily in data collection systems (such as hyperspectral scanners, XRF analyzers, automated mineralogy, and real-time sensors), but less in the institutional infrastructure and collaboration needed to manage and share these data (Olson & Frenzel, 2022). As a result, valuable information is often isolated within departments or sites, limiting its use for decision-making across the MVC.

This section presents four main integration challenges identified through the combined analysis of the GiM Map and expert interviews: (1) data infrastructure gaps, (2) technology implementation, (3) disconnects between long- and short-term models, and (4) limited environmental integration.

6.3.1 Data Infrastructure: The Hidden Bottleneck

Mining's data infrastructure has not kept pace with the rapid expansion of sensing and analytical technologies. Geological, metallurgical, and environmental datasets are often stored in incompatible or legacy systems, limiting accessibility and integration. These challenges are compounded by the long-lived nature of mining operations, where active mines may have operational histories exceeding 100 years and new developments routinely depend on data collected 20 years or more prior to production. Over such timeframes, fragmented data management leads to duplicated effort, erosion of institutional knowledge, and inconsistent decision-making across the mining value chain.

The diversity of all data (including empirical, interpretive, and calculated) creates additional complexity. Each type is often managed and stored in separate systems with different rules and access levels. Without integrated, interoperable databases, relationships between data types are lost, and the potential for cross-disciplinary analysis remains unrealized.

This problem increases at the corporate level. Large mining companies often struggle to access information from previous projects or sister operations. Data may be scattered across legacy databases, vendor systems, or departmental repositories with limited access. In addition, depending on the company's policies, some internal functions are outsourced to contractors, who manage the information they collect separately from official repositories, creating additional problems and introducing bias into data manipulation and interpretation. This leads to an inability to use existing information effectively, or even to recognize its presence, causing teams to redundantly collect information they already have.

Some companies are now building enterprise-wide data systems to centralize geological, geotechnical, and metallurgical data. However, these projects face major technical and organizational challenges, including standardizing data formats, reconciling domain terminology, and maintaining governance across multiple sites.

6.3.2 Technology implementation and challenges

A significant operational challenge is implementing new technologies and innovations. Existing systems often cannot process high-resolution models or machine-learning outputs. This limitation arises from software incompatibility, inadequate investment in workflow redesign, persistent legacy workflows, and, in some cases, limited understanding of how newly generated information connects with existing operational systems. As a result, technologies and high-

resolution models that perform well in controlled or pilot settings frequently fail to deliver value at the site level. Establishing clear implementation pathways is therefore essential to avoid significant financial losses from technologies misaligned with site-specific conditions or operational realities.

These implementation challenges are further compounded by organizational and communication challenges. In many operations, there is a clear separation between those responsible for data interpretation and those collecting data in the field. Data interpretation is typically carried out by vendors, contractors, or technical service providers, while data acquisition remains the responsibility of site-based teams. This separation often leads to the development of "black box" models that are difficult to replicate, validate, or update. Even when implemented, as operating conditions evolve, such models quickly lose operational relevance, limiting their long-term usefulness.

This disconnect is reinforced by the absence of systematic reconciliation and post-implementation assessment in many operations. Without consistent evaluation of model performance against operational outcomes, it becomes difficult to determine whether high-quality models justify changes to existing infrastructure or decision-making processes, or whether they deliver measurable improvements in practice.

In this context, geological information frameworks, such as the GiM framework, become particularly valuable. By explicitly mapping how data is generated, transformed, and consumed, these frameworks clarify the human, technical, and informational connections required for successful implementation. This enables teams to assess whether new information can be realistically integrated into existing workflows.

When significant implementation challenges arise, it is necessary to move beyond a narrow business-case perspective focused solely on problem definition, solution selection, and contractual delivery. Instead, a system-level view is required (Figure 6.1), one that maps the information pathway associated with a given technology. This includes identifying operational users of model outputs, relevant software platforms, the transformation of empirical data into calculated information, and the potential barriers to adoption. Direct engagement with operational teams is equally important to ensure that solutions address actual operational needs rather than assumed ones.



Figure 6.1 Conceptual implementation pathway for technology and model deployment. The flowchart illustrates an information-centric, people-aware approach that begins with identifying an operational problem and progresses through identifying disciplines, benchmarking technologies, and mapping information flows, teams, and workflows.

The pathway emphasizes aligning data inputs, software environments, and operational users before selecting technology and developing the model. Implementation is treated as an iterative process, incorporating feedback and reassessment to ensure continued operational relevance and model credibility.

With this understanding in place, models and technologies can be developed with clearer and more practical implementation pathways, increasing the likelihood of operational success. A critical factor in successful implementation is establishing a structured feedback loop. Implementation should extend beyond product delivery, with continuous feedback from operations used to evaluate performance, maintain confidence in the model, and guide timely updates as conditions change. In the absence of such feedback, even technically robust models often lose credibility and are ultimately abandoned.

6.3.3 The Long-Term and Short-Term Model Disconnect

Another important challenge in MVC involves the disconnect between long-term and short-term operational models. Both models should represent the same orebody but were developed by different teams with different objectives, software, and update cycles. This leads to inconsistencies that can have direct financial and operational impacts by propagating divergent assumptions into planning, scheduling, and performance forecasting (Carrasco et al., 2004; Parker, 2012; Rossi & Deutsch, 2014; Dominy, 2016).

Model weighting analysis (Table 5.2) highlights this gap in relative terms (e.g., Resources Block model Long Term and Block Model Short Term), showing that long-term resource block models receive high input support due to rich empirical datasets, while short-term operational models score lower despite being closer to real-time production. According to the connections identified in the GiM Map, long-term models (e.g., Resources Block, lithology, alteration models) serve as data inputs for short-term models and, most of the time, provide a basis for increasing

data density and resolution. However, the main geological or geotechnical features identified in short-term activities should feed back into long-term models (e.g., Lithology, structural, and clays), but this kind of interconnectivity often does not exist.

When conditions encountered during mining differ from model predictions, a clear trigger and feedback mechanism is needed. In practice, this trigger often arises when short-term reconciliation deviates beyond predefined thresholds (e.g., grade, hardness, fragmentation, dilution, recovery, or structural discrepancies). At this point, data from blast-hole assays, mapping, or plant performance should update the short-term model and initiate a structured review of long-term domains, assumptions, or estimation parameters.

In this context, three outcomes are commonly observed:

1. Reconciliation discrepancies are often interpreted as geological complexity, but frequently arise from inconsistencies between long-term and short-term models (Parker, 2012; Richard & Sulemana, 2015).
2. Metallurgical performance may deviate from expectations when short-term or grade-control models do not adequately represent the ore variability characterized in long-term or geometallurgical models, particularly where heterogeneity is spatially or temporally smoothed (Parker, 2012; Dominy, 2016; Navarra et al., 2018).
3. Geotechnical or structural issues may arise when structural and rock-mass information included in resource models is not fully incorporated into operational design and short-term planning (Potvin et al., 2020; Gholami et al., 2024)

In this context, it is clear that the two types of models are not intended to be equivalent, as they differ in data quality, spatial scale, and the nature of their underlying inputs. While the resulting domains are expected to be broadly comparable and to represent similar geological

variability, discrepancies can arise when the models diverge significantly. Issues become particularly evident when high-value attributes (e.g., clay content or alteration intensity) differ markedly between models, or when their spatial distributions are inconsistent with site observations and geological understanding. Such mismatches can indicate limitations in data density, scale compatibility, or modelling assumptions, and may propagate uncertainty into downstream interpretations and design decisions.

These challenges are primarily organizational rather than technical. Different reporting structures and performance indicators prevent shared responsibility for model accuracy. Resource geologists (who compile long-term resource models) focus on reserve classification and long-term accuracy, while mine geologists (who compile short-term models) prioritize daily grade control and production support. It has even been observed that this disconnection becomes more pronounced when third parties are outsourced to collect field data for short-term models. A more integrated approach would assign joint accountability between resource and mine geology teams, using shared reconciliation metrics and regular model updates.

6.3.4 Environmental Integration: From Reactive to Proactive

Environmental and geochemical data are relevant across all stages of mining; however, their integration into planning and operations remains limited. Environmental information is often introduced late in the process, after mine designs have been finalized, which reduces opportunities for prevention and increases remediation costs (Parbhakar-Fox & Lottermoser, 2015).

The GiM Map shows that technologies such as hyperspectral scanning and XRF can identify both acid-forming and neutralizing minerals early in the exploration stage (Frenzel et al., 2023). Despite this capability, these datasets rarely reach environmental teams in time to influence key decisions such as pit design, waste placement, or waste management strategies (Franks et al.,

2011). As a result, environmental control is typically reactive rather than proactive. Stronger integration of geochemical and mineralogical data with environmental models would allow earlier prediction of acid-forming materials and support improved waste segregation and blending strategies.

A more detailed analysis reveals that geoenvironmental models often exhibit a “silent gap”: they are not prioritized until adverse operational conditions emerge. In practice, Acid–Base Accounting (ABA) models are commonly developed reactively rather than proactively. This information has significant implications for mine planning. For instead, as early access to this information can influence waste dump locations based on the distribution of acid-generating materials and inform Acid Rock Drainage (ARD)- related inputs to hydrological models. On other hand, ABA models can inform metallurgical and tailings management decisions where acidic water may negatively affect flotation performance.

Beyond risk reduction, early integration of environmental data also provides clear economic benefits. Identifying and managing reactive materials during active mining, when equipment and material movement are already available, is substantially more cost-effective than post-closure remediation. Early use of on-site neutralizing materials can reduce the need for imported remediation materials and long-term monitoring. In this way, integrating environmental data from the exploration stage onward shifts environmental management from a compliance-driven cost to a value-generating component of the mining value chain.

6.4 Organizational Culture and Collaboration across MVC

Industry consultations revealed that many practitioners lack awareness of how their work connects to upstream data sources or downstream applications. This finding aligns with previous observations of persistent departmental silos in mining (Kunz et al., 2017; Lishchuk et al., 2015;

Dominy et al., 2018; Olson & Frenzel, 2022). Interviews with geologists, engineers, and metallurgists showed a bidirectional knowledge gap: data producers often could not trace where their information was used, while downstream users lacked understanding of the assumptions and limitations behind the data they relied on.

A significant contributor to this gap is the "language barrier" within mining operations. Differences in terminology, software, and disciplinary culture create parallel professional worlds that hinder collaboration. Geologists and engineers may describe the same material differently or prioritize distinct variables, limiting shared interpretation. Consequently, technically valuable initiatives (such as new mineralogical models or plant optimization systems) are sometimes abandoned because recipients cannot readily integrate them into established workflows. The perceived disruption and uncertainty of adopting new technologies often outweigh the expected benefits. Without clear communication of advantages and practical implementation pathways, even robust technical solutions fail to gain acceptance and adoption.

The underlying causes extend beyond data management to organizational design and incentive structures. Mining operations are organized by technical specialization, with separate departments (such as geology, mining, metallurgy, and environment), each evaluated and incentivized on independent performance metrics. These siloed objectives discourage collaboration and can even penalize time spent on integration efforts (Carrasco et al., 2004; Lund & Lamberg, 2014; Kunz et al., 2017; Lishchuk et al., 2020). Organizational studies have shown that technical alignment alone does not guarantee effective communication across departments. For example, a social network analysis conducted by Kunz et al. (2017) within a mine-site hydrogeological context demonstrated that, even when personnel shared similar professional backgrounds and were connected through multiple communication workflows (e.g., information

exchange, problem-solving, and informal networks), direct communication on water-related issues remained limited. The study further revealed that water management in the mine was primarily perceived as an environmental rather than a production-critical topic, with production objectives consistently taking precedence. As a result, water-related risks were often managed within specialized functions rather than being integrated into day-to-day operational decision-making, reinforcing organizational silos and constraining cross-departmental learning.

It is important to emphasize that, although the GiM Map reveals a high density of technical connections between models and processes, human and organizational barriers, including interdepartmental communication gaps, misaligned objectives, differing planning horizons, and cultural silos, cannot be inferred directly from the map itself. These dimensions emerged exclusively through the qualitative component of the research, demonstrating that information integration is not solely a technical challenge, but also an organizational one

Achieving truly integrated orebody knowledge, therefore, requires not only standardized data systems but also multidisciplinary cooperation, communication, systems thinking, and, more importantly, company policy changes. Better orebody understanding emerges from collective engagement and shared learning, rather than from data volume alone, positioning the integration of data across the MVC as both a cultural and operational necessity for responsible, knowledge-based mining.

Chapter 7: Conclusions

As the global transition toward electrification and low-carbon technologies accelerates, mining faces simultaneous demands for higher production and improved environmental and social performance. Meeting these dual challenges requires a shift from production-first optimization towards knowledge-based operations in which orebody understanding informs technical and managerial decisions. This thesis shows that mining performance, resilience, and sustainability are materially shaped by how effectively geological information is integrated, transferred, and reused across the MVC. By mapping information dependencies and connectivity, the GiM Map reframes geology from a static record into a decision-support system that links technical choices to operational, environmental, and financial outcomes.

By classifying information types, assigning ordinal certainty weights, and quantifying connectivity, the GiM Map provides a reproducible framework for applying VoI approaches across mining stages. The results indicate that models with stronger empirical and test-based input support are typically associated with more frequent validation and feedback, particularly in downstream operational contexts. In practice, strengthening orebody knowledge and improving information continuity can support more reliable planning, more stable processing performance, and more defensible environmental management decisions.

A very important component of this research involved industry interviews conducted during the development of the GiM Map. While the interviews provided qualitative insights into how information flows between disciplines, additional challenges became evident through the process of constructing and systematically quantifying the GiM Map itself. By mapping information nodes and tracking connections across the MVC, the GiM Map revealed structural gaps, limited cross-disciplinary linkages, and asymmetric data exchange patterns. These results

support the interview-based observation that the primary barriers to integration are often organizational and cultural rather than technological. Despite advances in sensing, automation, and analytics, the GiM Map quantitatively demonstrates that fragmented workflows and departmental silos continue to constrain information reuse and limit integrated decision-making across disciplines.

The GiM Map positions geological information as a strategic enterprise asset whose value increases when datasets, assumptions, and feedback loops are shared across disciplines. Improved interoperability and collaboration can act as a knowledge multiplier, often without requiring proportional increases in data acquisition, by increasing the reuse and decision leverage of existing information. The GiM Map has limitations. Certainty weights were applied using equal-interval ordinal categories, and the framework does not yet account for derivation depth (i.e., the interpretive distance from primary measurements or multiple stages of data transformation), uncertainty propagation, or differences in decision criticality across nodes. As a result, cumulative input scores represent informational support rather than statistical model reliability. Future work should incorporate probabilistic methods, site-calibrated reconciliation data, and alternative weighting schemes, and should adapt the framework to additional deposit types, mining methods, and processing routes.

Ultimately, integrated orebody knowledge is both an economic and a sustainability imperative. The development of a new generation of geoscientists and engineers trained in holistic, cross-disciplinary collaboration is needed. In parallel, internal mobility programs that allow employees to temporarily work within other departments within an operation can further strengthen cross-department collaboration and organizational learning. Mining operations that systematically connect people, processes, and data across geological, metallurgical, and

environmental domains will enhance productivity, reduce uncertainty, and maintain social license to operate, enabling more responsible, efficient, and future-ready mining.

7.1 Future work

While this study establishes the structural foundation of the GiM framework, further research is required to advance it into a fully quantitative uncertainty propagation and decision-support system. The GiM Map provides a systems-level representation of how geological information flows across the Mining Value Chain (MVC). However, additional work is needed to generalize, quantify, and operationalize the framework.

A first priority is extending the GiM Map beyond flotation-based sulfide systems to include heap leaching and hydrometallurgical operations. Heap leaching introduces additional parameters such as permeability, hydraulic conductivity, acid consumption, leach kinetics, and solution management. These variables create stronger interactions between geological, geotechnical, hydrogeological, metallurgical, and environmental domains. Expanding the framework to incorporate these processes would increase its applicability to oxide and mixed deposits and would require structured collaboration with hydrometallurgical specialists and site-based operators.

Second, the density uncertainty example presented in Appendix B demonstrates that small parameter errors can produce significant economic impacts. Future research should expand this approach by developing a multi-parameter Monte Carlo framework that simultaneously evaluates density, recovery, throughput, geotechnical variables, and commodity price variability. A system-level stochastic simulator would allow ranking of upstream uncertainties according to their economic sensitivity, thereby guiding prioritization of sampling programs, sensing technologies, and model refinement efforts.

Third, future research should focus on models that exhibit low input support but high operational consequence. The connectivity ranking developed in this thesis indicates that certain geotechnical models, particularly the Rock Quality Designation (RQD) model, receive limited empirical input relative to downstream models such as recovery and throughput. However, RQD influences slope stability, blast design, fragmentation, dilution, and indirectly, mill performance and energy consumption. A dedicated study should isolate the RQD information pathway and quantify how uncertainty in geotechnical parameters propagates into fragmentation distributions, throughput variability, and economic outcomes. Such work would strengthen the Value of Information (VoI) logic by identifying where targeted data acquisition could materially reduce operational risk.

Fourth, the stages of data transformation identified in this thesis—geological acquisition, modeling, domaining, interpolation, block model construction, secondary model development, and feedback loops—should be formalized into a quantitative framework. Each stage introduces assumptions, scale effects, and potential information distortion. Future work should evaluate how uncertainty is amplified, reduced, or redistributed across these transformation stages. Incorporating probabilistic methods and reconciliation data would strengthen the theoretical basis of the GiM framework and support more rigorous VoI assessments.

Finally, a longer-term research direction involves integrating the GiM Map into a dynamic digital environment linking block models, plant performance data, and reconciliation results. Such integration would support adaptive planning and continuous updating of uncertainty estimates, moving the framework from static mapping toward a real-time decision-support tool.

Overall, the next phase of research should focus on quantification, uncertainty propagation, and economic prioritization. By advancing from structural information mapping to stochastic

system modeling, the GiM framework can evolve into a practical tool for guiding information investment, improving cross-disciplinary integration, and supporting risk-informed decision-making across the mining value chain.

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Appendices

Appendix A

A 1 General Interview Questions

1. What information do you generate in your role?
2. What are your main inputs of information?
3. What are your main outputs of information?
4. What geological information do you receive that you actually use as input in your workflows?
5. How do you receive this information (e.g., 3D models, technical reports, Excel files, databases)?
6. Once you receive this information, how do you integrate it into your workflow?

   Do you work with raw data, or does the process include estimation, interpolation, or additional analysis?
7. What software do you use, and is it compatible with the software used by the geology, geotechnical, or hydrogeology teams?
8. What geotechnical information do you use in your workflow?

   In what format do you receive it (e.g., reports, models, databases), and how do you integrate it?
9. What information do you receive from the hydrogeology or environmental teams?
10. What information do you provide to metallurgy, and in what format?
11. What are the main challenges you face when working in an integrated manner with other technical teams?

12. Do you actually use the information provided by other departments, or is there limited information transfer between teams?
13. What are the main operational bottlenecks?
14. What is the biggest challenge you face when performing reconciliation?
15. What is your opinion of integrated operations systems?

    Are they truly integrated?

    Do they perform better than operating without such systems?
16. How do you hand off your information? How is reporting managed?
17. What types of permits are relevant to your work (social, environmental, mining-related, or a combination)?

    At what stage of the mining operation are these required?
18. What are the biggest challenges in connecting your information with other departments?

A 2 Example specific questions related to Hydrological parameters. This include initial conversations and follow-up questions.

I'm working on a flowchart summarizing the key information involved in mining operations. I have mapped the hydrogeological tests and their connections to the models (attached in the photos). I would like to understand **where depressurization is used in these models**. From the presentation, I noted that many people mentioned depressurization for slope stability. I understand that dewatering is a key part of depressurization, as removing water helps lower pressure. Effective depressurization planning requires strong integration of hydrogeology, geotechnical engineering, and mine planning.

1. Could you explain how you include this information in your models, and how it is delivered to the geotechnical or mine planning teams? Is it shared as a model, a report, or through another process?
2. From your perspective, what specific key information is essential here? I'm not asking for extensive detail—just enough to understand how to correctly link hydrogeological data with geotechnical and mine planning work.
3. Based on your work experience, how can current hydrogeological models be improved to enhance their integration and relevance for decision-making across other stages and stakeholders in the mining value chain?

Regarding the permitting constraints you mentioned: you noted that having good-quality baseline information available can make the permitting process easier.

1. Could you clarify what type of information you are referring to, models, hydrogeological tests, water chemistry? I've seen that much of the data used to build models requires hydrological campaigns, which themselves require permits.
2. On the other hand, what kind of permits do you mean: social, environmental, mine-related, or a combination? At what stage of the mining operation is this needed?
3. What are the consequences if the baseline information is lacking or if permitting is not completed on time and in the proper way?

Finally, I would like your opinion on the challenges of working in an industry with many specialized, siloed areas. What challenges have you experienced when working in multidisciplinary teams—such as differences in background, terminology between geologists and engineers, or in model and database structures?

Appendix B

Python code used to classify the information in the CSV files: Node file and Edge file. This code counts the number of connections for each listed model and separates the data and weight according to the classification methods explained in Chapter 3.

```
import pandas as pd
import re
import numpy as np
from pathlib import Path

# ================================
# Path Configuration
# ================================
path = "/content/nodes"
NODES_PATH = path + "/Nodes GiM map_complete_v.xlsx"
EDGES_PATH = path + "/Edges_GiM_map_filtered_v2.csv"

NODE_ID_COL = "Id"
NODE_LABEL_COL = "Label"
EDGE_SOURCE_COL = "Source"
EDGE_TARGET_COL = "Target"
NODE_DATA_CLASS_COL = "Data_Classification"
NODE_TEST_CLASS_COL = "Test_Classification"

MODEL_LIST = [
    "Lithological Model", "Structural Model", "Alteration/Proxies Model",
    "Ore/Minezone Model", "Calculated Mineralogical Models", "Clays Model",
    "Proxy Models", "ABA models", "Density Model", "DWI/BWI Models",
    "FF Model", "RQD Model", "PLT Model", "Geotechnical Models",
    "Geotechnical Block Model", "NP Model", "Flotation Model",
    "Hydrogeological Modeling", "Mine Geological Model", "Mining Block Model",
    "Geomet Models", "Recovery Models", "Throughput Model",
    "Resources Block Model Long Term", "Block Model Short Term", "Dynamic Block Model"
]

OUT_CSV = path + "/GiM3_Final_Report_Percentiles_75_25.csv"
```

```
# Manual ID Overrides to ensure key models are always counted
MODEL_ID_OVERRIDES = {
    "Recovery Models": ["384"],
    "Throughput Model": ["381"],
    "Resources Block Model Long Term": ["113"],
    "Block Model Short Term": ["213"],
    "Ore/Minezone Model": ["207"],
    "Structural Model": ["91"]
}

def clean_text(s):
    if pd.isna(s): return ""
    return re.sub(r"\s+", " ", str(s).strip())

def main():
    print("Starting processing with thresholds: 25% (Low) and 75% (High)...")

    # Load data
    nodes = pd.read_excel(NODES_PATH)
    edges = pd.read_csv(EDGES_PATH)

    # Clean column names
    nodes.columns = [c.strip() for c in nodes.columns]
    edges.columns = [c.strip() for c in edges.columns]

    # Standardize IDs to strings and remove .0 suffix
    nodes[NODE_ID_COL] = nodes[NODE_ID_COL].astype(str).str.replace(r'\.0$', '',
regex=True)
    edges[EDGE_SOURCE_COL] = edges[EDGE_SOURCE_COL].astype(str).str.replace(r'\.0$',
'', regex=True)
    edges[EDGE_TARGET_COL] = edges[EDGE_TARGET_COL].astype(str).str.replace(r'\.0$',
'', regex=True)

    # Map IDs to classifications
    id_to_data = dict(zip(nodes[NODE_ID_COL],
nodes[NODE_DATA_CLASS_COL].fillna("")))
    id_to_test = dict(zip(nodes[NODE_ID_COL],
nodes[NODE_TEST_CLASS_COL].fillna("")))
```

```
    # Count connections
    out_counts = edges[EDGE_SOURCE_COL].value_counts()
    in_counts = edges[EDGE_TARGET_COL].value_counts()

    def count_strictly_in_validated(model_ids):
        if not model_ids: return [0]*10
        m_set = set(model_ids)
        in_edges = edges[edges[EDGE_TARGET_COL].isin(m_set)]
        sources = in_edges[EDGE_SOURCE_COL].tolist()

        # Data and Test counters
        d_t, d_emp, d_calc, d_int = 0, 0, 0, 0
        t_t, t_emp, t_semi, t_int, t_rob, t_fld = 0, 0, 0, 0, 0, 0

        for sid in sources:
            d_val = str(id_to_data.get(sid, "")).strip()
            t_val = str(id_to_test.get(sid, "")).strip()

            if d_val == "" and t_val == "": continue

            if d_val != "":
                d_t += 1
                if "Empirical" in d_val: d_emp += 1
                elif "Calculated" in d_val or "Derived" in d_val: d_calc += 1
                elif "Interpretive" in d_val: d_int += 1
            elif t_val != "":
                t_t += 1
                if "Empirical" in t_val: t_emp += 1
                elif "Semi-Quant" in t_val: t_semi += 1
                elif "Interpretive" in t_val: t_int += 1
                elif "HighRobust" in t_val or "Robust" in t_val: t_rob += 1
                elif "Field" in t_val: t_fld += 1

        return d_t, d_emp, d_calc, d_int, t_t, t_emp, t_semi, t_int, t_rob, t_fld

    rows = []
    for model in MODEL_LIST:
        ids = MODEL_ID_OVERRIDES.get(model, nodes.loc[nodes[NODE_LABEL_COL] ==
clean_text(model), NODE_ID_COL].tolist())
```

```
        o_sum = sum(out_counts.get(i, 0) for i in ids)
        i_sum = sum(in_counts.get(i, 0) for i in ids)
        res = count_strictly_in_validated(ids)

        total_classified_in = res[0] + res[4]

        # Calculate Weighted Robustness Score
        score = (res[1]*5) + (res[2]*3) + (res[3]*1) + (res[5]*4) + (res[6]*3) + (res[7]*1) +
(res[8]*5) + (res[9]*2)

        rows.append({
            "Model_Label": model,
            "id": ", ".join(ids),
            "Total CN": int(o_sum + i_sum),
            "OUT": int(o_sum),
            "IN": int(i_sum),
            "IN (Classified)": int(total_classified_in),
            "Data class": int(res[0]),
            "Data_Emp": int(res[1]),
            "Data_Calc_Derived": int(res[2]),
            "Data_Int": int(res[3]),
            "Test class": int(res[4]),
            "Test_Emp": int(res[5]),
            "Test_Semi_Quant": int(res[6]),
            "Test_Interpret": int(res[7]),
            "Test_HighRobust": int(res[8]),
            "Test_Field_Ctrl": int(res[9]),
            "SCORE TOTAL": int(score)
        })

    result = pd.DataFrame(rows).sort_values(by="SCORE TOTAL", ascending=False)

    # ==========================================
    # NEW THRESHOLDS: 25% (P25) and 75% (P75)
    # ==========================================
    p75 = result["SCORE TOTAL"].quantile(0.75) # Top 25% of models
    p25 = result["SCORE TOTAL"].quantile(0.25) # Bottom 25% of models

    def classify_by_new_thresholds(s):
```

```
        if s >= p75: return "Highly Robust"
        if s >= p25: return "Moderately Robust"
        return "Low Robust"

    result["Robustness_Category"] = result["SCORE
TOTAL"].apply(classify_by_new_thresholds)

    # Export to CSV
    result.to_csv(OUT_CSV, index=False)

    print(f"Report generated with new thresholds (25/75) at: {OUT_CSV}")
    print(result[["Model_Label", "SCORE TOTAL", "Robustness_Category"]].head(10))

if __name__ == "__main__":
    main()
```

Appendix C

C 1 Total number of connections (Total CN) and the classification of each edge: OUT and IN.

Rank	Model_Label	id	Total CN	OUT	IN
1	Recovery Models	384	83	19	64
2	Throughput Model	381	93	23	70
3	Resources Block Model Long Term	113	97	34	63
4	Dynamic Block Model	229	80	19	61
5	Mining Block Model	198	81	23	58
6	Flotation Model	464	61	18	43
7	ABA models	362	51	13	38
8	Block Model Short Term	213	69	21	48
9	Geotechnical Block Model	170	78	26	52
10	Geomet Models	95	46	14	32
11	Hydrogeological Modeling	247	37	9	28
12	Mine Geological Model	187	44	15	29
13	Clays Model	100	67	49	18
14	NP Model	92	32	11	21
15	Geotechnical Models	93	51	24	27
16	Calculated Mineralogical Models	94	68	55	13
17	Ore/Minezone Model	207	45	26	19
18	Proxy Models	218	38	26	12
19	Structural Model	91	54	42	12
20	DWI/BWI Models	101	29	18	11
21	Lithological Model	88	47	36	11
22	Alteration/Proxies Model	89	56	45	11
23	RQD Model	97	41	31	10
24	PLT Model	99	27	19	8
25	FF Model	98	25	19	6
26	Density Model	118	18	12	6

id= model identifier in the node file.

C 2 Results of the data classification

Rank	Model_Label	id	Data class	Data_Emp	Data_Calc	Data_Int	Data Total Score
1	Recovery Models	384	57	22	29	6	203
2	Throughput Model	381	60	21	27	12	198
3	Resources Block Model Long Term	113	53	14	28	11	165
4	Dynamic Block Model	229	52	12	29	11	158
5	Mining Block Model	198	55	11	39	5	177
6	Flotation Model	464	36	22	13	1	150
7	ABA models	362	33	9	24	0	117
8	Block Model Short Term	213	47	4	34	9	131
9	Geotechnical Block Model	170	46	4	26	16	114
10	Geomet Models	95	23	6	11	6	69
11	Hydrogeological Modeling	247	25	9	12	4	85
12	Mine Geological Model	187	28	5	21	2	90
13	Clays Model	100	16	11	4	1	68
14	NP Model	92	18	5	13	0	64
15	Geotechnical Models	93	23	3	11	9	57
16	Calculated Mineralogical Models	94	12	8	3	1	50
17	Ore/Minezone Model	207	18	3	10	5	50
18	Proxy Models	218	11	8	2	1	47
19	Structural Model	91	12	4	4	4	36
20	DWI/BWI Models	101	6	1	3	2	16
21	Lithological Model	88	10	2	5	3	28
22	Alteration/Proxies Model	89	10	2	5	3	28
23	RQD Model	97	8	1	5	2	22
24	PLT Model	99	7	1	4	2	19
25	FF Model	98	6	1	3	2	16
26	Density Model	118	5	1	1	3	11

C 3 Results of the Test-data based classification

Rank	Model_Label	id	Test Score	Data Total Score	Score
1	Recovery Models	384	28	203	231
2	Throughput Model	381	29	198	227
3	Resources Block Model Long Term	113	33	165	198
4	Dynamic Block Model	229	29	158	187
5	Mining Block Model	198	2	177	179
6	Flotation Model	464	26	150	176
7	ABA models	362	19	117	136

8	**Block Model Short Term**	**213**	4	131	135
9	**Geotechnical Block Model**	**170**	12	114	126
10	**Geomet Models**	**95**	36	69	105
11	**Hydrogeological Modeling**	**247**	9	85	94
12	**Mine Geological Model**	**187**	1	90	91
13	**Clays Model**	**100**	8	68	76
14	**NP Model**	**92**	12	64	76
15	**Geotechnical Models**	**93**	10	57	67
16	**Calculated Mineralogical Models**	**94**	4	50	54
17	**Ore/Minezone Model**	**207**	4	50	54
18	**Proxy Models**	**218**	4	47	51
19	**Structural Model**	**91**	0	36	36
20	**DWI/BWI Models**	**101**	20	16	36
21	**Lithological Model**	**88**	4	28	32
22	**Alteration/Proxies Model**	**89**	4	28	32
23	**RQD Model**	**97**	2	22	24
24	**PLT Model**	**99**	4	19	23
25	**FF Model**	**98**	0	16	16
26	**Density Model**	**118**	4	11	15

Appendix D

D 1 Scenario 1: 3% Density Error

Stepwise calculation procedure to calculate density error.

Step 1: Tonnage calculation

The first point at which density uncertainty enters the evaluation workflow is during the conversion of modeled rock volume to mass. For a fixed geological volume, tonnage is calculated as the product of volume and bulk density. As a result, any systematic bias in density translates directly into an equivalent proportional bias in tonnage.

Estimated tonnage (using assumed density):

Tonnage = Volume × Measured Density

Tonnage = 37.3 Mm^3 × 2.68 t/m^3 = 100.0 Mt

Actual tonnage (using true density): Tonnage = 37.3 Mm^3 × 2.60 t/m^3 = 97.0 Mt

Tonnage error: Error = Estimated - Actual = 100.0 - 97.0 = **3.0 Mt**

The tonnage error directly mirrors the density error because the modeled volume remains fixed while mass calculations depend linearly on bulk density. The initially estimated tonnage of 100 Mt is reduced to 97 Mt, reflecting an overestimation of 3 Mt due to density bias.

Step 2: Contained Cu calculation

Tonnage estimates are subsequently combined with grade to calculate the quantity of contained metal. At this stage, the previously introduced tonnage bias directly propagates into the contained Cu estimate, assuming the grade remains unchanged.

Estimated contained Cu:

Contained Cu = Tonnage × Grade Cu

Contained Cu = 100.0 Mt × 0.005 = 500,000 tonnes Cu

Actual contained Cu:

Contained Cu = 97.0 Mt × 0.005 = 485,000 tonnes Cu

Contained Cu error: Error = 500,000 - 485,000 = 15,000 tonnes Cu overestimation

Alternative calculation: 3.0 Mt × 0.005 = 15,000 tonnes Cu

The initially estimated Cu content of 500,000 tonnes decreases to 485,000 tonnes, representing a **15,000-tonne reduction in Cu** within the 100 Mt resource.

Step 3: Recoverable Cu calculation

Contained metal estimates are then adjusted for metallurgical recovery to determine the quantity of recoverable metal. As recovery is applied as a constant proportion, the contained metal bias remains unchanged relative to recoverable Cu.

Estimated recoverable Cu:

Recoverable Cu = Contained Cu × Recovery

Recoverable Cu = 500,000 tonnes × 0.85 = 425,000 tonnes Cu

Actual recoverable Cu:

Recoverable Cu = 485,000 tonnes × 0.85 = 412,250 tonnes Cu

Recoverable Cu error: Error = 425,000 - 412,250 = 12,750 tonnes Cu overestimation

Alternative calculation: 15,000 tonnes × 0.85 = 12,750 tonnes Cu

The estimated recoverable Cu of 425,000 tonnes Cu is reduced to 412,250 tonnes, corresponding to an overestimation of **12,750 tonnes of Cu** that would not be recovered.

Step 4: Revenue impact calculation

The final step in the propagation chain involves translating recoverable metal into projected revenue using an assumed commodity price. At this stage, the cumulative effect of density uncertainty is expressed directly in economic terms.

Gross revenue equals the quantity of recoverable metal multiplied by the commodity price:

Estimated gross revenue:

Revenue = Recoverable Cu × Price Revenue

Revenue = 425,000 tonnes × US$8,000/tonne = US$3,400,000,000

Actual gross revenue: Revenue = 412,250 tonnes × US$8,000/tonne = US$3,298,000,000

Gross revenue error (overestimation): Error = US$3,400M - US$3,298M = US$102,000,000 Alternative calculation: 12,750 tonnes × US$8,000/t = US$102,000,00

D 2 Scenario 2: 5% Density Error

This scenario evaluates the impact of a moderate increase in density bias, consistent with the upper end of the systematic error range commonly reported in the literature for porphyry systems. Using the same procedure detailed in C 1, the 5% density error is evaluated.

Parameters:

- Assumed density: 2.68 t/m³
- Actual density: 2.546 t/m³ (5% lower)
- Density error: 0.134 t/m³

Propagation of density error:

Using the same fixed modeled volume (37.3 Mm³), the application of the lower actual density results in a reduction in estimated tonnage:

- Actual tonnage: 37.3 Mm³ × 2.546 t/m³ = 95.0 Mt
- Tonnage error: 100.0 Mt − 95.0 Mt = 5.0 Mt (overestimation)

This tonnage discrepancy propagates directly into contained metal estimates:

- Contained Cu error: 5.0 Mt × 0.005 = 25,000 tonnes Cu

Applying the assumed metallurgical recovery (85%) yields:

- Recoverable Cu error: 25,000 tonnes × 0.85 = 21,250 tonnes Cu

  At the assumed long-term Cu price, this corresponds to a substantial economic impact:

- Gross revenue impact: 21,250 tonnes × US$8,000/t = **US$170 million**

D 3 Tonnage calculations for cascading effects of density error.

Using the assumed design density described in appendix D1, the planned volume requirement is calculated as:

Planned volume (based on assumed density):

Volume = Mass ÷ Density

Planned volume = 106.5 Mt ÷ 2.70 t/m³ = 39.44 Mm³

Actual volume = 106.5 Mt ÷ 2.60 t/m³ = 40.96 Mm³

Volume underestimation:

Error = 40.96 - 39.44 = 1.52 Mm³ ≈ 1.5 million m³

Relative error = (1.52 ÷ 39.44) × 100 = 3.9% ≈ 3.8%

Appendix E

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
# --- Thesis Base Parameters [cite: 757-768] ---
n_simulations = 10000
volume = 37.3e6          # m3 [cite: 759]
cu_grade = 0.005         # 0.50% Cu [cite: 762]
recovery = 0.85          # 85% [cite: 765]
cu_price = 8000          # USD/t [cite: 768]
measured_density = 2.68  # t/m3 (Current biased assumption) [cite: 760]
# --- Simulate Density Variability ---
# Normal distribution centered on 2.68 to test error sensitivity
actual_density_dist = np.random.normal(measured_density, 0.005, n_simulations)
# --- Value Chain Calculations (Millions USD) [cite: 838-842] ---
# Revenue = (Volume * Density * Grade * Recovery * Price) / 1e6
revenue_dist = (volume * actual_density_dist * cu_grade * recovery * cu_price) / 1e6
# --- Calculating Static Scenario Markers ---
threshold_densities = {
    "Under-5%": measured_density * 1.05,
    "Under-3%": measured_density * 1.03,
    "Baseline": measured_density,
    "Over-3%": measured_density * 0.97,
    "Over-5%": measured_density * 0.95
}
threshold_revenues = {k: (volume * v * cu_grade * recovery * cu_price) / 1e6 for k, v in
threshold_densities.items()}
# --- Visualization ---
plt.rcParams.update({'font.size': 11})
sns.set_style("whitegrid")
fig, ax1 = plt.subplots(figsize=(15, 8)) # Slightly wider figure
# 1. Primary Axis (Revenue)
sns.histplot(revenue_dist, bins=80, kde=True, color='lightsteelblue', alpha=1, label='Revenue Probability',
ax=ax1)
```

```
# Scenario Lines with detailed labels [cite: 847, 871]
colors = ['darkgreen', 'forestgreen', 'black', 'darkorange', 'firebrick']
labels = [
    f"-5% Underest. ({threshold_densities['Under-5%']:.2f} t/m³): ${threshold_revenues['Under-5%']:,.0f}M",
    f"-3% Underest. ({threshold_densities['Under-3%']:.2f} t/m³): ${threshold_revenues['Under-3%']:,.0f}M",
    f"Baseline ({measured_density} t/m³): ${threshold_revenues['Baseline']:,.0f}M",
    f"+3% Overest. ({threshold_densities['Over-3%']:.2f} t/m³): ${threshold_revenues['Over-3%']:,.0f}M",
    f"+5% Overest. ({threshold_densities['Over-5%']:.2f} t/m³): ${threshold_revenues['Over-5%']:,.0f}M"
]
for val, color, label in zip(threshold_revenues.values(), colors, labels):
    ax1.axvline(val, color=color, linestyle='--', linewidth=2, label=label)
# Highlight Risk Zones [cite: 843, 869]
ax1.axvspan(threshold_revenues['Over-5%'], threshold_revenues['Baseline'], color='red', alpha=0.1,
label='Overestimation Risk (Value Loss)')
ax1.axvspan(threshold_revenues['Baseline'], threshold_revenues['Under-5%'], color='green', alpha=0.1,
label='Underestimation Opp. (Found Value)')
# 2. Secondary Axis (Density)
ax2 = ax1.twiny()
rev_min, rev_max = ax1.get_xlim()
conversion_factor = (volume * cu_grade * recovery * cu_price) / 1e6
ax2.set_xlim(rev_min / conversion_factor, rev_max / conversion_factor)
# Formatting
ax1.set_title('Monte Carlo Simulation: Value at Risk vs. Density Uncertainty', fontsize=16, fontweight='bold',
pad=40)
ax1.set_xlabel('Projected Gross Revenue (Millions USD)', fontsize=13)
ax1.set_ylabel('Simulation Frequency', fontsize=13)
ax2.set_xlabel('Equivalent Bulk Density (t/m³)', fontsize=13, labelpad=15)
# ADJUSTED LEGEND POSITION: Moved further right to avoid overlap
ax1.legend(loc='upper left', bbox_to_anchor=(1.05, 1), frameon=True, borderaxespad=0.)
# Use subplots_adjust to make room for the legend on the right
plt.subplots_adjust(right=0.75)
plt.savefig('dual_axis_density_monte_carlo_final.png', dpi=300, bbox_inches='tight')
plt.show()
```